

Virtuoso[®] Layout Pro: T6 Constraint-Driven Flow and Power Routing

Course Version IC 25.1

Lecture Manual

Revision 1.0

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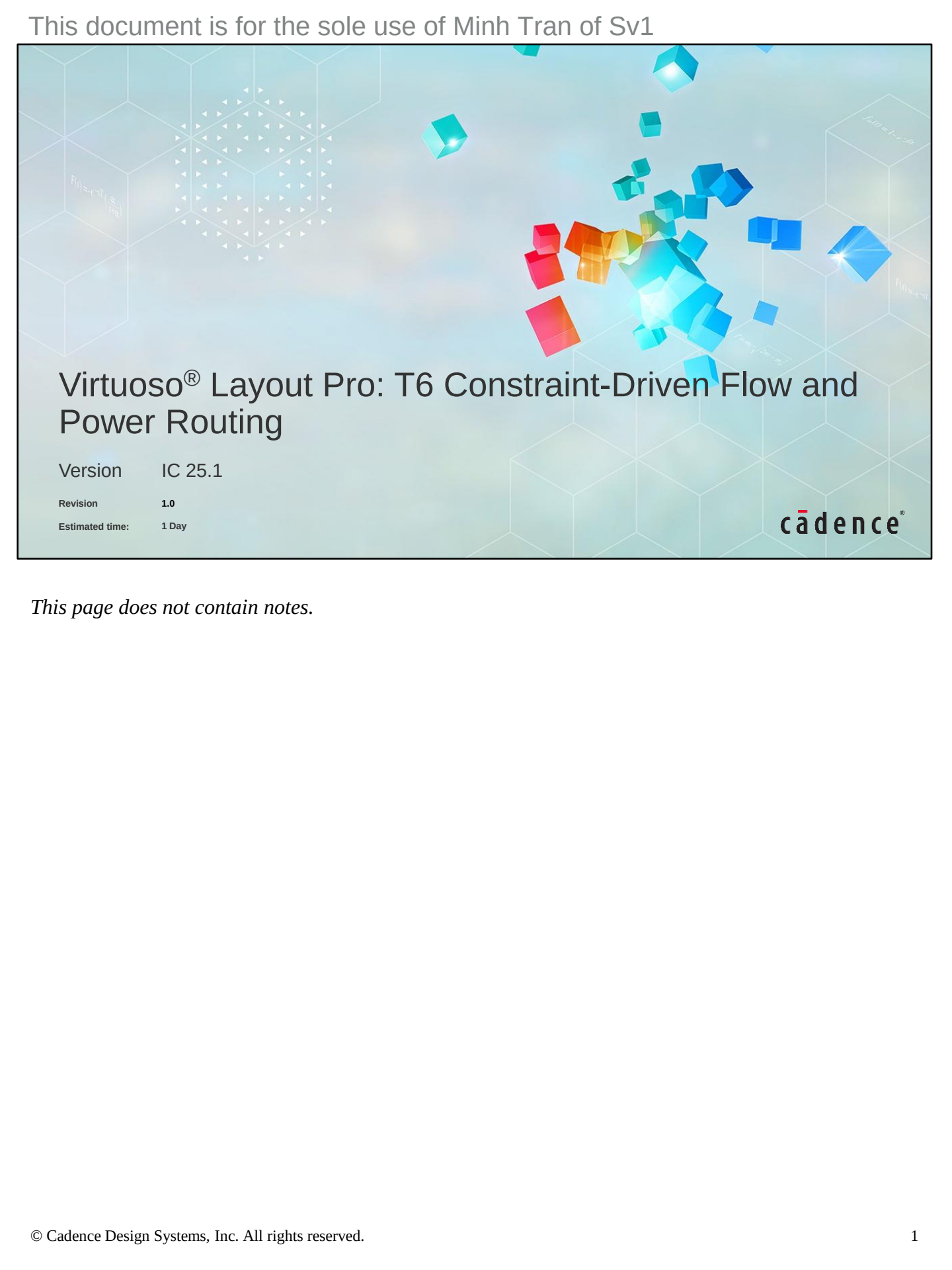
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Virtuoso® Layout Pro: T6 Constraint-Driven Flow and Power Routing

Version IC 25.1

Revision 1.0

Estimated time: 1 Day

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Module 1

About This Course

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Course Prerequisites

Before taking this course, you need to have already completed the following courses:

- [Virtuoso® Layout Pro: T1 Environment and Basic Commands](#)
- [Virtuoso Layout Pro: T2 Create and Edit Commands](#)
- [Virtuoso Layout Pro: T3 Basic Commands](#)
- [Virtuoso Layout Pro: T4 Advanced Commands](#)
- [Virtuoso Layout Pro: T5 Interactive Routing](#)

Or you need to have

- Experience with layout design
- Knowledge of schematic symbols and MOS devices
- A basic knowledge of UNIX/Linux



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Course Objectives

In this course, you

- Analyze the constraint-driven flow
- Set up and run power routing



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Course Agenda

- About This Course
- Constraint-Driven Flow
 - Analyzing VSE XL and VLS XL Features: Usability Improvements in the Constraint-Driven Flow
 - Performing Hierarchical Constraint Propagation
 - Analyzing the Rapid Analog Prototype (RAP) Category in the Circuit Prospector and Modgen
- Power Routing
 - Power Routing Using VSR
 - Signal Routing
 - Connectivity Verification and DRCs
- Course Conclusions
- Next Steps



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Software and Licenses

For the software and licenses used in the labs for this course, go to:

https://www.cadence.com/en_US/home/training/all-courses/86120.html

If there is additional information regarding the specific software, it is detailed in the lab document and/or the README file of the database provided with this course.



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How do I register to take the exam?

- Log in to our [Learning Management System](#), click on the course in your transcript, and go to the **Content** tab to locate the exam.

How long will it take to complete the exam?

- Most exams take 45 to 90 minutes to complete. You may retake the exam multiple times to pass the exam.

How do I access and use the digital badge?

- After you pass the exam, you get a digital badge and instructions on how to place it on social media sites.

How is the digital badge validated?

- [Credly](#) validates the digital badge as issued to you by Cadence and includes the details of the criteria you completed to earn the badge.

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Icons Used in This Class



Best Practice



Language/
Command Syntax



Concept/
Glossary



Frequently Asked
Questions/
Quiz



Error Message

Throughout this class, we use icons to draw your attention to certain kinds of information. Here are the icons we use, and what they mean.



Problem & Solution



Quick Reference



GUI and Command



How To



Lab List

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Module 2

Constraint-Driven Flow

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Module Objectives

In this module, you

- Create Rail constraint supported by AAP and CAE
- Create a Cell Boundary constraint that includes Area Utilization
- Identify usability improvements in the Constraint Manager
- Propagate net-based constraints across the schematic hierarchy
- Create Process Rule Overrides (PRO) constraint
- Edit Process Rule Overrides (PRO) constraint
- Set up and run Rapid Analog Prototype (RAP) flow and modgen



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Useful Terms and Definitions



Terms	Definition
Analog Auto Placer (AAP)	Automatically places the layout objects using the Analog Auto Placer.
Constraint-Aware Editing (CAE)	CAE mode ensures that the Layout XL manual editing commands respect physical constraints that may be set on your design components.
Level-1 Editing (L1 editing)	You are allowed to create pin and boundary constraints in the soft blocks in the L1 editing mode.
Constraint Manager (CM)	The Constraint Manager dockable assistant pane allows for the addition, modification, and deletion of constraints from/to the Cadence® Constraint Storage System.
User Interface (UI)	It refers to the interface of the particular assistant/canvas. For example, Circuit Prospector User Interface (UI) refers to the Circuit Prospector assistant.



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Useful Terms and Definitions (continued)



Virtuoso Schematic Editor (VSE)	The Virtuoso® Schematic Editor is a design entry tool that supports the work of logic and circuit design engineers, including drafters. Physical layout designers and printed circuit board designers can use the information as background material to support their work.
Virtuoso Layout Suite (VLS)	The Virtuoso Layout Suite is an editing tool that provides the basic creation and editing capabilities to work with layout designs. The layout editor lets you create custom layouts or edit any existing layouts. It lets you work with multipart paths and hierarchical designs. You can also use the layout editor to view the connected shapes in a design.
Process Rule Overrides (PRO)	The Process Rule Overrides (PRO) can be used to override, on a given object, the generic foundry and design process rules with specific, more stringent rules as defined in Constraint Groups (which are stored in designs or technology files and act as a collection of process constraints that are applicable to the objects in a design).
Rapid Analog Prototype (RAP)	The Rapid Analog Prototype (RAP) category in the Circuit Prospector contains all the finders/generators required to automatically constrain a design with just a few mouse clicks.
Virtuoso Space-based Router (VSR)	The Virtuoso Space-based Router enables high-speed shape-based routing for physical designs.

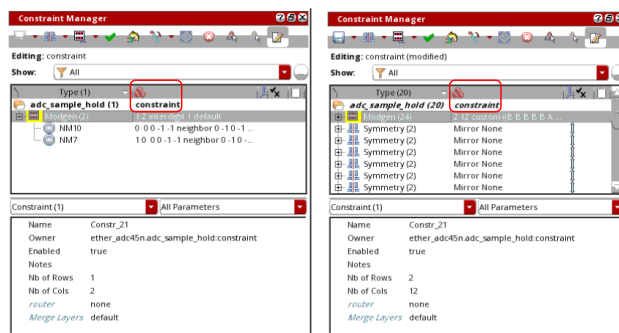
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What Are Constraints?



A **constraint** is an objective that can be used to influence the possible implementation, placement, or routing in a system. A constraint can include predefined name-value pairs that clarify the design intent and status of that constraint.

- In the context of layouts and schematics, this would be a design object with respect to its goal.
- Constraints are therefore requirements that need to be followed during design implementation.
- Constraints are stored in the layout database or in the constraint view (not the schematic view) for schematics.



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What Are Constraints?

A simple way to define a constraint is to consider it as an objective that can be used to influence the possible implementation, placement, or routing in a system. In the context of layouts and schematics, this would be a design object with respect to its goal. Constraints are therefore requirements that need to be followed during design implementation.

A constraint can have parameters that are pre-defined name-value pairs that further define the design intent and the status of that constraint.

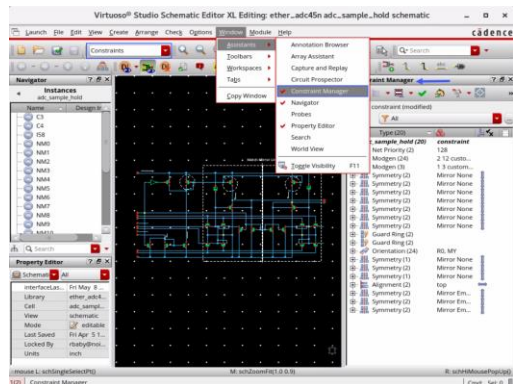
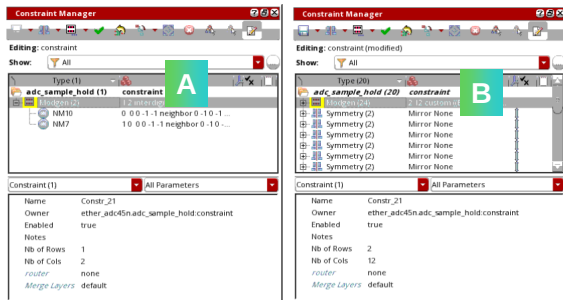
Note: Constraints are stored in the layout database or in the constraint view (not the schematic view) for schematics.

Accessing the Constraint View



From schematic, select **Window – Assistants – Constraint Manager** to dock the Constraint Manager assistant in the schematic window with its constraints in the constraint view.

- A. If the constraint view is in read-only or edit mode, the constraint names and related information is displayed in black. The name of the constraint view appears in **bold**.
- B. If the constraint view has been edited and not saved yet, its name appears as **bold italic**.



The constraint view name (default constraint) is displayed in the *Parameters* column in the *Constraint Manager*.

Accessing the Constraint View

From schematic, select **Window – Assistants – Constraint Manager** to dock the Constraint Manager assistant in the schematic window with its constraints in the constraint view.

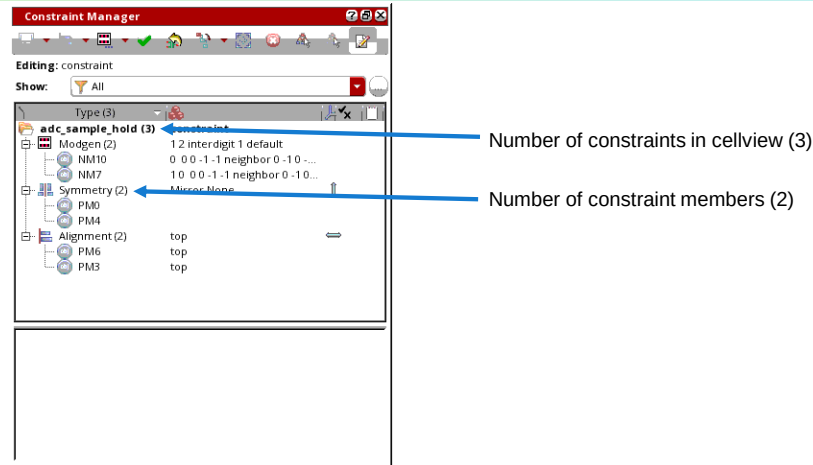
If the constraint view is in read-only or edit mode, the constraint names and related information is displayed in black. The name of the constraint view appears in **bold**.

If the constraint view has been edited and not saved yet, its name appears as **bold italic**.

The constraint view name (default constraint) is displayed in the *Parameters* column in the *Constraint Manager*.

The Constraint View: The Constraints and Their Members

You see the constraints and their members in the constraint view.



The Constraint View – The Constraints and Their Members

You see the constraints and their members in the constraint view.

In this example, the number of constraints in the cellview are three, namely, Modgen constraint, Symmetry constraint, and Alignment constraint.

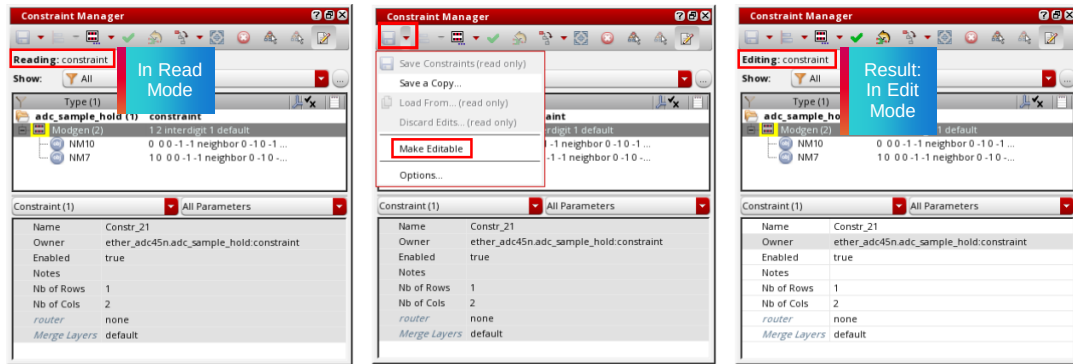
You see here that the number of constraint members in the Symmetry constraint is two.

Making the Constraint View Editable



From the Constraint Manager toolbar, select the **Save Constraints > Make Editable** option to make the constraint view editable.

Checking out a schematic view does not check out the (companion) constraint view automatically, and the constraints remain grayed out in the Constraint Manager assistant.



Result: The constraint view is opened in edit mode.



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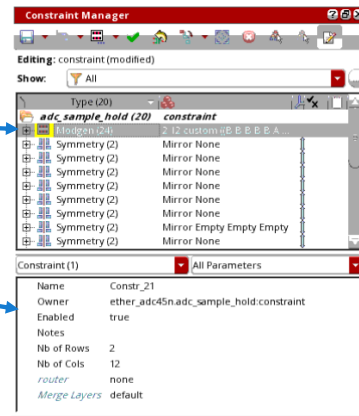
What Is the Constraint Manager Assistant?



The **Constraint Manager** dockable assistant pane allows for the addition, modification, and deletion of constraints from/to the Cadence Constraint Storage system.

The *Constraint Manager* user interface comprises two main component parts:

- *Constraint Browser*
 - The *Constraint Manager* displays design constraints (in creation order), along with their parameters in the *Constraint Browser*.
- *Constraint Parameter Editor*



What Is Constraint Manager Assistant?

The **Constraint Manager** dockable assistant pane allows for the addition, modification, and deletion of constraints from/to the Cadence Constraint Storage system.

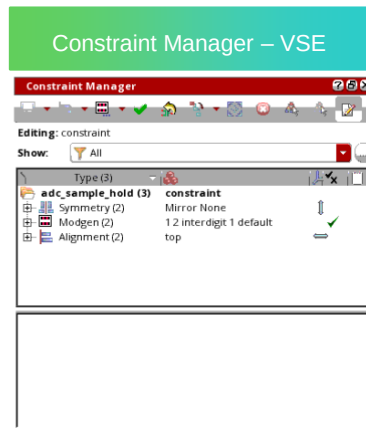
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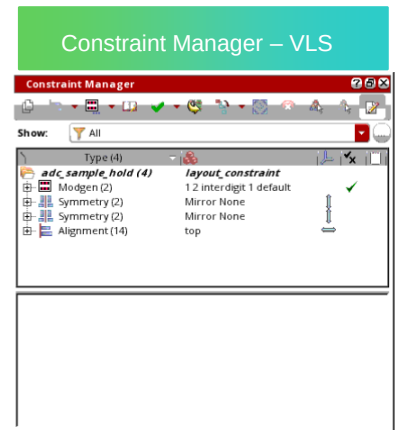
Refer to the *Virtuoso Unified Custom Constraints User Guide, Product Version IC23.1* for more details.

The Constraint Manager

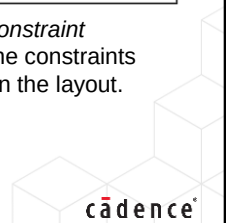
The *Constraint Manager* is available as a docked assistant inside both the *Virtuoso Schematic Editor* (VSE) and the *Virtuoso Layout Suite* (VLS) at the XL tier and above, displaying specification and implementation constraints respectively.



In Schematic XL, the *Constraint Manager* displays the content of the constraint view being used as per a given configuration.



In Layout XL, the *Constraint Manager* displays the constraints that currently exist in the layout.



The Constraint Manager

The *Constraint Manager* is available as a docked assistant inside both the *Virtuoso Schematic Editor* (VSE) and the *Virtuoso Layout Suite* (VLS) at the XL tier and above, displaying specification and implementation constraints respectively.

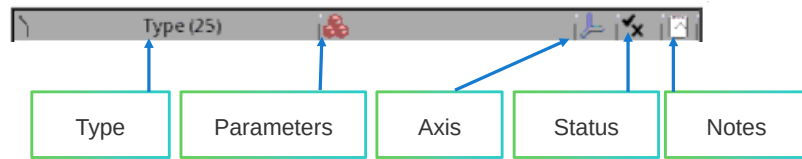
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In Layout XL, the *Constraint Manager* displays the constraints that currently exist in the layout.

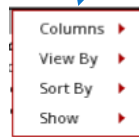
The Constraint Manager: Constraint Browser Columns

The *Constraint Manager* has the following sortable and re-positionable columns:

- Type
- Parameters
- Axis
- Status
- Notes



Right-clicking over the *Constraint Browser* column headers displays the *Constraint Browser Column Header Context-Menu*.



The Constraint Manager – Constraint Browser Columns

The *Constraint Manager* has the following sortable and re-positionable columns:

- Type
- Parameters
- Axis
- Status
- Notes

Note: Right-clicking over the *Constraint Browser* column headers displays the *Constraint Browser Column Header Context-Menu*.

The Constraint Manager: Constraint Storage System

The Constraint storage system differs from schematic to layout.

- In the schematic, the constraints are stored in an OpenAccess database (hierDesign.oa) that resides in a view called constraint view.
- In the layout, the constraints are stored in the layout database itself.

When you open a design in Schematic XL that does not have associated constraints, a constraint view is not created. A constraint view is created when you perform the following tasks:

Create the first constraint by using:	Constraint Manager assistant and Circuit Prospector assistant
Run the SKILL® functions:	ciConCreate and ciConCreateExpanded
Transfer constraints:	From layout to schematic
Load constraints:	From another view
Create an axis by using:	ciAxisCreate SKILL function
Use Process Rule editor to create:	A process rule and/or a design Constraint Group



Constraint Storage

Constraint storage differs from schematic to layout. In the schematic, constraints are stored in an OpenAccess database (hierDesign.oa) that resides in a view called constraint view; whereas, in the layout, the constraints are stored in the layout database itself.

When you open in Schematic XL a design that does not have associated constraints, a constraint view is not created.

A constraint view is created when you:

- Create the first constraint by using: Constraint Manager assistant and Circuit Prospector assistant.
- Run the following SKILL functions: ciConCreate & ciConCreateExpanded.
- Transfer constraints from layout to schematic.
- Load constraints from another view.
- Create an axis by using the ciAxisCreate SKILL function.
- Use Process Rule Editor to create a process rule and to create a design Constraint Group.

Refer to the *Virtuoso Unified Custom Constraints SKILL Reference*, Product Version IC23.1 for more details on the SKILL functions.

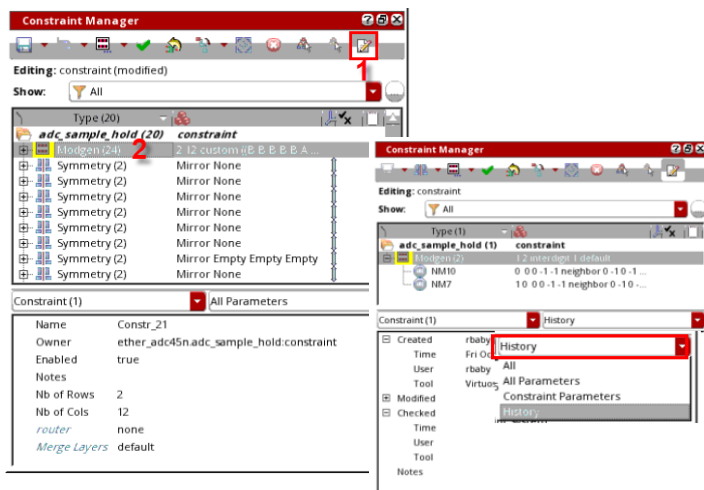
Displaying the Constraint Parameter Editor



The lower section of the *Constraint Manager* has an integrated *Constraint Parameter Editor*.

You can use this editor to add and/or edit parameters and parameter values for one or more constraints or constraint members, that have been selected in the *Constraint Browser*.

You can use this editor to view the *Constraint History*.



The *Constraint Parameter Editor* pane can be displayed by using these 2 options:

1. Selecting the *Constraint Editor* option in the *Constraint Manager* toolbar.
2. Double-clicking a constraint in the *Constraint Browser*.

Displaying the Constraint Parameter Editor

The lower section of the *Constraint Manager* has an integrated *Constraint Parameter Editor*. You can use this editor to add and/or edit parameters and parameter values for one or more constraints or constraint members, that have been selected in the *Constraint Browser* as well as to view the *Constraint History*.

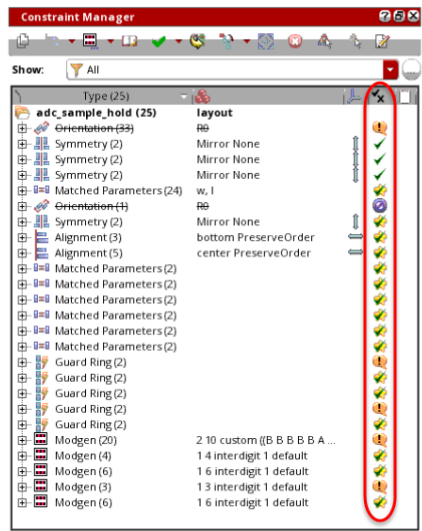
The *Constraint Parameter Editor* pane can be displayed by using these 2 options:

1. Selecting the *Constraint Editor* option in the *Constraint Manager* toolbar.
2. Double-clicking a constraint in the *Constraint Browser*.

The Constraint Status: Constraint Name and Definition



A constraint's current status is indicated using symbols and strikethrough text.



Constraint Status Icon	Constraint Name and Definition
	New Informs that the constraint is new in the current view (for example, the schematic view), but does not yet have a corresponding constraint in the other view (for example, the layout view).
	Different Informs that there is a difference between this constraint and the corresponding constraint in the other view. The related constraint status tooltip describes what the differences are, which could be any or all of the constraint parameters, member parameters, member order, or axis parameters. Note: The Constraint Comparison Report can be used to examine any differences in more detail.
	Passed Informs that the constraint has been met successfully.
	Inactive Informs that the constraint is not currently active. This means that either it has enabled=false or it is out-of-context (that is, the constraint is in the same hierarchy, but as part of a different traversal), or it has been overridden. Again, the related tooltip provides more details.
	Failed Informs that the constraint has not been met (validated). For example, depending on the constraint, it is possible the parameters do not match, are not aligned, are not symmetrical, and so on.
	New and Passed Informs that the constraint is new in this view, and has also been marked as passed.
	New and Failed Informs that the constraint only exists in the current view and that it has also failed in this view.
(No icon)	A constraint that is blank, with no icon, represents a constraint that has not yet been checked.

Constraint Status: Constraint Name and Definition

The Constraint Status and its name and definition are given here.

A constraint's current status is indicated using symbols and strikethrough text.

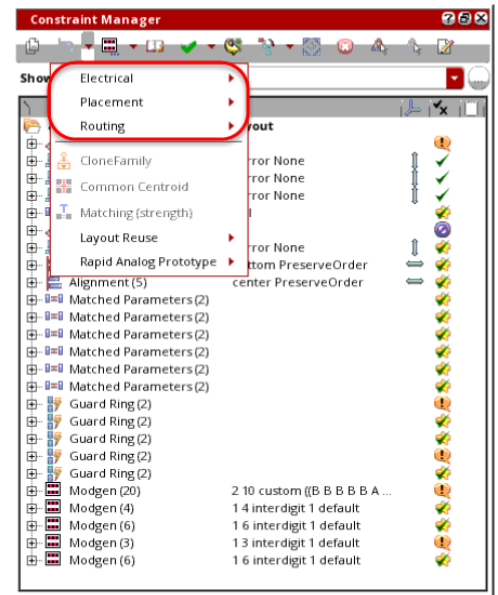
What Is a Constraint Type?



A **constraint type** is a formal definition of a design annotation that describes the required electrical or physical relations between a set of design objects (constraint members) through a set of pre-defined constraint parameters.

Constraints in the Virtuoso Schematic and Layout Editing Environment are:

- Electrical
- Placement
- Routing



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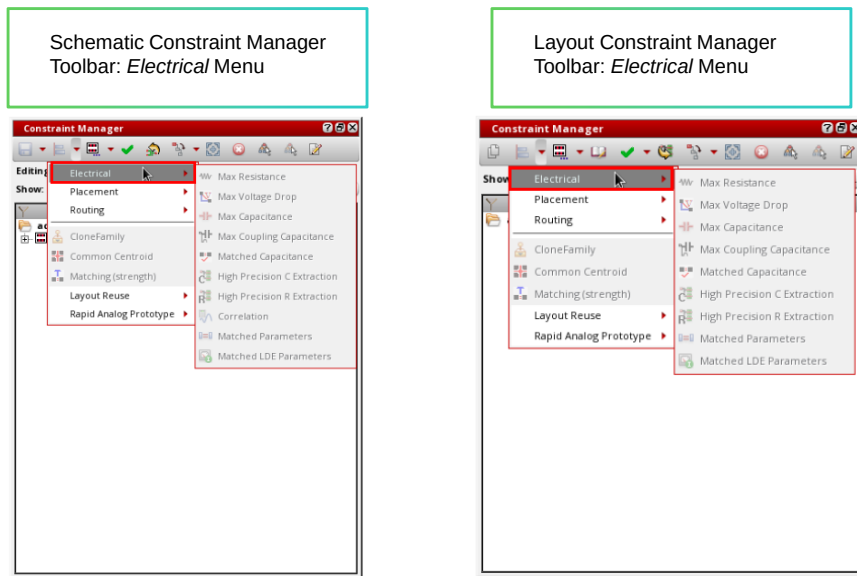
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Constraints in Virtuoso Schematic and Layout Editing Environment are:

- Electrical
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- Routing

Constraints in Virtuoso Schematic and Layout Editing Environment: *Electrical Menu*



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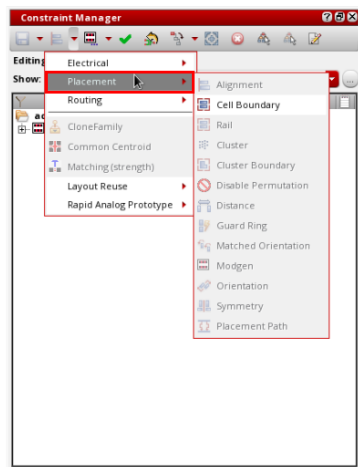


Constraints in Virtuoso Schematic and Layout Editing Environment: *Electrical Menu*

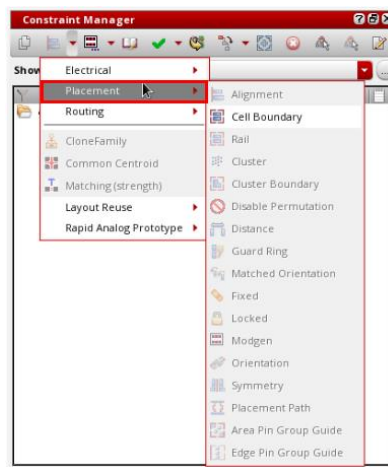
Electrical menu in Schematic Constraint Manager Toolbar and Layout Constraint Manager Toolbar are shown here.

Constraints in Virtuoso Schematic and Layout Editing Environment: *Placement Menu*

Schematic Constraint Manager
Toolbar: *Placement Menu*



Layout Constraint Manager
Toolbar: *Placement Menu*

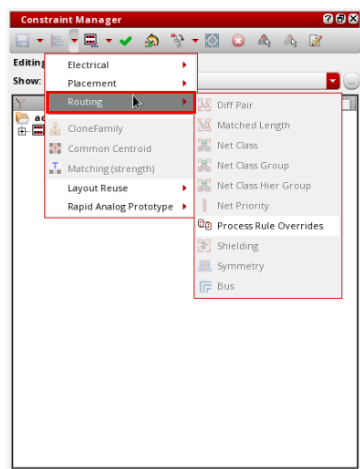


Constraints in Virtuoso Schematic and Layout Editing Environment: *Placement Menu*

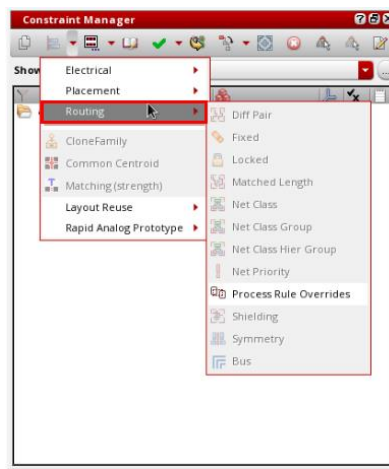
Placement menu in Schematic Constraint Manager Toolbar and Layout Constraint Manager Toolbar are shown here.

Constraints in Virtuoso Schematic and Layout Editing Environment: *Routing Menu*

Schematic Constraint Manager
Toolbar: *Routing Menu*



Layout Constraint Manager
Toolbar: *Routing Menu*



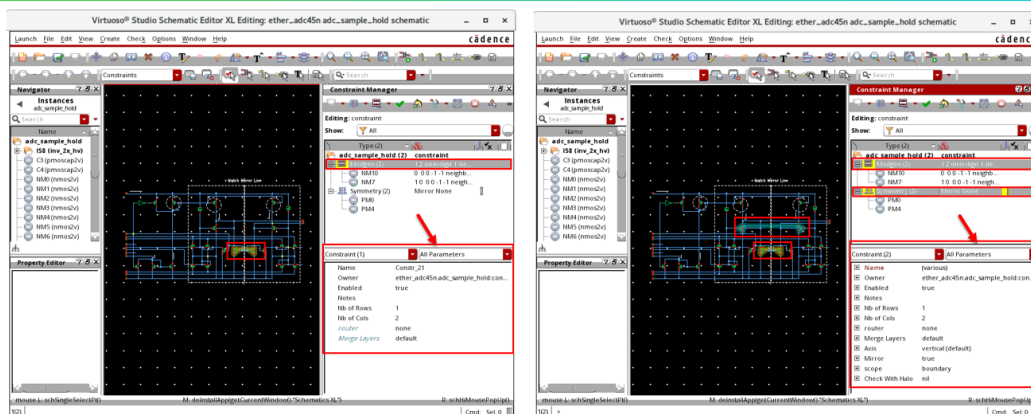
Constraints in Virtuoso Schematic and Layout Editing Environment: *Routing Menu*

Routing menu in Schematic Constraint Manager Toolbar and Layout Constraint Manager Toolbar are shown here.

Selecting and Editing Constraints in the Constraint Manager



- Select the constraint in the Constraint Browser.
- The parameters and parameter values of the selected constraint are presented in the Constraint Parameter Editor.
- You can edit the constraint's parameters in the Constraint Parameter Editor.



Select multiple constraints from the Constraint Browser to edit multiple constraints' parameters at the same time.

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Selection and Editing of Constraints

Whenever one or more constraints are selected in the *Constraint Browser*, *Constraint Parameter Editor* will update to present the parameters and parameter values of the selected constraints.

Note: If you select multiple constraints from the *Constraint Browser*, you can edit multiple constraints' parameters at the same time.

Accessing the Constraint Manager Assistant from VSE/VLS



To view/access the constraints, you can use the different methods available to invoke the *Constraint Manager* assistant from both the *Virtuoso Schematic Editor* (VSE) and the *Virtuoso Layout Suite* (VLS) at the XL tier and above.

Different methods available are:

1. Using the *Assistants* menu in VSE/VLS
2. Using the *Workspaces* menu in VSE/VLS
3. Using the *Menu Bar/Toolbars* context-sensitive menu in VSE/VLS
4. Using the *Workspace Configuration* pull-down menu on the toolbar in VSE/VLS



Accessing the Constraint Manager Assistant from VSE/VLS

To view/access the constraints, you can use the different methods available to invoke the *Constraint Manager* assistant from both the *Virtuoso Schematic Editor* (VSE) and the *Virtuoso Layout Suite* (VLS) at the XL tier and above.

The options available are:

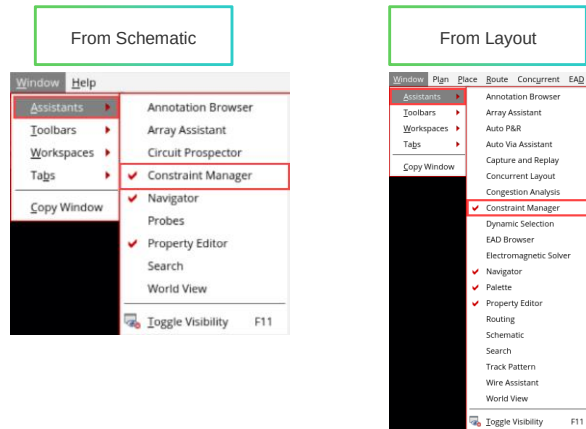
- Using the *Assistants* Menu in VSE/VLS
- Using the *Workspaces* Menu in VSE/VLS
- Using the *Menu Bar/Toolbars* Context-Sensitive Menu in VSE/VLS
- Using the *Workspace Configuration* Pull-Down Menu on the toolbar in VSE/VLS

1. Using the *Assistants* Menu in VSE/VLS

To access the dockable *Constraint Manager* assistant from both VSE/VLS:

- Select **Window – Assistants – Constraint Manager**.

Result: The *Constraint Manager* assistant is added as a docked assistant pane in the VSE/VLS windows, respectively.



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Step 1: Using the *Assistants* Menu in VSE/VLS

To access the dockable *Constraint Manager* assistant from both VSE/VLS:

- Select **Window – Assistants – Constraint Manager**.

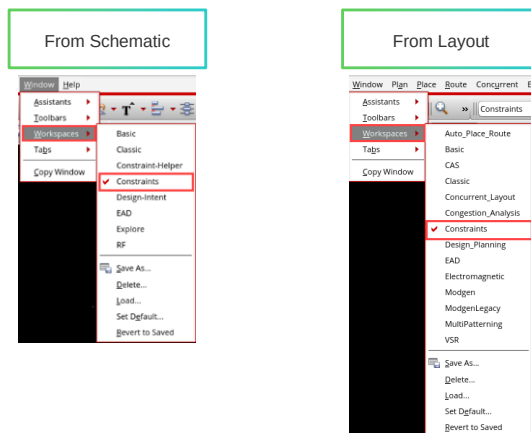
Result: The *Constraint Manager* assistant is added as a docked assistant pane in the VSE/VLS windows respectively.

2. Using the *Workspaces* Menu in VSE/VLS

To access the dockable *Constraint Manager* assistant from both VSE/VLS:

- Select **Window – Workspaces – Constraints**.

Result: The *Constraint Manager* assistant is added as a docked assistant pane in the VSE/VLS windows, respectively.



Step 2: Using the *Workspaces* Menu in VSE/VLS

To access the dockable *Constraint Manager* assistant from both VSE/VLS:

- Select **Window – Workspaces – Constraints**.

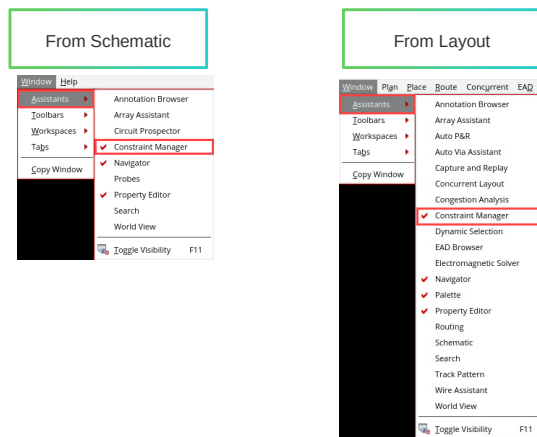
Result: The *Constraint Manager* assistant is added as a docked assistant pane in the VSE/VLS windows respectively.

3. Using the *Menu Bar/Toolbars* Context-Sensitive Menu in VSE/VLS

To access the dockable *Constraint Manager* assistant from both VSE/VLS:

- From the *Menu Bar/Toolbars* context-sensitive menu, select **Assistants – Constraint Manager**.

Result: The *Constraint Manager* assistant is added as a docked assistant pane in the VSE/VLS windows, respectively.



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Step 3: Using the *Menu Bar/Toolbars* Context-Sensitive Menu in VSE/VLS

To access the dockable *Constraint Manager* assistant from both VSE/VLS:

- From the *Menu Bar/Toolbars* Context-Sensitive Menu, select **Assistants – Constraint Manager**.

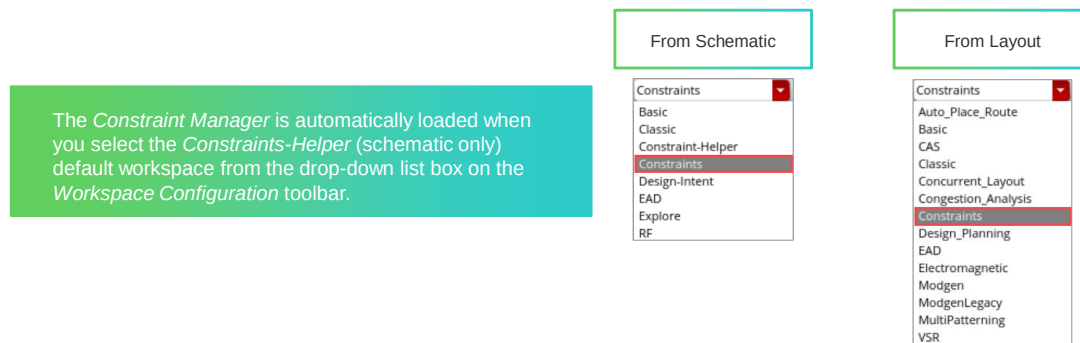
Result: The *Constraint Manager* assistant is added as a docked assistant pane in the VSE/VLS windows respectively.

4. Using the *Workspace Configuration* Pull-Down Menu on the Toolbar in VSE/VLS

To access the dockable *Constraint Manager* assistant from both VSE/VLS:

- From the *Workspace Configuration* pull-down menu on the toolbar, select the **Constraints** default workspace.

Result: The *Constraint Manager* assistant is added as a docked assistant pane in the VSE/VLS windows, respectively.



Step 4: Using the *Workspace Configuration* Pull-Down Menu on the Toolbar in VSE/VLS

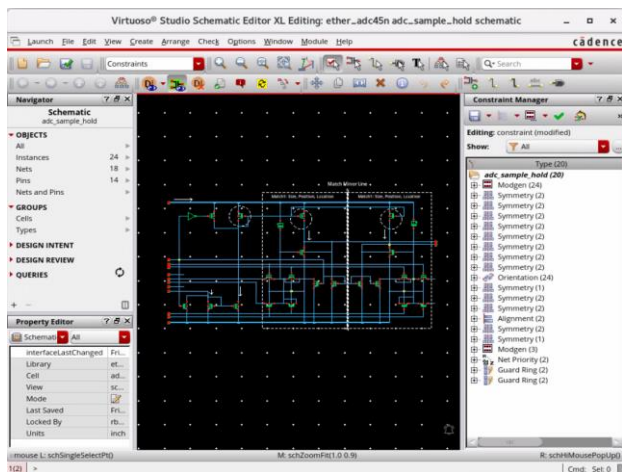
To access the dockable *Constraint Manager* assistant from both VSE/VLS:

- From the *Workspace Configuration* pull-down menu on the toolbar, select the **Constraints** default workspace.

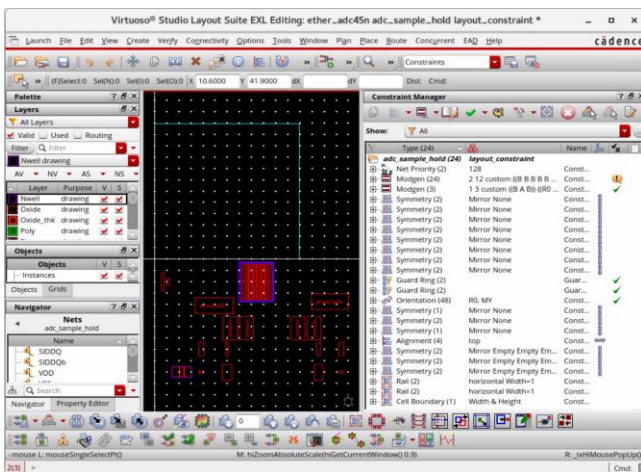
Result: The *Constraint Manager* assistant is added as a docked assistant pane in the VSE/VLS windows respectively.

Result: Accessing the Constraint Manager Assistant from VSE/VLS

Result: The *Constraint Manager* assistant is added as a docked assistant pane in VSE.



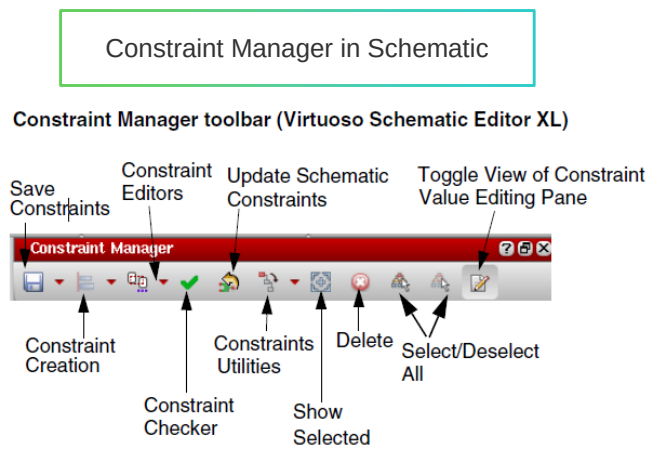
Result: The *Constraint Manager* assistant is added as a docked assistant pane in VLS.



Result: Accessing the Constraint Manager Assistant from VSE/VLS

The *Constraint Manager* assistant is added as a docked assistant pane in VSE/VLS.

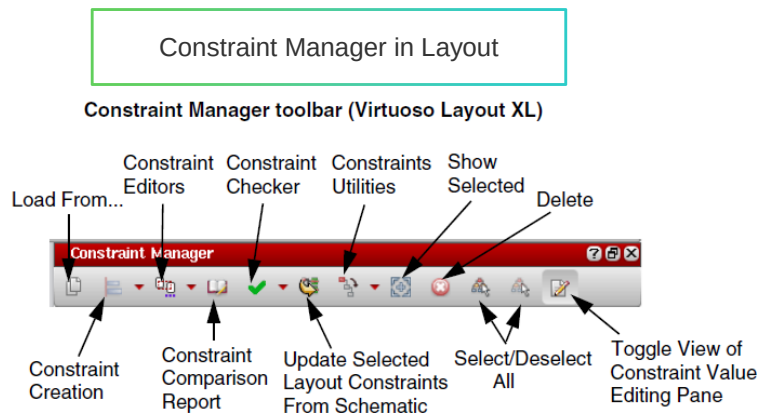
Constraint Manager Toolbar: VSE



Constraint Manager Toolbar – VSE

The Constraint Manager Toolbar in the Virtuoso Schematic Editor.

Constraint Manager Toolbar: VLS



Constraint Manager Toolbar – VLS

The Constraint Manager Toolbar in the Virtuoso Layout Suite.

Creating Constraints in the Constraint Manager

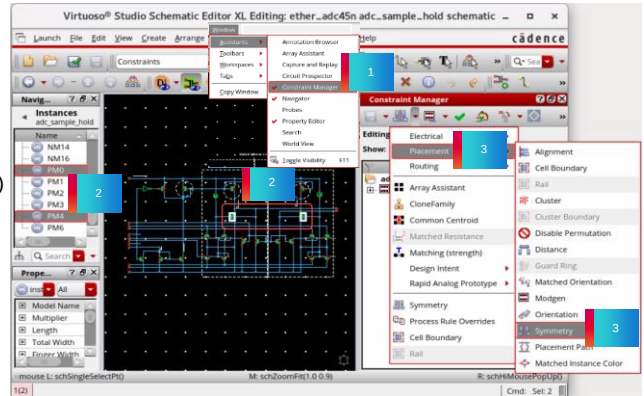


Use the **Constraint Manager** to create and add constraints to the Cadence Constraint Storage System.

To create a new constraint using the *Constraint Manager*:

1. First, if not already on display, access the Constraint Manager.
2. Then, in the current design session window, or in the Navigator assistant, select the design objects (instances, nets, pins, or terminals) that you want to include as constraint members in the new constraint.
3. Next, choose the type of constraint that is to be created by selecting it from the *Electrical*, *Placement*, *Routing*, and *Rapid Analog Prototype* sub-menus available within the **Constraint Creation** menu in the Constraint Manager toolbar.

The constraints are created in the current design session window.



Note: Optionally, descend to the level of the design hierarchy in which you want to create the constraint. You can descend the design hierarchy using the **Edit – Hierarchy – Descend Edit** menu command.

Only those constraints that can be created for the current selected object set will be shown as selectable. Those constraints that cannot currently be created will be grayed (filtered) out.

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Creating Constraints in the Constraint Manager

Use the **Constraint Manager** to create and add the constraints to the Cadence Constraint Storage system.

To create a new constraint using the *Constraint Manager*:

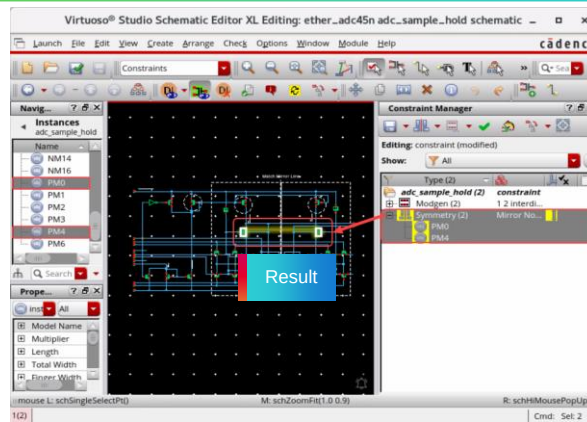
1. First, if not already on display, access the Constraint Manager. Note: Optionally, descend to the level of the design hierarchy that you want to create the constraint in. You can descend the design hierarchy using the **Edit – Hierarchy – Descend Edit** menu command.
2. Then, in the current design session window, or in the Navigator assistant, select the design objects (instances, nets, pins, or terminals) that you want to include as constraint members in the new constraint.
3. Next, choose the type of constraint that is to be created by selecting it from the *Electrical*, *Placement*, *Routing*, and *Rapid Analog Prototype* sub-menus available within the **Constraint Creation** menu in the Constraint Manager toolbar.

The constraints are created in the current design session window.

Only those constraints that can be created for the current selected object set will be shown as selectable. Those constraints that cannot currently be created will be grayed (filtered) out.

Result: Creating Constraints in the Constraint Manager

Result: A **Symmetry** constraint shows up in the Constraint Manager and a halo appears in the canvas around the constrained instances.



Clicking on the '+' sign in the Constraint Manager expands the constraint to display its members.

Result: Creating Constraints in the Constraint Manager

Result: A **Symmetry** constraint shows up in the Constraint Manager and a halo appears in the canvas around the constrained instances.

Clicking on the '+' sign in the Constraint Manager expands the constraint to display its members.

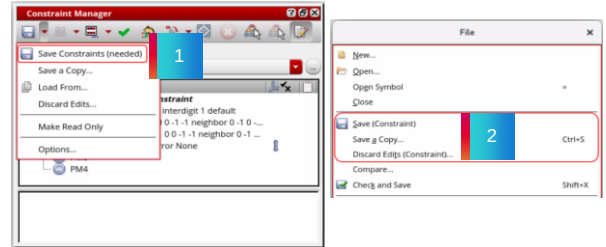
Saving and Storing the Constraints (Schematics Only)



Use the **Constraint Manager** to save the constraints to the Cadence Constraint Storage System (schematics only).

To save the constraints in the schematic window:

1. Select the **Save Constraints** option on the Constraint Manager toolbar (schematics only).
2. Alternatively, to handle any updates done for a constraint, you can use the following options:
 - a. Choose the **File – Save (Constraint)** menu command.
 - b. Choose the **File – Save a Copy** menu command.
 - c. Choose the **File – Discard Edits (Constraint)** menu command to save the constraints.



Saving and Storing the Constraints (Schematics Only)

Use the **Constraint Manager** to save the constraints to the Cadence Constraint Storage System (Schematics Only).

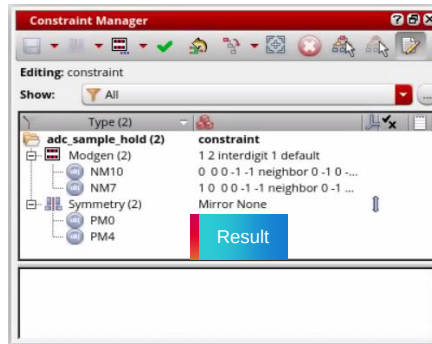
To save the constraints in the schematic window:

1. Select the **Save Constraints** option on the Constraint Manager Toolbar (schematics only).
2. Alternatively, to handle any updates done for a constraint, you can use the below options:
 - a. Choose the **File – Save (Constraint)** menu command.
 - b. Choose the **File – Save a Copy** menu command.
 - c. Choose the **File – Discard Edits (Constraint)** menu command.

The constraints detailed in the Constraint Manager are saved.

Result: Saving and Storing the Constraints (Schematics Only)

Result: The constraints detailed in the Constraint Manager are saved.



Result: Saving and Storing the Constraints (Schematics Only)

Result: The constraints detailed in the Constraint Manager are saved.

Deleting the Constraints from VSE/VLS

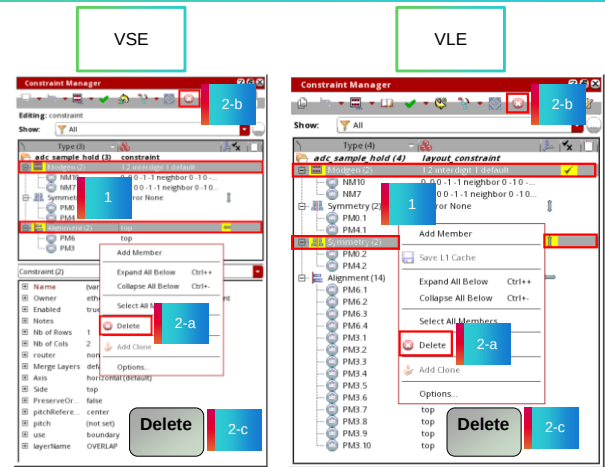


Use the **Constraint Manager** to delete the constraints from the Cadence Constraint Storage System.

To delete the constraints from both VSE/VLS:

1. First, select one or more constraints from the **Constraint Browser** in the Constraint Manager.
2. Now use one of the following options:
 - a. **Right-click** in the **Constraint Browser** and select the **Delete** option.
 - b. Select the **Delete** button from the Constraint Manager toolbar.
 - c. Press the **Delete** key.

The selected constraints are deleted from the Constraint Manager.

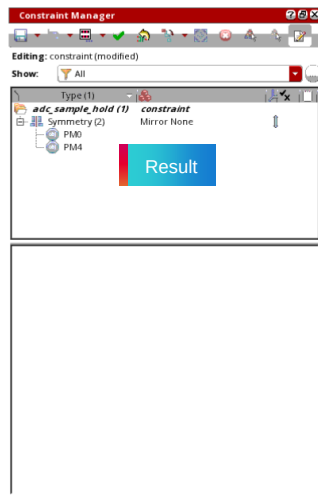


The selected constraints are deleted from the CM in VSE/VLS.

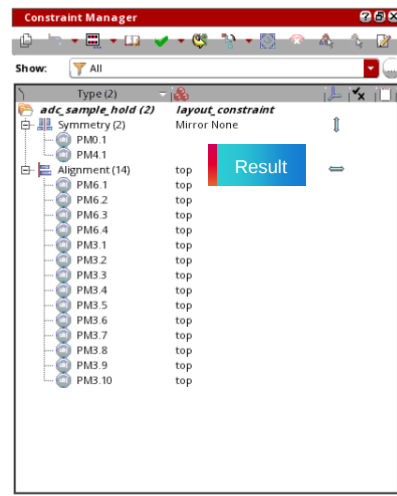
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Result: Deleting the Constraints from VSE/VLS

Result: The selected constraints are deleted from the CM in VSE.



Result: The selected constraints are deleted from the CM in VLS.



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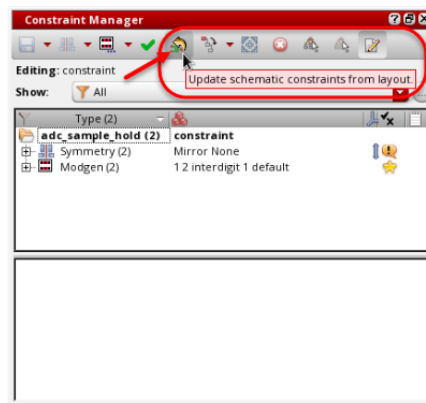
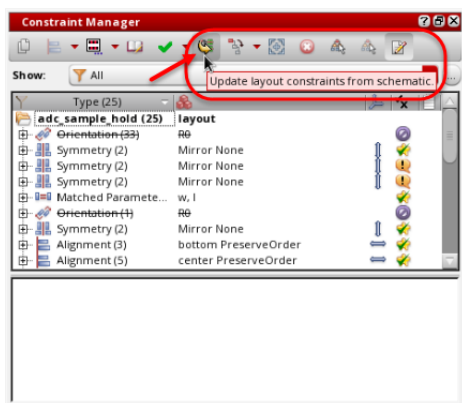
Transferring Constraints Between Schematic and Layout



In the forward dialog (schematic to layout transfer), constraints in the schematic can be transferred or updated in the layout on demand by using the **Update layout constraints from schematic** command displayed in the Constraint Manager menu in Layout XL.



In the backward dialog (layout to schematic transfer), constraints in the layout can be transferred or updated in the schematic on demand by using the **Update schematic constraints from layout** command displayed in the Constraint Manager menu in Schematic XL.



Transferring Constraints Between Schematic and Layout

In the forward dialog (schematic to layout transfer), constraints in the schematic can be transferred or updated in the layout on demand by using the **Update layout constraints from schematic** command displayed in the Constraint Manager menu in Layout XL.

In the backward dialog (layout to schematic transfer), constraints in the layout can be transferred or updated in the schematic on demand by using the **Update schematic constraints from layout** command displayed in the Constraint Manager menu in Schematic XL.

Constraint-Driven Flow: New Features



The following are new features in the constraint-driven flow.

- **Constraint Support**
 - Rail constraint supported by AAP and CAE
 - Cell Boundary constraint that includes Area Utilization
 - Ability to create/edit L1 constraints in the Layout Constraint Manager and support for interactive L1 editing by CAE
 - Usability improvements in the CM: Show Selected, Constraint Name Display
 - Hierarchical Constraint Propagation in the schematic
 - Improved Constraint Propagation UI
 - Hierarchy display and support in the Layout Constraint Manager
 - New Circuit Prospector Finders to create Modgens and other constraints automatically for the RAP flow with enhanced template support
 - Many infrastructure improvements (performance, initialization, transfer)



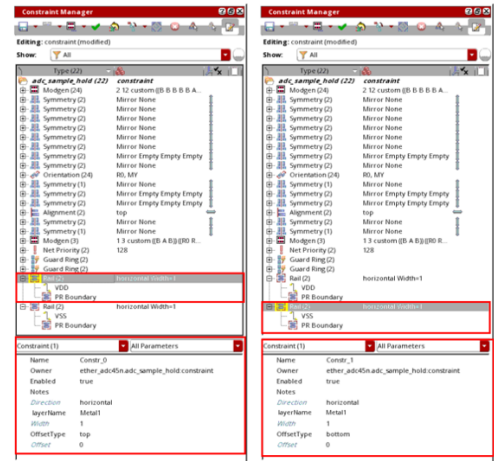
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Using the Rail Constraint Features



Use the **Rail** constraint to specify rails in the layout for nets such as *power* and *ground*.

- The **Rail** constraint can be specified in the Constraint Manager in the schematic or layout canvas.
- To specify a **Rail** constraint, the user simply needs to select a net in the design.
 - The constraint is created with the net and the prBoundary as members.
- Once a **Rail** constraint has been created, the user can specify the following parameters (change from default values):
 - *Direction* (horizontal or vertical), *layerName*, *Width*, *OffsetType* (top/bottom for horizontal and left/right for vertical), and *Offset* (distance from the boundary edge).



Using the Rail Constraint Features

Use the **Rail** constraint to specify rails in the layout for nets such as *power* and *ground*.

- The **Rail** constraint can be specified in the Constraint Manager in the schematic or layout canvas.
- To specify a **Rail** constraint, the user simply needs to select a net in the design.
 - The constraint is created with the net and the prBoundary as members.
- Once a **Rail** constraint has been created, the user can specify the following parameters (change from default values):
 - *Direction* (horizontal or vertical), *layerName*, *Width*, *OffsetType* (top/bottom for horizontal and left/right for vertical), and *Offset* (distance from the boundary edge).

Creating the Rail Constraint

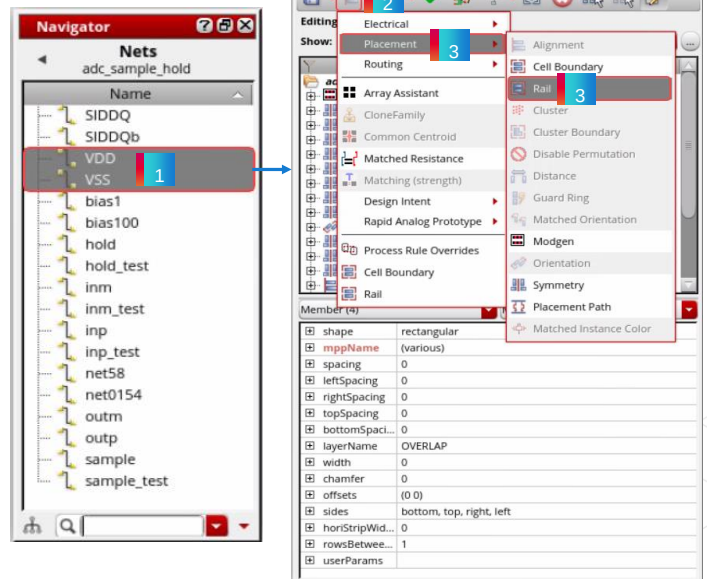


Use the **Rail** constraint to specify rails in the layout for nets such as *power* and *ground*. It can be specified in the Constraint Manager either in the schematic or in the layout canvas.

To create the **Rail** constraint in the schematic:

1. First, in the schematic window, select the nets **VDD** and **VSS** in the Navigator assistant.
2. Then, click on the **Constraint Creation** menu in the Constraint Manager toolbar.
3. Next, from the **Placement** sub-menu, select the **Rail** constraint.

Now, the newly created constraint is displayed in the Constraint Manager with the nets and prBoundary as members.



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Creating the Rail Constraint

Use the **Rail** constraint to specify rails in the layout for nets such as *power* and *ground*.

It can be specified in the CM either in the schematic or in the layout canvas.

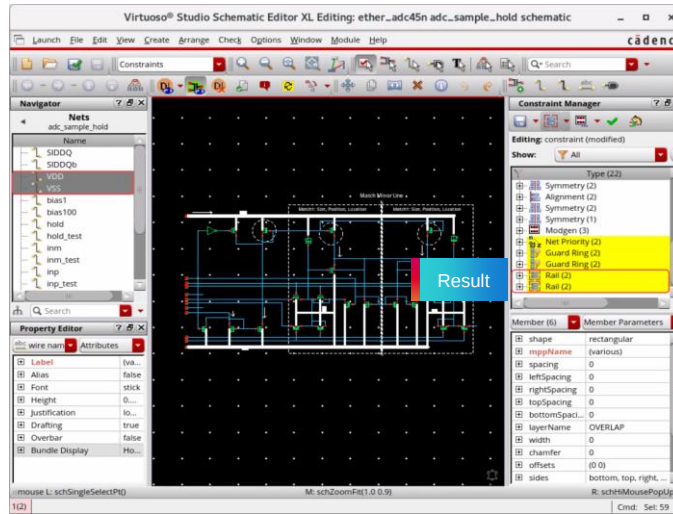
To create the **Rail** constraint in the schematic:

1. First, in the schematic window, select the nets **VDD** and **VSS** in the Navigator assistant.
2. Then, click on the **Constraint Creation** menu in the Constraint Manager toolbar.
3. Next, from the **Placement** sub-menu, select the **Rail** constraint.

Now, the newly created constraint is displayed in the Constraint Manager with the nets and prBoundary as members.

Result: Creating the Rail Constraint

Result: The **Rail** constraint is created on both the nets **VDD** and **VSS**.



Result: Creating the Rail Constraint

Result: The **Rail** constraint is created on both the nets **VDD** and **VSS**.

Modifying the Rail Constraint Parameters



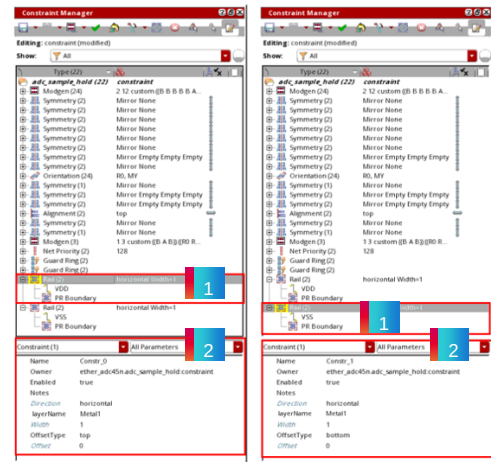
Once the **Rail** constraint is created, you can modify the constraint's parameters as needed.

To modify the **Rail** constraint in the schematic:

1. First, in the Constraint Manager, expand the **Rail** constraint to view its members.
2. Next, modify the parameters as required.

Now, in the Constraint Manager, the **Rail** constraint's parameters are modified.

The **Rail** constraint's parameters are modified.



Modifying the Rail Constraint Parameters

Once the **Rail** constraint is created, you can modify the constraint's parameters as needed.

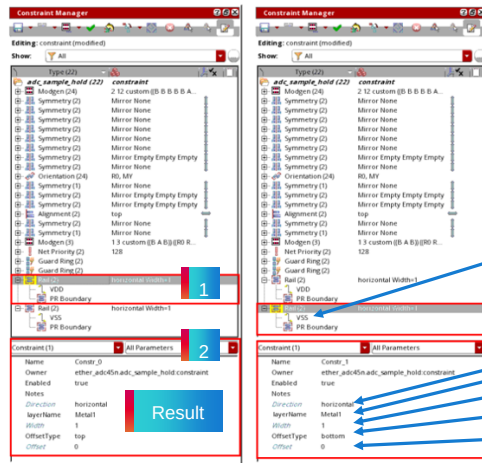
To modify the **Rail** constraint in the schematic:

1. First, in the Constraint Manager, expand the **Rail** constraint to view its members.
2. Next, modify the parameters as required.

Now, in the Constraint Manager, the **Rail** constraint's parameters are modified.

Result: Modifying the Rail Constraint Parameters

Result: The Rail constraint's parameters are modified.



Specifying the Rail constraint

Members

Expanding the Rail constraint shows its members (net and boundary)

Modify the Parameters as required:

Direction (horizontal or vertical)

LayerName (from techfile)

Width

OffsetType (reference edge: top/bottom/left/right)

Offset (offset value spacing from reference edge)



Result: Modifying the Rail Constraint Parameters

Result: The Rail constraint's parameters are modified.

Using the Cell Boundary Constraint Features



Use the **Cell Boundary** constraint to define the aspect ratio for the floorplanning tools for your design.

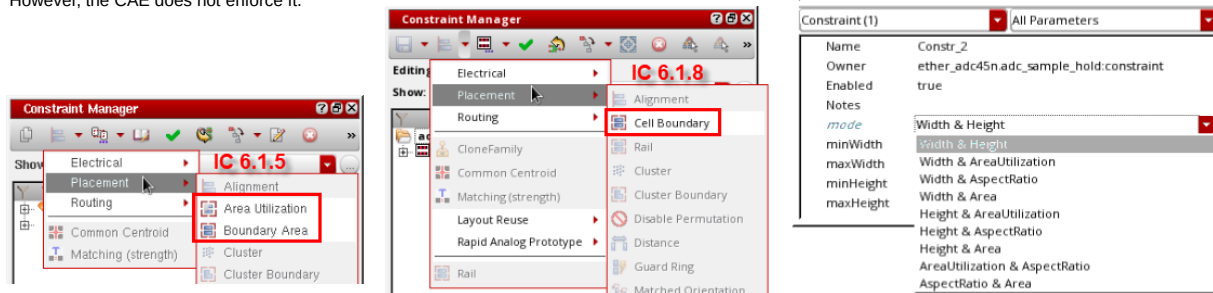
The **Boundary Area** constraint has been revamped and consolidated with the **Area Utilization** constraint to the new **Cell Boundary** constraint from IC 6.1.6 onwards.

The *mode* parameter allows users to specify feasible combinations of the following as a range (min/max):

- *Width, Height, Area, Aspect Ratio, and Area Utilization.*

The constraint is followed by Analog Placer (and the Block Placer), and the Constraint Aware Editing (CAE) flags any violation.

- However, the CAE does not enforce it.



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Using the Cell Boundary Constraint Features

The **Boundary Area** constraint has been revamped and consolidated with the **Area Utilization** constraint to the new **Cell Boundary** constraint from IC 6.1.6 onwards.

The *mode* parameter allows users to specify feasible combinations of the following as a range (min/max):

- *Width, Height, Area, Aspect Ratio, and Area Utilization.*

The constraint is followed by Analog Placer (and the Block Placer) and the Constraint Aware Editing (CAE) flags any violation.

- However, the CAE do not enforce it.

Creating the Cell Boundary Constraint



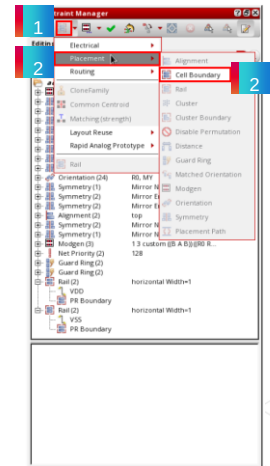
A *Cell Boundary* constraint can be created without any selection having been made in the design canvas in both the schematic and layout.

To create the **Cell Boundary** constraint in the schematic:

Note: Do not make any selection in the schematic window.

1. First, click on the **Constraint Creation** menu in the Constraint Manager toolbar.
2. Next, from the **Placement** sub-menu, select the **Cell Boundary** constraint.

Now, the newly created constraint is displayed in the Constraint Manager with one member, PR Boundary.



Use the **Cell Boundary** constraint to define implicitly the aspect ratio for the floorplanning tools using the following parameters: *mode*, *minWidth*, *maxWidth*, *minHeight*, and *maxHeight*.

Creating the Cell Boundary Constraint

Use the **Cell Boundary** constraint to define implicitly the aspect ratio for the floorplanning tools using the following parameters: *mode*, *minWidth*, *maxWidth*, *minHeight*, and *maxHeight*.

To create the **Cell Boundary** constraint in the schematic:

Note: Do not make any selection in the schematic window.

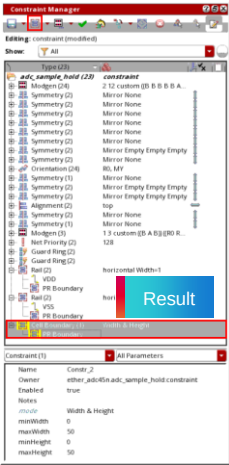
1. First, click on the **Constraint Creation** menu in the Constraint Manager toolbar.
2. Next, from the **Placement** sub-menu, select the **Cell Boundary** constraint.

Now, the newly created constraint is displayed in the Constraint Manager with one member, PR Boundary.

A *Cell Boundary* constraint can be created without any selection having been made in the design canvas in both the schematic and layout.

Result: Creating the Cell Boundary Constraint

Result: The **Cell Boundary** constraint is created with one member, PR Boundary.



Result: Creating the Cell Boundary Constraint

Result: The **Cell Boundary** constraint is created with one member, PR Boundary.

Modifying the Cell Boundary Constraint Parameters



Once the **Cell Boundary** constraint is created, you can modify the constraint's parameters as needed.

To modify the **Cell Boundary** constraint in the schematic:

1. First, in the Constraint Manager, expand the **Cell Boundary** constraint to view its members.
2. Next, modify the parameters as required.

Now, in the Constraint Manager, the **Cell Boundary** constraint's parameters are modified.

Members

Expanding the Cell Boundary Constraint shows its member (boundary)

Modify the parameters as required

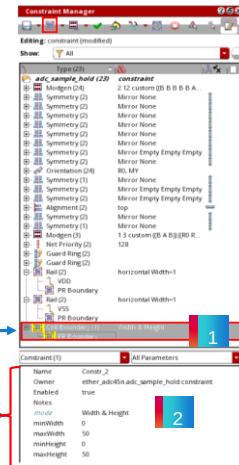
mode

minWidth

maxWidth

minHeight

maxHeight



Modifying the Cell Boundary Constraint Parameters

Once the **Cell Boundary** constraint is created, you can modify the constraint's parameters as needed.

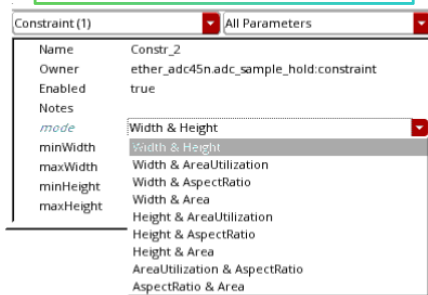
To modify the **Cell Boundary** constraint in the schematic:

1. First, in the Constraint Manager, expand the **Cell Boundary** constraint to view its members.
2. Next, modify the parameters as required.

Now, in the Constraint Manager, the **Cell Boundary** constraint's parameters are modified.

Allowable Parameter Combinations for the *mode* Parameter

Allowable parameter combinations for the *mode* parameter.



Allowable Parameter Combinations for *mode*

- *Width & Height*
- *Width & AreaUtilization*
- *Width & AspectRatio*
- *Width & Area*
- *Height & AreaUtilization*
- *Height & AspectRatio*
- *Height & Area*
- *AreaUtilization & AspectRatio*
- *AspectRatio & Area*



Allowable Parameter Combinations for the *mode* Parameter

Here you see the allowable parameter combinations for the *mode* parameter.

Result: Modifying the Cell Boundary Constraint Parameters

Result: The **Cell Boundary** constraint's parameters are modified.

Set the Cell Boundary constraint as follows:

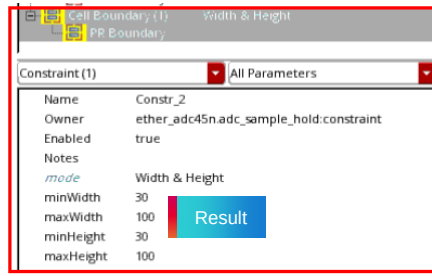
mode: Width & Height

minWidth: 30

maxWidth: 100

minHeight: 30

maxHeight: 100



Result: Modifying the Cell Boundary Constraint Parameters

In this example, the Cell Boundary constraint's parameters: mode, minWidth, maxWidth, minHeight, and maxHeight are modified, as needed.

Transferring the Constraints from Schematic to Layout

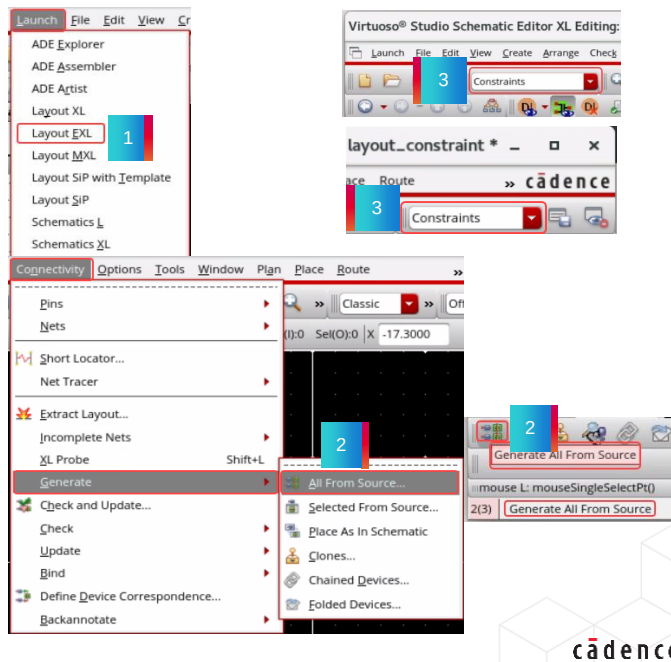


The constraints added in the schematic can be transferred to the layout which can be used for your layout design tasks.

To transfer the constraints from the schematic to the layout window:

1. First, from the schematic, use the **Launch – Layout EXL** command and open a new layout view.
2. Then, generate all from the source by choosing either the **Connectivity – Generate – All From Source** menu entry or by clicking the **Generate All From Source (GFS)** icon from the Layout XL Toolbar.
3. Next, switch the **Workspace** to **Constraints** to turn on the Constraint Manager assistant in schematic/layout.

Now, all constraints are transferred from the schematic to the layout.



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Transferring the Constraints from Schematic to Layout

The constraints added in the schematic can be transferred to the layout which can be used for your layout design tasks.

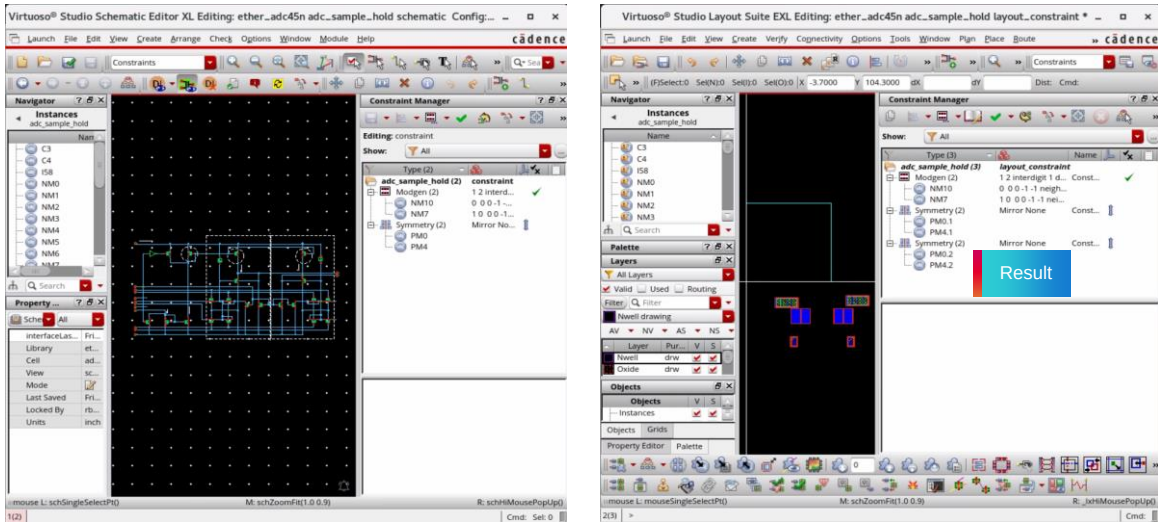
To transfer the constraints from the schematic to the layout window:

1. First, from the schematic, use the **Launch – Layout EXL** command and open a new layout view.
2. Then, generate all from the source by choosing either the **Connectivity – Generate – All From Source** menu entry or by clicking the **Generate All From Source (GFS)** icon from the Layout XL Toolbar.
3. Next, switch the **Workspace** to **Constraints** to turn on the Constraint Manager assistant.

Now, all the constraints are transferred from the schematic to the layout.

Result: Transferring the Constraints from Schematic to Layout

Result: All the constraints are transferred from the schematic to the layout.



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Result: Transferring the Constraints from Schematic to Layout

Result: All the constraints are transferred from the schematic to the layout.

Implementing the Rail Constraint Using Constraint Aware Editing



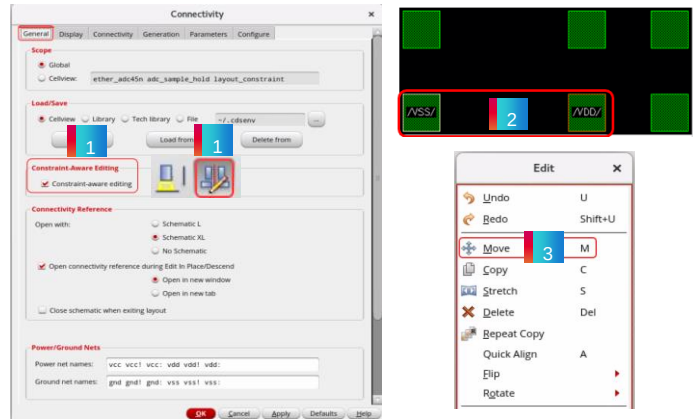
Use the **Constraint Aware Editing (CAE)** feature to implement the **Rail** constraint to specify rails in the layout for nets such as *power* and *ground*.

To implement the **Rail** constraint:

1. Firstly, from the **Options – Connectivity** menu command or from the **Options** toolbar, turn on the **Constraint Aware Editing** option in the layout window.
2. Next, select the **VDD** and **VSS** pins using the Navigator assistant or directly from the layout canvas.
3. Then, in the design canvas, choose the **Edit – Move** menu command or use the **Move** icon in the Edit toolbar or use the bindkey **M** to invoke the Move command and move the pins.

Now, rails are created when the pins are moved, and the constraints show up as passing.

Editing the pins in CAE mode implements the specified rails.



Implementing the Rail Constraint Using Constraint Aware Editing

Use the **Constraint Aware Editing (CAE)** feature to implement the **Rail** constraint to specify rails in the layout for nets such as *power* and *ground*.

To implement the **Rail** constraint:

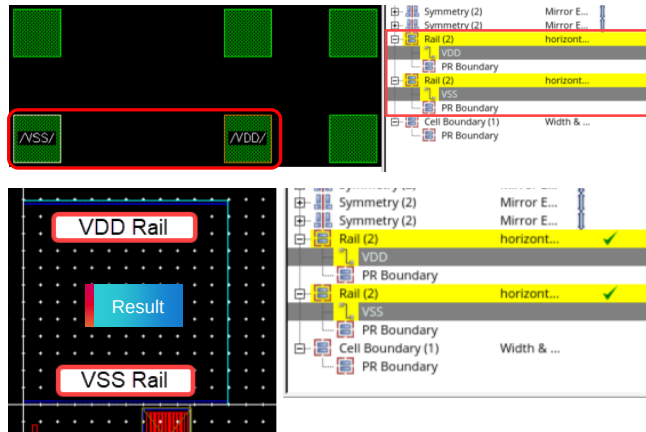
1. Firstly, from the **Options – Connectivity** menu command or from the **Options** toolbar, turn on the **Constraint Aware Editing** option in the layout window.
2. Next, select the **VDD** and **VSS** pins using the Navigator assistant or directly from the layout canvas.
3. Then, in the design canvas, choose the **Edit – Move** menu command or use the **Move** icon in the Edit toolbar or use the bindkey **M** to invoke the Move command and move the pins.

Now, rails are created when the pins are moved and the constraints show up as passing.

Editing the pins in CAE mode implements the rails as specified.

Result: Implementing the Rail Constraint Using Constraint Aware Editing

Result: Rails are created when the pins are moved and the constraints show up as passing which can be verified in the CM.



Result: Implementing the Rail Constraint Using Constraint Aware Editing

Result: Rails are created when the pins are moved and the constraints show up as passing which can be verified in the CM.

Implementing the Rail Constraint Using Analog Auto Placer

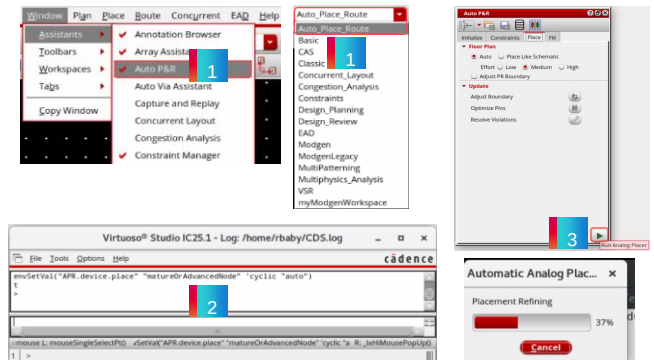


Use the **Analog Auto Placer** to implement the **Rail** constraint to specify rails in the layout for nets such as *power* and *ground*.

To implement the **Rail** constraint:

1. First, in the layout window, choose **Window – Assistants – Auto P&R** or select the workspace **Auto_Place_Route** from the Workspace Configuration to turn on the Auto P&R assistant.
2. Next, to run the Analog Automatic Placement, execute the following command in the CIW window:
`envSetVal("APR.device.place" "matureOrAdvancedNode" 'cyclic "auto")`
3. Then, in the **Auto P&R** form, select the options as per your requirement and run the placer.

Now, since the nets **VDD** and **VSS** are having the **Rail** constraint as you see in the Constraint Manager, after the **Analog Auto Placer** runs, rails are created for these 2 pins, and the constraints show up as passing.



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Implementing the Rail Constraint Using Analog Auto Placer

Use the **Analog Auto Placer** to implement the **Rail** constraint to specify rails in the layout for nets such as *power* and *ground*.

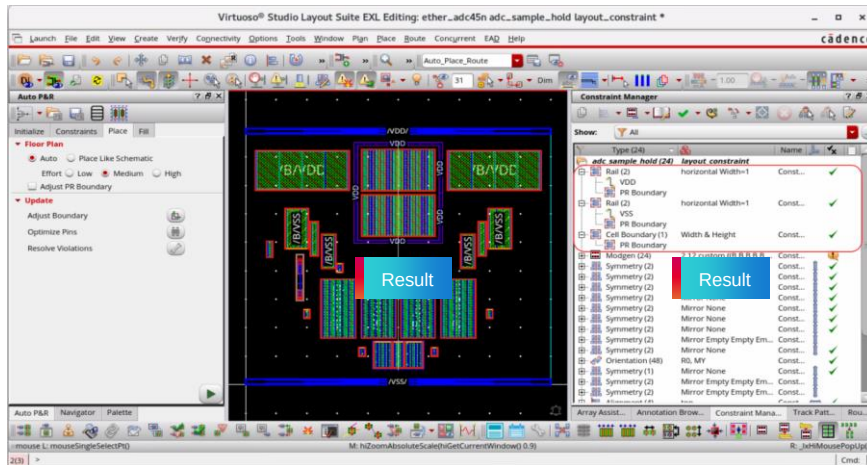
To implement the **Rail** constraint:

1. First, in the layout window, choose **Window – Assistants – Auto P&R** or select the workspace **Auto_Place_Route** from the Workspace Configuration to turn on the Auto P&R assistant.
2. Next, to run the Analog Automatic Placement, execute the following command in the CIW window:
`envSetVal("APR.device.place" "matureOrAdvancedNode" 'cyclic "auto")`
3. Then, in the **Auto P&R** form, select the options as per your requirement and run the placer.

Now, since the nets **VDD** and **VSS** are having the **Rail** constraint as you see in the Constraint Manager, after the **Analog Auto Placer** runs, rails are created for these 2 pins and the constraints show up as passing.

Result: Implementing the Rail Constraint Using Analog Auto Placer

Result: After the **Analog Auto Placer** runs, rails are created for the pins **VDD** and **VSS** and the constraints show up as passing which can be verified in the CM.



Result: Implementing the Rail Constraint Using Analog Auto Placer

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Modifying and Reimplementing the Rail Constraint

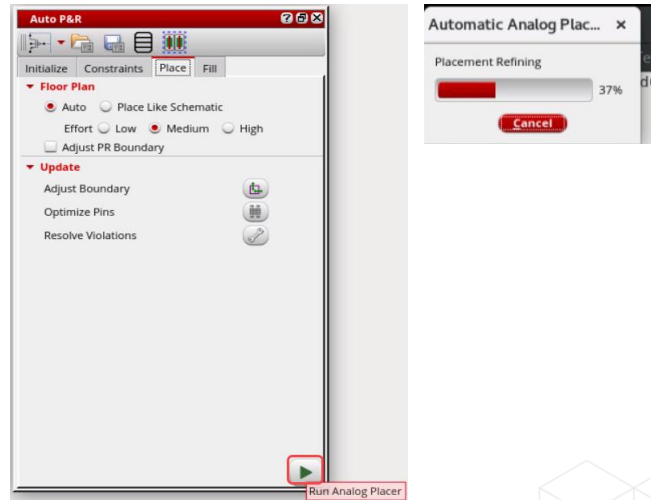


Modifying the **Rail** constraint in the Constraint Manager dynamically updates its status from *checked* to *not checked*.

To reimplement the modified **Rail** constraint:

- Rerun the **Analog Auto Placer**.

It is always a good practice to edit in **Constraint Aware Editing (CAE)** mode to honor the constraints.



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Modifying and Reimplementing the Rail Constraint

Modifying the **Rail** constraint in the Constraint Manager dynamically updates its status.

To reimplement the modified **Rail** constraint:

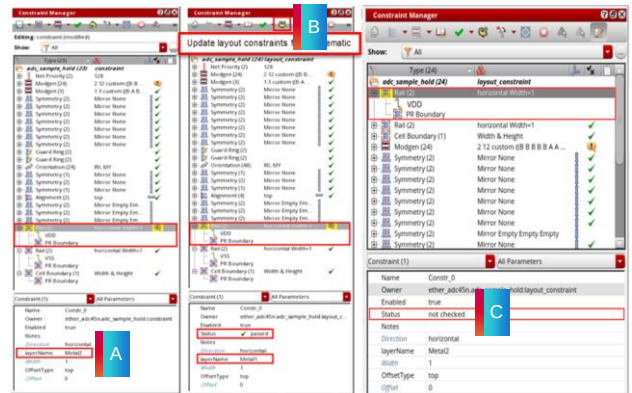
- Rerun the **Analog Auto Placer**.

It is always a good practice to edit in **Constraint Aware Editing (CAE)** mode to honor the constraints.

Example: Modifying the Rail Constraint

Modify the *Rail* constraint as follows:

- A. Change the *Rail* layer in the Constraint Manager on the schematic side, say for example from *Metal1* to *Metal2*.
- B. Followed by *Update layout constraints from schematic* on the layout side. 
- C. Constraint *Status* now shows up as *not checked* in the layout side.



Example: Modifying the Rail Constraint

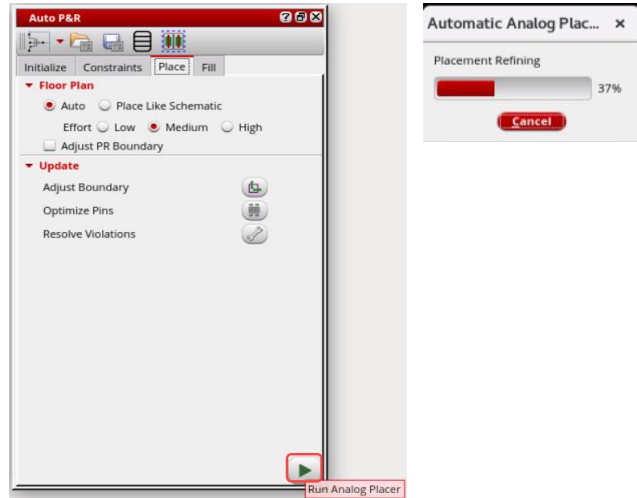
Modify the *Rail* constraint as follows:

- Change the *Rail* layer in the Constraint Manager on the schematic side, say for example from *Metal1* to *Metal2*.
- Followed by *Update layout constraints from schematic* on the layout side.
- Constraint *Status* now shows up as *not checked* in the layout side.

Example: Reimplementing the Rail Constraint

The modified *Rail* constraint can be reimplemented and fixed using the following method:

- In the layout canvas, rerun the **Analog Auto Placer**.



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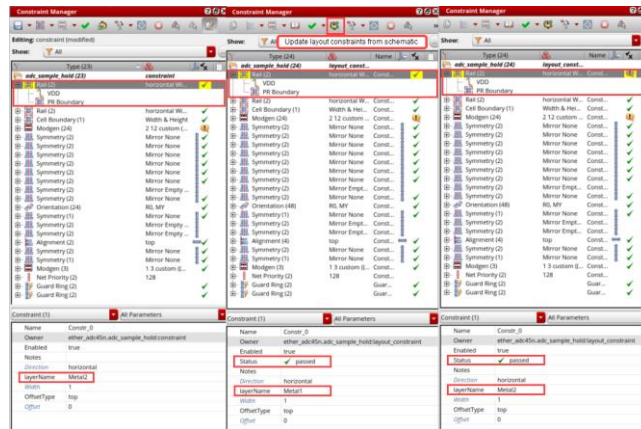


Example: Reimplementing the Rail Constraint

The modified *Rail* constraint can be reimplemented and fixed by rerunning the Analog Auto Placer.

Result: Modifying and Reimplementing the Rail Constraint

The modified **Rail** constraint is reimplemented after running the **Analog AutoPlacer** command.



Constraint *Status* now shows up as *passed* in the layout side.



Result: Modifying and Reimplementing the Rail Constraint

The modified **Rail** constraint is reimplemented after running the **Analog Auto Placer** command.

Constraint *Status* now shows up as *passed* in the layout side.

Modifying and Reimplementing the Cell Boundary Constraint

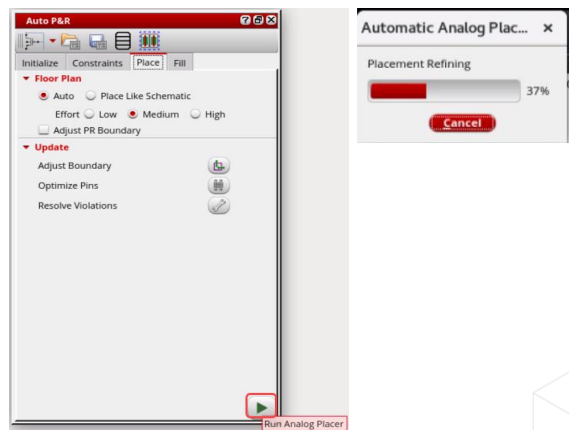


Modifying the **Cell Boundary** constraint in the Constraint Manager dynamically updates its status from *passed* to *failed*.

To reimplement the modified **Cell Boundary** constraint:

- Rerun the **Analog Auto Placer**.

It is always a good practice to edit in **Constraint Aware Editing (CAE)** mode to honor the constraints.



Modifying and Reimplementing the Cell Boundary Constraint

Modifying the **Cell Boundary** constraint in the Constraint Manager dynamically updates its status.

To reimplement the modified **Cell Boundary** constraint:

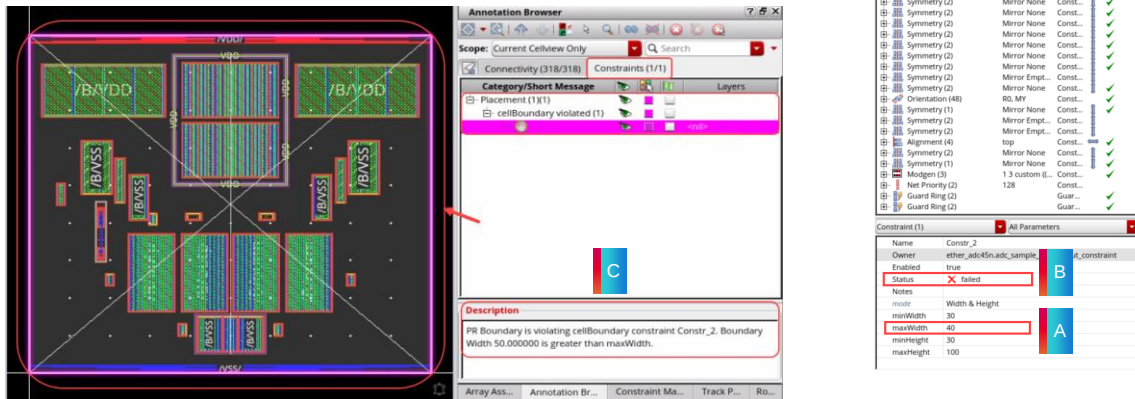
- Rerun the **Analog Auto Placer**.

It is always a good practice to edit in **Constraint Aware Editing (CAE)** mode to honor the constraints.

Example: Modifying the Cell Boundary Constraint

Modify the *Cell Boundary* constraint as follows:

- Change the *maxWidth* of the *Cell Boundary* constraint on the layout side from 100 to 40 and run the placer again.
- Constraint *Status* is dynamically updated as *failed* in the Constraint Manager assistant.
- The violation is reported by the Annotation Browser assistant.



Example: Modifying the Cell Boundary Constraint

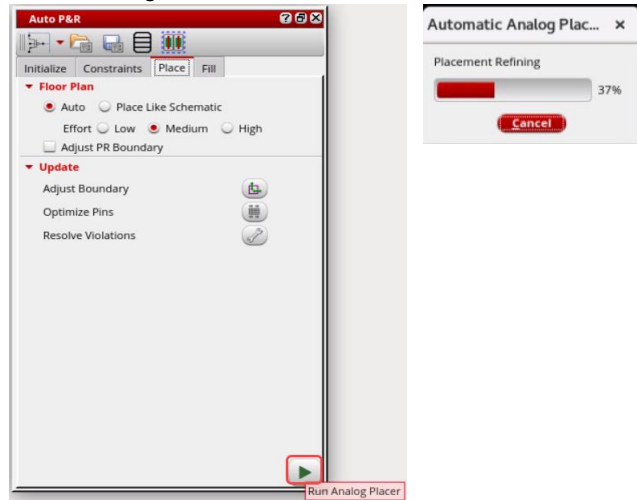
Modify the *Cell Boundary* constraint as follows:

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- The violation is reported by the Annotation Browser assistant.

Example: Reimplementing the Cell Boundary Constraint

The modified *Cell Boundary* constraint can be reimplemented and fixed using the following method:

- In the layout canvas, rerun the Analog Auto Placer.



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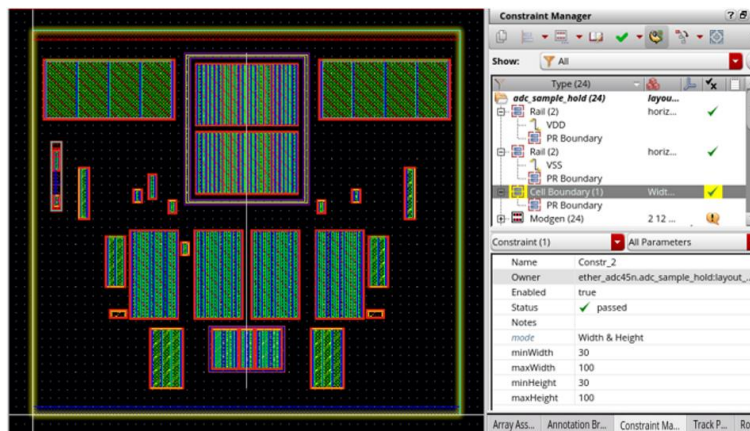


Example: Reimplementing the Cell Boundary Constraint

The modified *Cell Boundary* constraint can be reimplemented and fixed by rerunning the Analog Auto Placer.

Result: Modifying and Reimplementing the Cell Boundary Constraint

The modified **Cell Boundary** constraint is reimplemented after running the **Analog Auto Placer**.



Constraint *Status* now shows up as *passed* in the layout side.



Result: Modifying and Reimplementing the Cell Boundary Constraint

The modified **Cell Boundary** constraint is reimplemented after running the **Analog Auto Placer**.

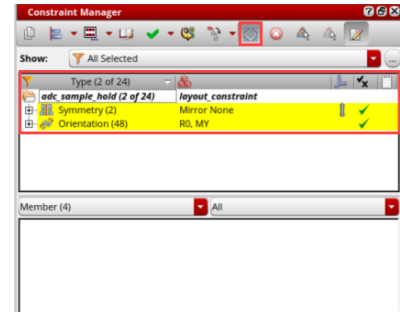
Constraint *Status* now shows up as *passed* in the layout side.

Usability Improvements in the Constraint Manager: Show Selected



You can quickly check whether there is already a constraint set on a selected object or not.

- Constraints on selected objects can be viewed by clicking the **Show Selected** toggle button in the Constraint Manager toolbar.
- This in turn controls the corresponding filter.



Usability Improvements in the Constraint Manager: Show Selected

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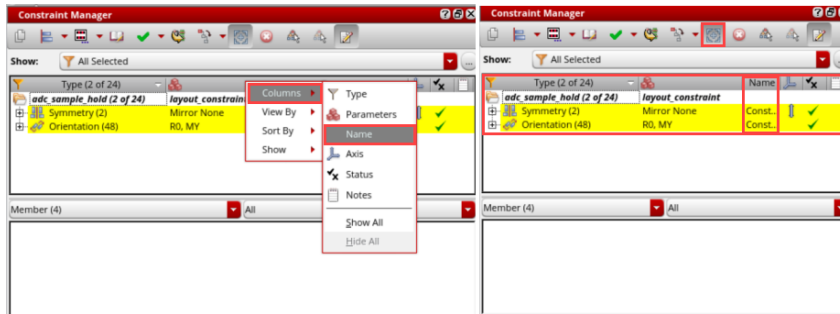
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- This in turn controls the corresponding filter.

Usability Improvements in the Constraint Manager: Constraint Names



You can display the constraint names as a column in the Constraint Manager through the **Columns** customization menu to improve the user experience.

- From the Constraint Manager assistant, turn on **Columns - Name**.
- The constraint names are displayed in the new *Name* column in the Constraint Manager assistant now.



Usability Improvements in the Constraint Manager: Constraint Names

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- From the Constraint Manager assistant, turn on **Columns - Name**.
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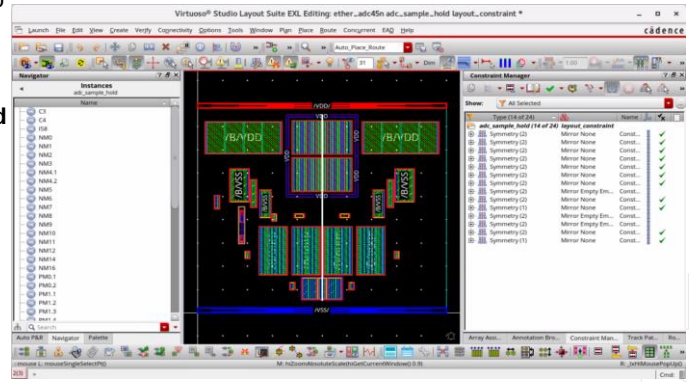
Usability Improvements in the Constraint Manager: Constraint Axis



You can use the **Axis** column in the *Constraint Browser* to display the information on axis direction, either vertical or horizontal.

- A constraint axis is a virtual reference support object used by *Symmetry*, *Alignment*, and *Distance* constraint types to define the direction and location of symmetrical pairs and aligned objects.
- Selecting **Axis** in the layout canvas displays the associated *Symmetry* constraints with the **Show Selected** filter.
- In this example, the Constraint Manager gets populated with the related *Symmetry* constraints with their axis direction (vertical) displayed in the **Axis** column.
- **Axis** is highlighted in the layout when a *Symmetry* constraint is selected in the Constraint Manager.

Symmetry, Distance, and Alignment constraint types should always reference an axis.



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Note: *Symmetry, Distance and Alignment* constraint types should always reference an axis.

Creating the Process Rule Overrides (PRO) Constraint



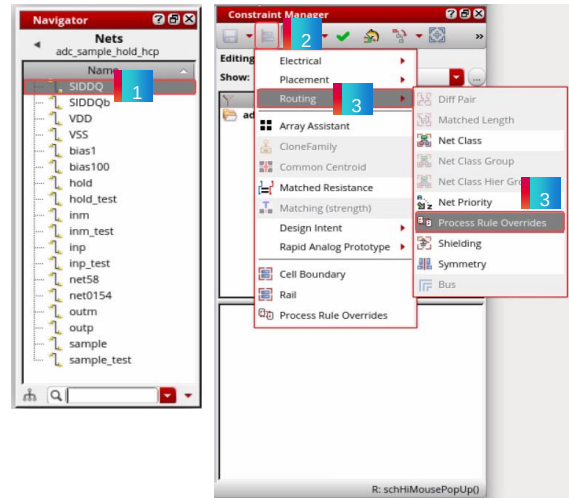
The **Process Rule Overrides (PRO)** can be used to override on a given object, the generic foundry and design process rules with specific, more stringent rules as defined in Constraint Groups.

They are stored in designs or technology files and act as a collection of process constraints that are applicable to the objects in a design.

To create the PRO constraint in the schematic:

1. First, in the schematic window, select the net **SIDDQ** in the Navigator assistant.
2. Then, click on the **Constraint Creation** menu in the Constraint Manager toolbar.
3. Next, from the **Routing** submenu, select the **Process Rule Overrides** constraint.

Now, the **PRO** constraint is created on the net **SIDDQ**, which is displayed in the Constraint Manager.



Creating the Process Rule Overrides (PRO) Constraint

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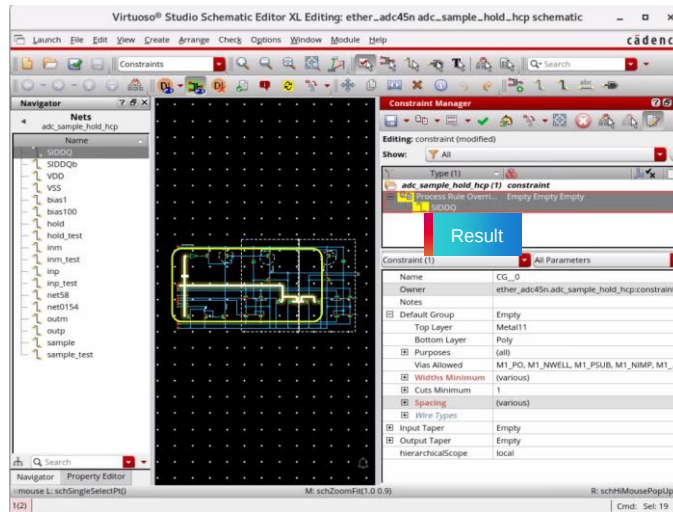
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Now, the **PRO** constraint is created on the net **SIDDQ** which is displayed in the Constraint Manager.

Result: Creating the Process Rule Overrides (PRO) Constraint

Result: The **PRO** constraint is created on the net **SIDDQ**.



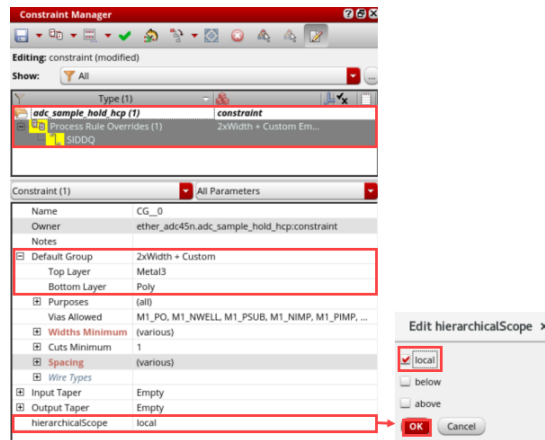
Result: Creating the Process Rule Overrides (PRO) Constraint

Result: The **PRO** constraint is created on the net **SIDDQ**.

PRO: Using the *hierarchicalScope* Parameter

In the Process Rule Overrides constraint:

- In the Parameters section, clicking the **hierarchicalScope** brings up the *Edit hierarchicalScope* form.
- Constraints can be tagged automatically for propagation, without having to set explicitly the scope parameter.



PRO: Using the *hierarchicalScope* Parameter

In the Process Rule Overrides constraint:

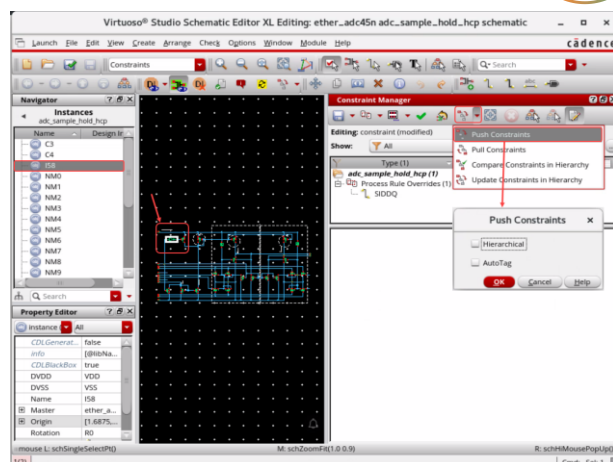
- In the Parameters section, clicking the **hierarchicalScope** brings up the *Edit hierarchicalScope* form.
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Hierarchical Constraint Propagation in Schematic



Propagation of net-based constraints across the hierarchy now also works in schematic from IC 6.1.6 onwards. In IC 6.1.5, this feature was available only in the layout.

- An identical pull-down menu for Constraint Propagation is available in the *Schematic Constraint Manager*, and the use model is the same as that in the layout.
- Transfer of propagated constraints in lower levels of hierarchy from schematic to layout follows the same use model (transfer occurs during editing of the lower-level cell in VLS-XL).



Hierarchical Constraint Propagation in Schematic

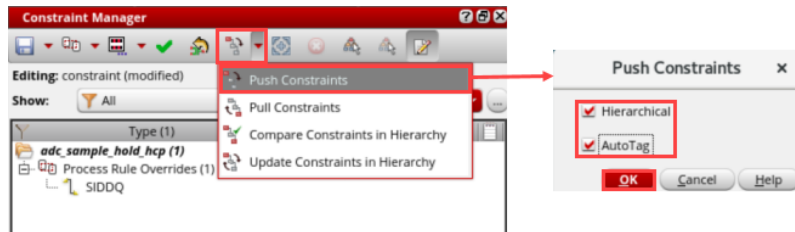
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Improved UI for Constraint Propagation

- **Hierarchical:** Constraints can now be propagated recursively all the way to the bottom or to the top of the hierarchy from the Constraint Manager.
 - Earlier, this was possible only through SKILL.
- **AutoTag:** Constraints can be tagged automatically for propagation, without having to set the scope parameter explicitly.



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Performing Hierarchical Constraint Propagation in Schematic



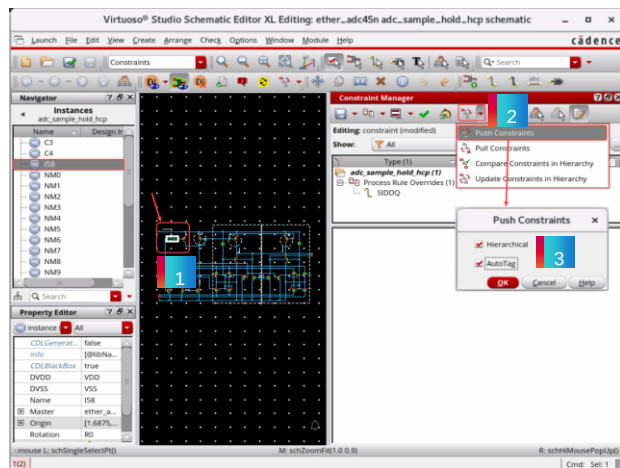
The Hierarchical Constraint Propagation feature allows propagating net-based constraints up and down the hierarchy (to the top level or to the lower-level blocks).

It is often required in a top-down or concurrent design flow for floorplanning and block integration.

To perform hierarchical constraint propagation in the schematic:

1. First, select an instance, say, **I58** (an instance of an inverter) in the schematic.
2. Then, from the Constraint Manager toolbar, click on the *Constraint Propagation* pull-down and select the **Push Constraints** option to display the *Push Constraints* form.
3. Next, turn on both **Hierarchical** and **AutoTag** options in the *Push Constraints* form and click **OK**.

Now, messages in CIW indicate that constraints are pushed into the selected lower-level block. Note that, *hierarchicalScope* parameter of the PRO constraint gets updated to local, below.



Performing Hierarchical Constraint Propagation in Schematic

Hierarchical Constraint Propagation feature allows propagating net-based constraints up and down the hierarchy (to the top level or to the lower-level blocks).

It is often required in a top-down or concurrent design flow for floorplanning and block integration.

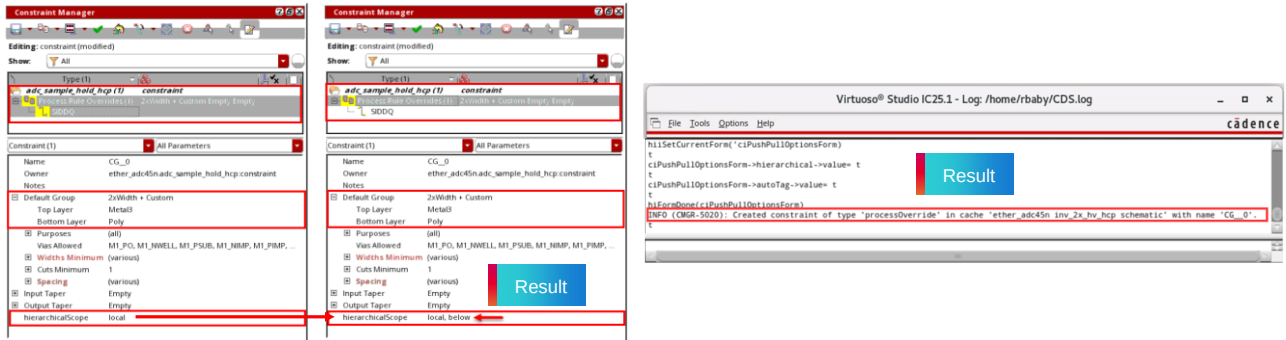
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Result: Performing Hierarchical Constraint Propagation in Schematic

Result: Messages in CIW: *INFO (CMGR-5020): Created constraint of type 'processOverride' in cache 'ether_adc45n inv_2x_hv_hcp schematic' with name 'CG__0'.*



Result: *hierarchicalScope* parameter of the PRO constraint gets updated to **local, Below**.



Result: Performing Hierarchical Constraint Propagation in Schematic

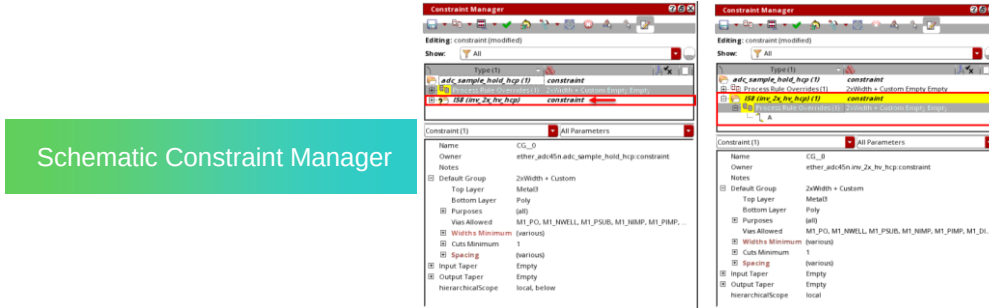
Result: Messages in CIW: *INFO (CMGR-5020): Created constraint of type 'processOverride' in cache 'ether_adc45n inv_2x_hv_hcp schematic' with name 'CG__0'.*

Result: *hierarchicalScope* parameter of the PRO constraint gets updated to **local, below**.

Hierarchical Constraint Manager in Schematic and Layout



- Constraints in cells in the layout hierarchy are now shown in folders in the *Layout Constraint Manager*.
- The lower-level constraints are not displayed until the user expands the folder to avoid any performance impact.
- The folders are categorized by block type (soft-block, hard-macro) as per the floorplanning configuration and the icons reflect the types as well.
 - Only soft-block (L1) constraints are editable from the top level.
- Display of hierarchical constraints in the *Schematic Constraint Manager* follow the same use model.



Schematic Constraint Manager

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Hierarchical Constraint Manager in Schematic and Layout

- Constraints in cells in the layout hierarchy are now shown in folders in the *Layout Constraint Manager*.
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- Only soft-block (L1) constraints are editable from the top level.
- Display of hierarchical constraints in the *Schematic Constraint Manager* follow the same use model.

Displaying the Hierarchical Constraints in Folders



You can use the **Show Folders** checkbox to see the directories for sub-cell instances as well as the constraints at the current level schematic/layout.

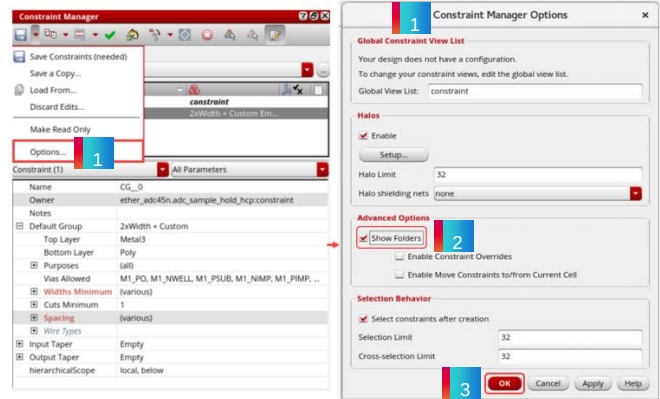
These directories will be shown corresponding to all the children of the current cellview, initially with a question mark icon.

To display the hierarchical constraints in folders in the schematic:

1. From the Constraint Manager, click the **Options** button to invoke the *Constraint Manager Options* form.
2. Then, in the *Constraint Manager Options* form, enable the **Show Folders** button.
3. Click **OK**.

Now, the constraints are displayed in folders in CM.

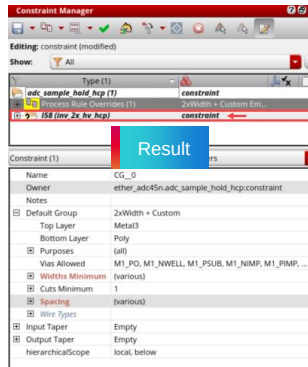
You can display the hierarchical constraints in folders in both schematic and layout.



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Result: Displaying the Hierarchical Constraints in Folders

Result: The constraints are displayed in the folders with a question mark icon.



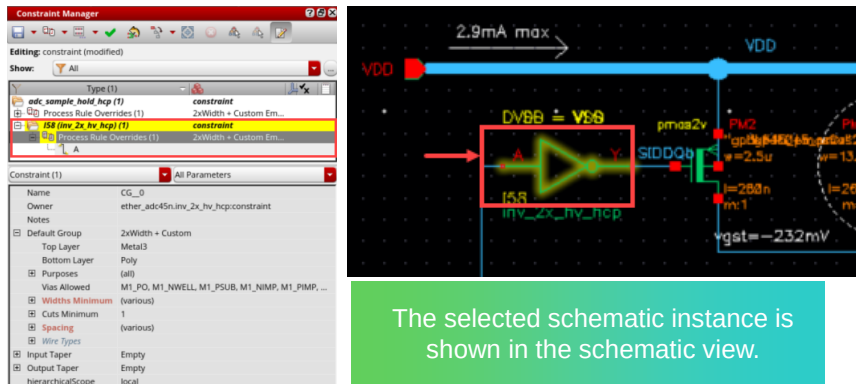
The lower-level constraints are not displayed until you expand the folder to avoid any performance impact.



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Result: Displaying the Hierarchical Constraints in Folders (continued)

Result: Expand the folder to see the pushed constraint.



The selected schematic instance is shown in the schematic view.

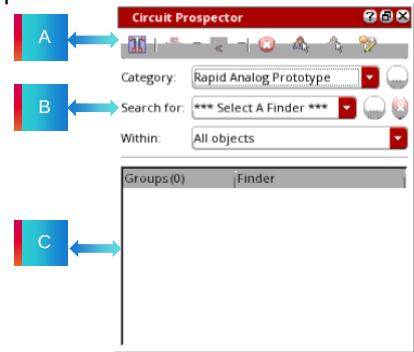
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What Is the Circuit Prospector Assistant?



The **Circuit Prospector** assistant can be used to detect circuit structures, devices, and nets based on certain criteria and optionally apply a default set of constraints to them.

- The *Circuit Prospector* can be used to set properties or to visualize various groups of objects, acting as an additional *search* facility.
- The *Circuit Prospector* can also be used for debugging and to apply specific actions to selected results.
- The **Circuit Prospector** user interface has three main components:
 - A. Circuit Prospector toolbar
 - B. Set of drop-down list boxes
 - C. Results table



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What Is Circuit Prospector Assistant?

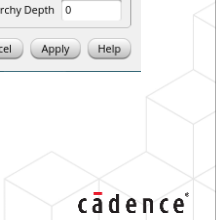
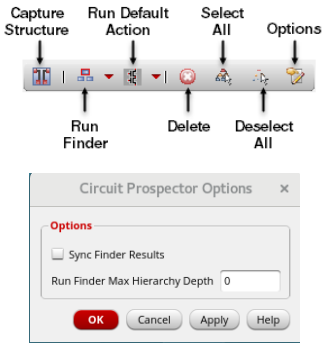
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- The *Circuit Prospector* can also be used for debugging and to apply specific actions to selected results.

The Circuit Prospector UI: Circuit Prospector Toolbar

The first component of the **Circuit Prospector** is the toolbar at the top. It contains the following options: *Capture Structure*, *Run Finder*, *Run Default (Constraint Generation) Action*, *Delete*, *Select All*, *Deselect All*, and *Options*.

Capture Structure	<i>Capture Structure</i> based on the settings and selections made in the <i>Circuit Prospector</i> .
Run Finder	<i>Run finder...</i> based on the finder selection in the <i>Search for</i> drop-down list box.
Run Default (Constraint Generation) Action	The options available in this menu (default constraint generation) are dependent upon the finder selected in the <i>Search for</i> drop-down list box and the <i>Groups</i> selected in the <i>Circuit Prospector</i> results table.
Delete	Use this menu to delete result entries from the results table.
Select All	Use this menu to select all result entries in the results table.
Deselect All	Use this menu to deselect all result entries from the results table.
Options	Clicking this menu item displays the <i>Circuit Prospector Options</i> form.



The Circuit Prospector UI: Circuit Prospector Toolbar

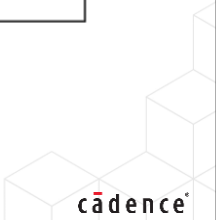
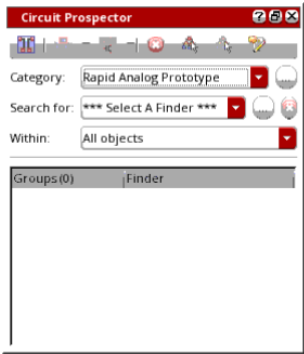
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The Circuit Prospector UI: Set of Drop-Down List Boxes B

The second component of the **Circuit Prospector**, right under the toolbar, are a set of drop-down list boxes to select categories of constraints, apply finders, and choose those objects to be constrained.

The drop-down list boxes are: *Category*, *Search for*, and *Within*.

Category	The first thing as part of the <i>Circuit Prospector</i> constraint creation flow is to select a <i>Category</i> from the list of available assistants.
Search for	Select a finder to search for relevant design objects and finders.
Within	Optionally Filter / select the found objects to apply constraints to.



The Circuit Prospector UI: Set of Drop-Down List Boxes

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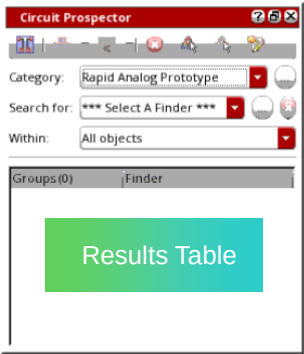
The drop-down list boxes are: *Category*, *Search for*, and *Within*.

The Circuit Prospector UI: Results Table

The third component is a results table that lists, in tree display format, all the constrained groups and the finder that they belong to.

The results table contains: *Groups* and *Finder*.

Groups	The <i>Groups</i> column displays the number of constrained groups and lists the current objects that will be included when you choose to <i>Capture Structure</i> .
Finder	The <i>Finder</i> column details the particular finder a group belongs to.



The Circuit Prospector UI: Results Table

The third component is a results table that lists, in tree display format, all the constrained groups and the finder that they belong to.

The results table contains: *Groups* and *Finder*.

Accessing the Circuit Prospector Assistant from Schematic



Virtuoso Unified Custom Constraints offers the *Circuit Prospector* assistant in VSE XL to automate or assist with the constraint creation process.

The *Circuit Prospector* Assistant is capable of finding circuit sub-structures in the design and generating the right set of constraints.

You can access the dockable *Circuit Prospector* assistant by using the following methods.

Different methods available are:

1. Using the *Assistants* menu in VSE
2. Using the *Workspaces* menu in VSE
3. Using the *Menu Bar/Toolbars* context-sensitive menu in VSE
4. Using the *Workspace Configuration* pull-down menu on the toolbar in VSE



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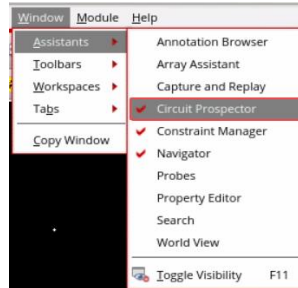
- Using the Assistants Menu in VSE
- Using the Workspaces Menu in VSE
- Using the Menu Bar/Toolbars Context-Sensitive Menu in VSE
- Using the Workspace Configuration Pull-Down Menu on the Toolbar in VSE

1. Using the *Assistants* Menu in VSE

To access the dockable *Circuit Prospector* assistant from schematic:

- Select **Window – Assistants – Circuit Prospector**.

Result: Now, the *Circuit Prospector* assistant is added as a docked assistant pane in the schematic window.



Step 1: Using the *Assistants* Menu in VSE

To access the dockable *Circuit Prospector* assistant from schematic:

- Select **Window – Assistants – Circuit Prospector**.

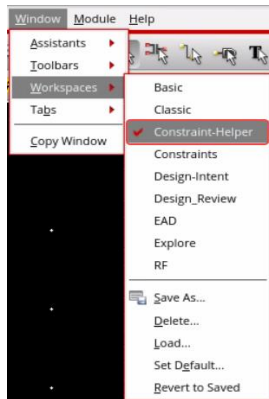
Result: Now, the *Circuit Prospector* assistant is added as a docked assistant pane in the schematic window.

2. Using the *Workspaces* Menu in VSE

To access the dockable *Circuit Prospector* assistant from schematic:

- Select **Window – Workspaces – Constraint-Helper**.

Result: Now, the *Circuit Prospector* assistant is added as a docked assistant pane in the schematic window.



Step 2: Using the *Workspaces* Menu in VSE

To access the dockable *Circuit Prospector* assistant from schematic:

- Select **Window – Workspaces – Constraint-Helper**.

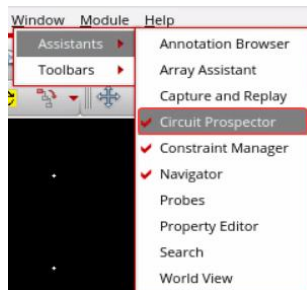
Result: Now, the *Circuit Prospector* assistant is added as a docked assistant pane in the schematic window.

3. Using the *Menu Bar/Toolbars* Context-Sensitive Menu in VSE

To access the dockable *Circuit Prospector* assistant from schematic:

- From the *Menu Bar/Toolbars* Context-Sensitive Menu, select **Assistants – Circuit Prospector**.

Result: Now, the *Circuit Prospector* assistant is added as a docked assistant pane in the schematic window.



Step 3: Using the *Menu Bar/Toolbars* Context-Sensitive Menu in VSE

To access the dockable *Circuit Prospector* assistant from schematic:

- From the *Menu Bar/Toolbars* Context-Sensitive Menu, select **Assistants – Circuit Prospector**.

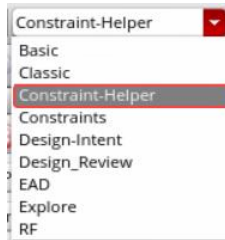
Result: Now, the *Circuit Prospector* assistant is added as a docked assistant pane in the schematic window.

4. Using the *Workspace Configuration* Pull-Down Menu on the Toolbar in VSE

To access the dockable *Circuit Prospector* assistant from schematic:

- From the *Workspace Configuration* pull-down menu on the toolbar, select the **Constraint-Helper** default workspace.

Result: Now, the *Circuit Prospector* assistant is added as a docked assistant pane in the schematic window.



Step 4: Using the *Workspace Configuration* Pull-Down Menu on the Toolbar in VSE

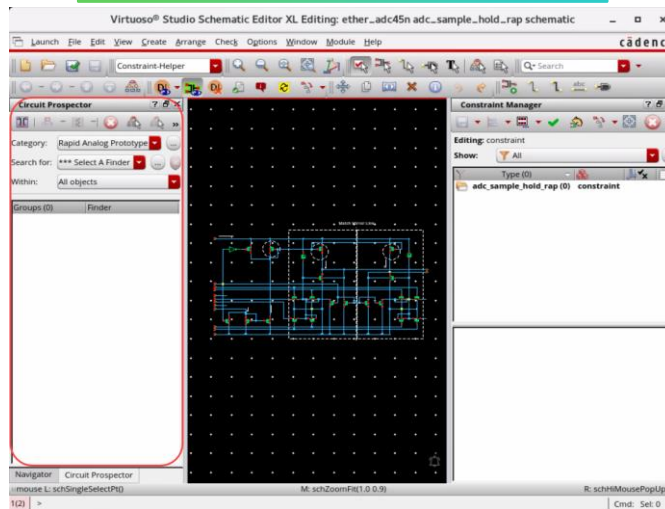
To access the dockable *Circuit Prospector* assistant from schematic:

- From the *Workspace Configuration* pull-down menu on the toolbar, select the **Constraint-Helper** default workspace.

Result: Now, the *Circuit Prospector* assistant is added as a docked assistant pane in the schematic window.

Result: Accessing the Circuit Prospector Assistant from Schematic

Result: The *Circuit Prospector* assistant is added as a docked assistant pane in the schematic window.



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Result: Accessing the Circuit Prospector Assistant from Schematic

Result: The *Circuit Prospector* assistant is added as a docked assistant pane in the schematic window.

What Is Rapid Analog Prototyping (RAP)?



The **Rapid Analog Prototype (RAP)** category in the **Circuit Prospector** contains all the finders/generators required to automatically constrain a design.

A dedicated new category has been introduced in the Circuit Prospector for rapid prototyping.

Associated with each finder is a generator to create relevant constraints for the found structures, including Modgens.

- Combining these devices into Modgens reduces the number of devices and constraints that the Analog Place and Route tools have to work with, leading to faster run times, better results, and more predictability.
- The finders/generators in the RAP category will generate all the constraints required to achieve a good initial (prototype) layout.
- The generators can be modified, and new finders/generators can be defined by the user to better suit their design requirements.



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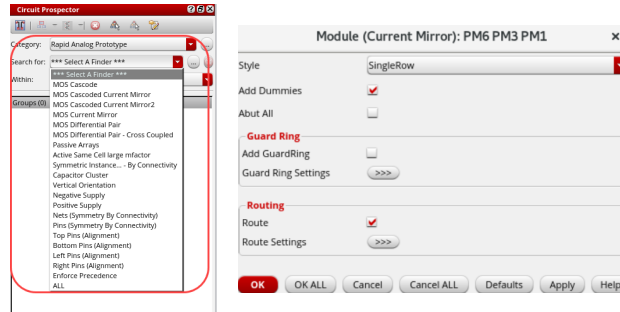
Accessing the RAP Constraint Generator from Circuit Prospector



In the schematic, from the Circuit Prospector, choose **Category > Rapid Analog Prototype** to access the RAP Constraint Generator.

Under the Search for cyclic field, you can review all the available finders.

You can run the RAP Constraint Generator on the selected finders.



Rapid Analog Prototyping is fully customizable which allows you to generate Modgens to your particular requirements.

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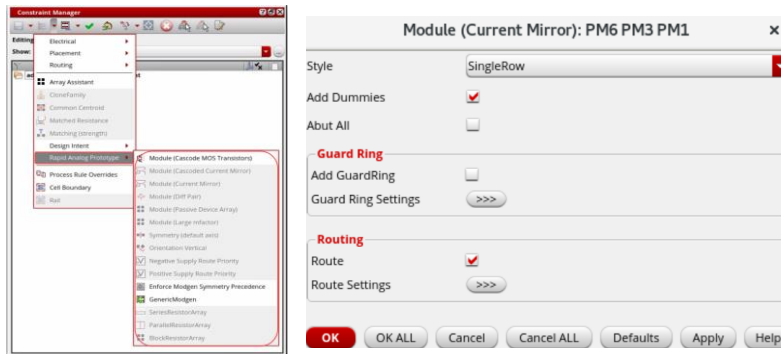
Rapid Analog Prototyping is fully customizable which allows you to generate Modgens to your particular requirements.

Accessing the RAP Constraint Generator from Constraint Manager



In the schematic, from the Constraint Manager, choose **Constraints > Rapid Analog Prototype** to access the RAP Constraint Generator.

You can manually select the structures and run the RAP Constraint Generator on them.



Accessing the RAP Constraint Generator from Constraint Manager

In schematic, from the Constraint Manager, choose **Constraints > Rapid Analog Prototype** to access the RAP Constraint Generator.

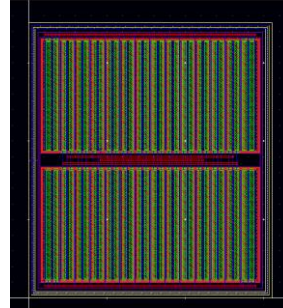
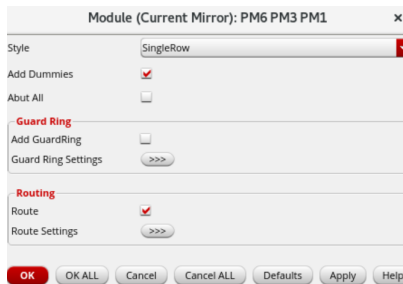
You can manually select the structures and run the RAP Constraint Generator on them.

Using Rapid Analog Prototyping to Create the Modgen Constraints



Use Rapid Analog Prototyping to create the Modgen constraints automatically for structures such as differential pairs, current mirrors, arrays of resistors/capacitors, large M-factored devices, etc.

- In the Constraint Generator, users can specify important parameters such as rows, abutment, dummies, guard rings, routing, etc.
 - The graphic shows a Modgen for a current mirror with two rows, abutted devices, dummies, guard ring, and balanced pin-to-trunk routing.
- The Modgen generators are fully customizable through a set of APIs to control how they get generated and what is exposed to users.



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Refer the *Virtuoso Layout Suite SKILL Reference, Product Version IC25.1* for the details on Module Generator SKILL Functions.

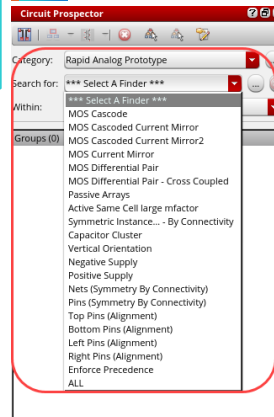
Using the Rapid Analog Prototype (RAP) Flow



The **Circuit Prospector** has a dedicated new category called **Rapid Analog Prototype (RAP)** for rapid prototyping.

Use **RAP** category to create a basic set of constraints to drive placement and routing.

- 1 Run (All/Selected) Finders in the Category
- 2 Run all respective actions for the Current Cellview
- 3 Specify Modgen Settings



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The **Circuit Prospector** has a dedicated new category called **Rapid Analog Prototype (RAP)** for rapid prototyping.

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In the RAP flow, you can perform the following tasks, namely:

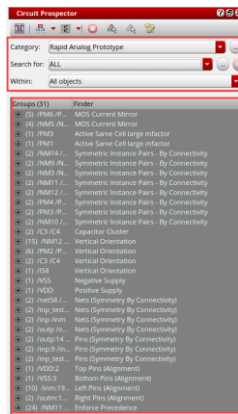
- Run All or Selected Finders in the Category
- Run all respective actions for the Current Cellview
- Specify the Modgen Settings

Step 1: Run (All/Selected) Finders in the Category

In the Circuit Prospector, choose the following options:

- Choose the **Category** as *Rapid Analog Prototype*.
- Under **Search for**, select *ALL*.
- Under **Within**, select *All objects*.
 - Ensure that all the groups are selected from the **Finder** results.

- 1 Run (All/Selected) Finders in the Category
- 2 Run all respective actions for the Current Cellview
- 3 Specify Modgen Settings



Step 1: Run (All/Selected) Finders in the Category

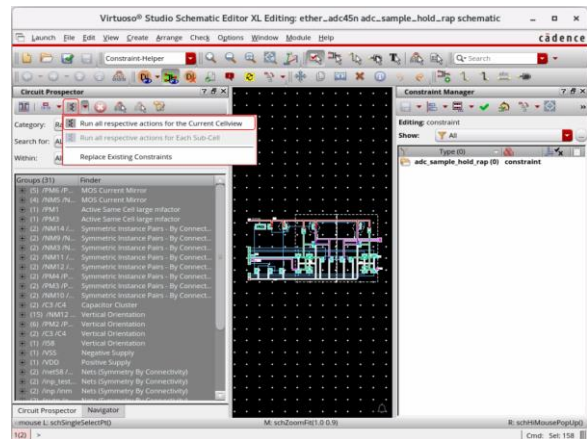
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- Choose the **Category** as *Rapid Analog Prototype*.
- Under **Search for**, select *ALL*.
- Under **Within**, select *All objects*.
- Ensure that all the groups are selected from the **Finder** results.

Step 2: Run All Respective Actions for the Current Cellview

In the Circuit Prospector, click on the **Run all respective actions for the Current Cellview** option to add the constraints.

- 1 Run (All/Selected) Finders in the Category
- 2 Run all respective actions for the Current Cellview
- 3 Specify Modgen Settings



Step 2: Run All Respective Actions for the Current Cellview

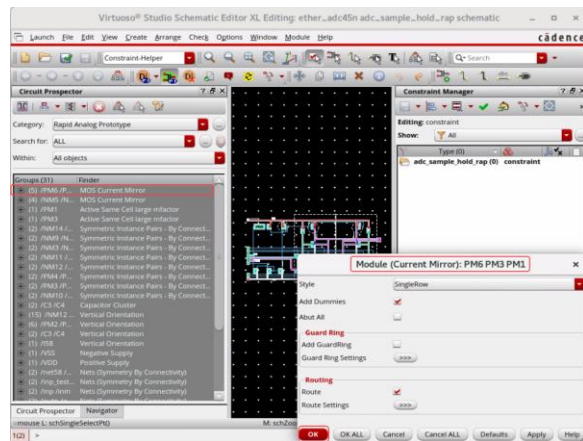
In the Circuit Prospector, click on the **Run all respective actions for the Current Cellview** option to add the constraints.

Step 3: Specify Modgen Settings

The Constraint Generator dialogs will pop up for each RAP constraint generator allowing you to configure the generator parameters for the associated set of devices.

- The Constraint Generator forms expose to users the most important parameters of the Modgen and their default settings.
- Users can configure them if needed e.g., whether to add dummies or guard rings.
- The Modgens are created as templates in the Constraint Manager that can be modified without launching the Modgen Editor.

- 1 Run (All/Selected) Finders in the Category
- 2 Run all respective actions for the Current Cellview
- 3 Specify Modgen Settings



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Step 3: Specify Modgen Settings (continued)

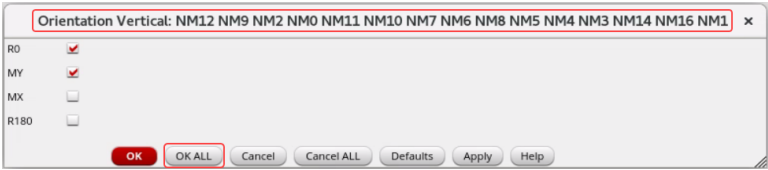
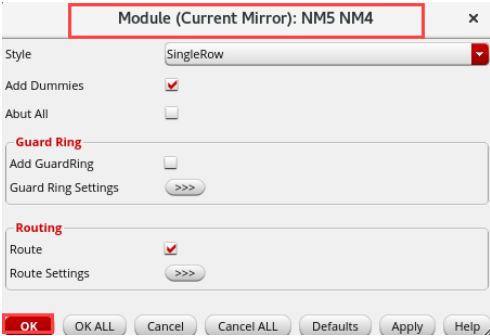
A few more Constraint Generator forms of an example are given here.

- 1

Run (All/Selected) Finders in the Category
- 2

Run all respective actions for the Current Cellview
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Specify Modgen Settings

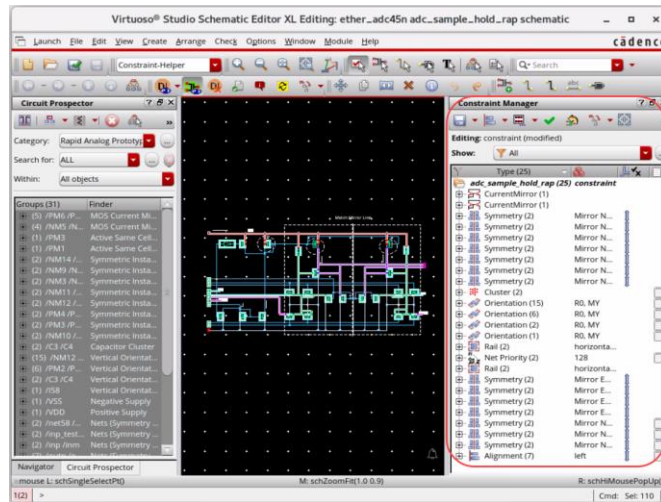


Step 3: Specify Modgen Settings

A few more Constraint Generator forms of an example are given here.

Result: Using the Rapid Analog Prototype (RAP) Flow

Result: After the RAP run, the constraints are added to the schematic and appear in the Constraint Manager.



Result: Using the Rapid Analog Prototype (RAP) Flow

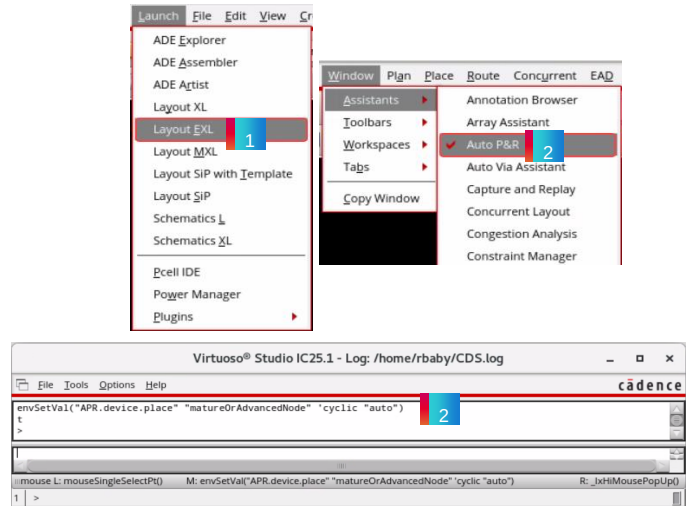
Transferring the RAP Constraints from Schematic to Layout



The RAP constraints added in the schematic can be transferred to the layout which can be used for your layout design tasks.

To transfer the RAP constraints from the schematic to the layout window:

1. First, from the schematic, use the **Launch – Layout EXL** command and open a new layout view.
2. Then, generate all from the source by entering the command `envSetVal("APR.device.place" "matureOrAdvancedNode" 'cyclic "auto")` in the CIW window then **Window – Assistants – Auto P&R** menu.



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Transferring the RAP Constraints from Schematic to Layout

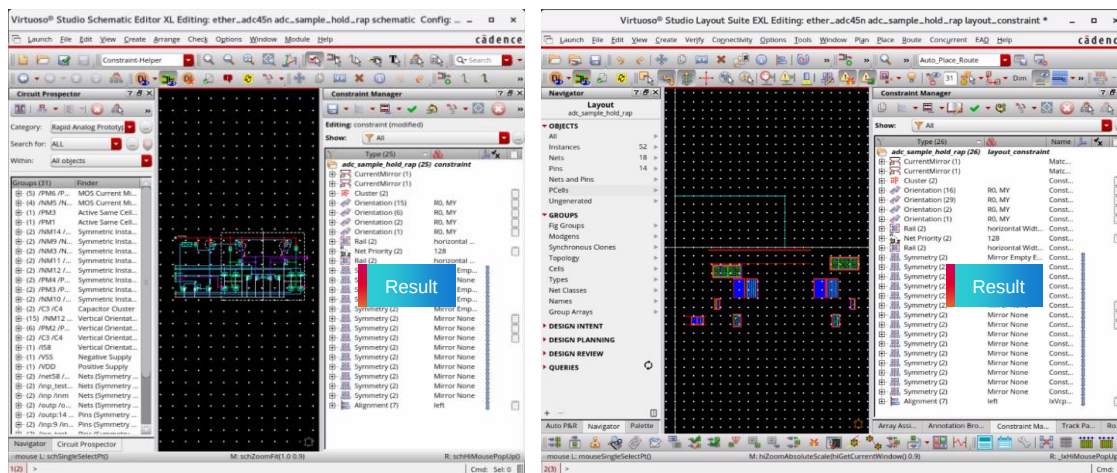
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2. Then, generate all from the source by entering the command `envSetVal("APR.device.place" "matureOrAdvancedNode" 'cyclic "auto")` in the CIW window then **Window – Assistants – Auto P&R** menu.
3. Next, in the layout canvas, switch the **Workspace** to **Constraints** to turn on the Constraint Manager. Now, all the RAP constraints are transferred from the schematic to the layout.

Result: Transferring the RAP Constraints from Schematic to Layout

Result: All the RAP constraints are transferred from the schematic to the layout.



Result: Transferring the RAP Constraints From Schematic To Layout

Result: All the RAP constraints are transferred from the schematic to the layout.

Invoking the Module Generator



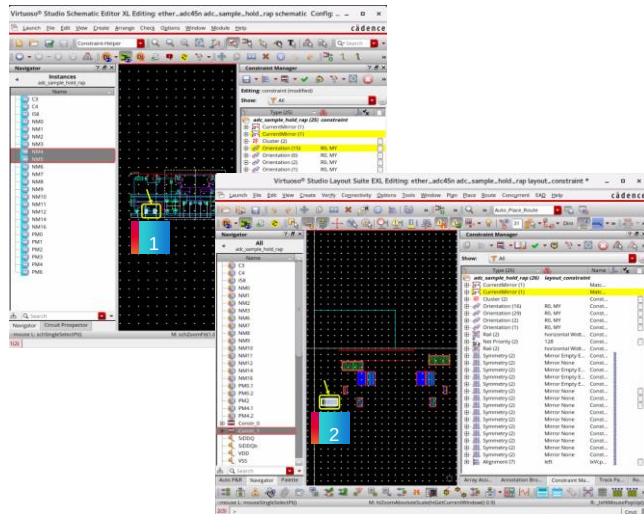
The **Module Generator** provides a simple and fast method of generating multiple Pcell instances into a complex, highly matched array based on defined Pcell leaf structures.

Use the **Module Generator** to run the constraint editing modgen functionality to edit the Modgens.

To invoke the Module Generator:

1. First, select the instances, for example, **/NM5** and **/NM4**, in the Navigator on the schematic side.
2. Next, in the layout window, **double-click** to descend.

Now, the selected **figGroup** is opened in the Modgen Editing window.



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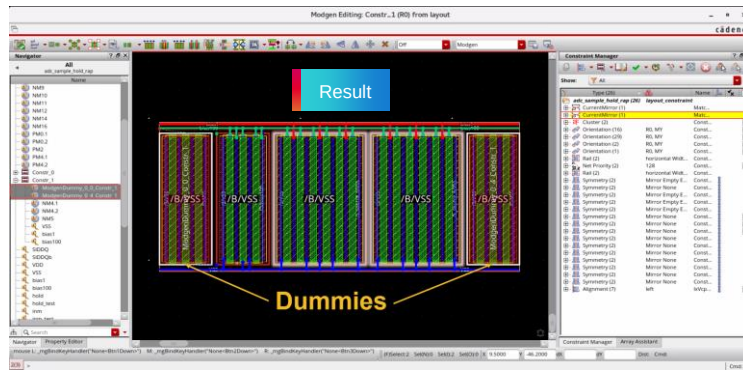
To invoke the Module Generator:

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2. Next, in the layout window, **double-click** to descend.

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Result: Invoking the Module Generator

Result: The selected **figGroup** is opened in the **Modgen Editing** window.



When running the **Constraint Generator**, dummies are specified.



Result: Invoking the Module Generator

Result: The selected **figGroup** is opened in the **Modgen Editing** window.

When running the **Constraint Generator**, dummies are specified.

Setting the Default Workspace for Modgen Editor Using the modgenWindowConfigFile



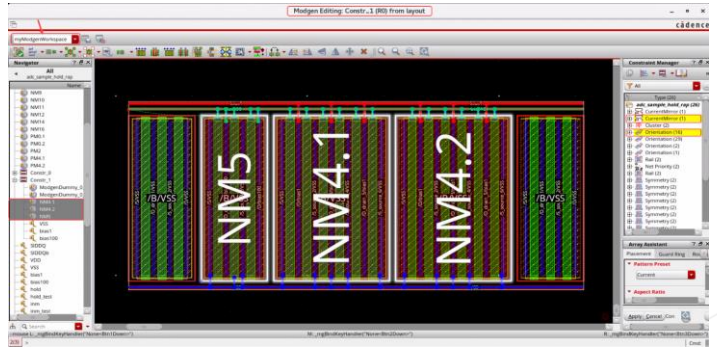
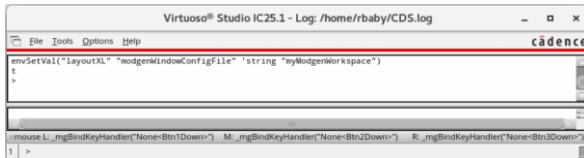
To specify the default workspace for the Modgen Editing session to start with:

- In the CIW, enter the following SKILL command and reinvoke the Modgen Editor.
 - `envSetVal("layoutXL" "modgenWindowConfigFile" 'string "myModgenWorkspace")`

Note: Here, `myModgenWorkspace` is the required workspace that the Modgen needs to load during startup.

Use the `modgenWindowConfigFile` environment variable to specify the default workspace to be loaded when the Modgen is invoked.

Result: After entering this SKILL command in the CIW window, when the Modgen Editor is reinvoked, it opens with `myModgenWorkspace` in the **Workspace Configuration** menu.



Use the `.cdsinit` file to specify the default workspace to be loaded automatically when Modgen Editor is invoked.

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Invoking the Array Assistant



In the layout canvas, choose the **Window > Assistants > Array Assistant** to invoke the Array Assistant.

Use the **Array Assistant** to quickly create and edit Modgens without invoking the Modgen Editor.

In Layout EXL and higher tiers, use the tabs in the **Array Assistant** to specify the following:

- **Placement:** Specifies the Modgen pattern settings.
- **Guard Ring:** Creates guard rings around Modgens.
- **Routing:** Defines Modgen topology patterns and routing preferences.
- **Reuse:** Loads and saves Modgen settings to template files.



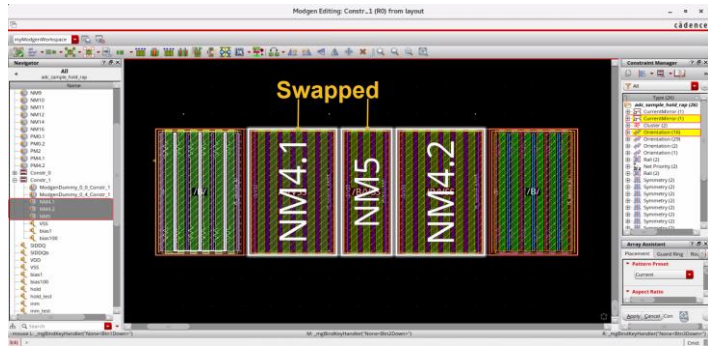
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Example: Using the Array Assistant Swapping the Pattern

In the Array Assistant, in the **Pattern** field-**Text** tab, swap the **Pattern** as follows and click **Apply**.

* A B B * into * B A B *

The instances get swapped.

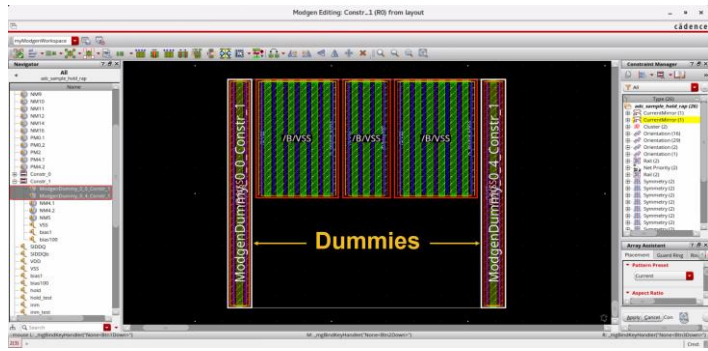


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Example: Using the Array Assistant: Modifying the Dummy Control Section

In the Array Assistant, modify the **Dummy Control** section as shown here and click **Apply**.

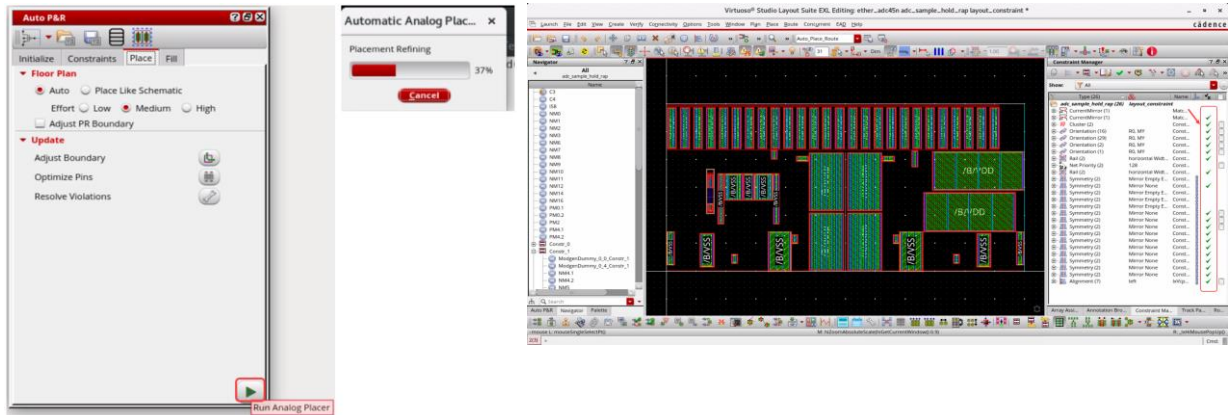
The dummies get updated.



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Example: Auto Placed Design

After running the *Analog Automatic Placement*, the devices get placed in the layout, meeting all the constraints.



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Quiz: The Rail Constraint



The Rail constraint can be specified in the Schematic Constraint Manager and NOT in the Layout Constraint Manager.

- A. True
- B. False

Answer: B

The Rail constraint can be specified in both the Schematic and the Layout Constraint Managers.



Answer: B

The Rail constraint can be specified in both the Schematic and the Layout Constraint Managers.

Quiz: The Constraint



The Boundary Area constraint has been revamped and consolidated with the Area Utilization constraint to the new ---- constraint.

Enter the answer in the blank field.

Answer: Cell Boundary

The Boundary Area constraint has been revamped and consolidated with the Area Utilization constraint to the new Cell Boundary constraint.



Answer: Cell Boundary

The Boundary Area constraint has been revamped and consolidated with the Area Utilization constraint to the new Cell Boundary constraint.

Quiz: Viewing the Constraints



Which toggle button in the Constraint Manager toolbar can you use to view the constraints on selected objects?

Select all that apply.

- A. View Selected
- B. Show Selected
- C. Show Constraints
- D. View Constraints
- E. None of the above

Answer: B

Constraints on selected objects can be viewed by clicking the **Show Selected** toggle button in the Constraint Manager toolbar.



Answer: B

Constraints on selected objects can be viewed by clicking the Show Selected toggle button in the Constraint Manager toolbar.

Quiz: Propagating the Constraints



What are the different methods can you use to propagate the constraints recursively all the way to the bottom or the top of the hierarchy?

Select all that apply.

- A. Constraint Manager
- B. SKILL
- C. Configure Physical Hierarchy
- D. Annotation Browser
- E. All of the above

Answer: A and B

You can use either the Constraint Manager assistant or SKILL to propagate the constraints recursively all the way to the bottom or the top of the hierarchy.



Answer: A and B

You can use either the Constraint Manager assistant or SKILL to propagate the constraints recursively all the way to the bottom or the top of the hierarchy.

Quiz: The Constraint



Which constraint can you use to override on a given object, the generic foundry and design process rules with specific, more stringent rules as defined in Constraint Groups?

Select all that apply.

- A. Cell Boundary
- B. Rail
- C. Process Rule Overrides (PRO)
- D. Symmetry
- E. None of the above

Answer: C

You can use the Process Rule Overrides (PRO) constraint to override on a given object, the generic foundry and design process rules with specific, more stringent rules as defined in Constraint Groups.



Answer: C

You can use the Process Rule Overrides (PRO) constraint to override on a given object, the generic foundry and design process rules with specific, more stringent rules as defined in Constraint Groups.

Quiz: Accessing the RAP Constraint Generator



Which assistants can you use to access the RAP Constraint Generator?

Select all that apply.

- A. Navigator
- B. Annotation Browser
- C. Circuit Prospector
- D. Constraint Manager
- E. All of the above

Answer: C and D

You can use either the Circuit Prospector or the Constraint Manager to access the RAP Constraint Generator.



Answer: C and D

You can use either the Circuit Prospector or the Constraint Manager to access the RAP Constraint Generator.

Module Summary

This module covered:

- The Rail Constraint Creation and Implementation supported by Constraint Aware Editing (CAE) and Analog Auto Placer (AAP).
- The Cell Boundary Constraint Creation and Implementation with Area Utilization.
- Usability Enhancements in Constraint Manager to improve User Experience through Show Selected, Name Column, Axis Column, and Folder-Based Hierarchy.
- Hierarchical Constraint Propagation to propagate constraints up or down through schematic and layout levels.
- Creating Process Rule Overrides (PRO) Constraints and editing parameters such as hierarchicalScope to control propagation.
- Use the Rapid Analog Prototyping (RAP) to automate the constraint generation using the Circuit Prospector finders.
- Use the Module Generator to build structured, matched device arrays.
- Use the Array Assistant for fast tuning of patterns, dummies, guard rings, routing, and reuse.



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Labs



- Lab 2-1 Analyzing VSE XL and VLS XL Features: Usability Improvements in the Constraint-Driven Flow
- Lab 2-2 Performing Hierarchical Constraint Propagation
- Lab 2-3 Analyzing the Rapid Analog Prototype (RAP) Category in the Circuit Prospector and Modgen



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Module 3

Power Routing

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Module Objectives

In this module, you

- Implement a power structure on a given design using the Virtuoso® Space-based Router (VSR)
- Run signal routing
- Extract the design
- Fix the connectivity errors
- Verify the design



VSR - Virtuoso Space-based Router

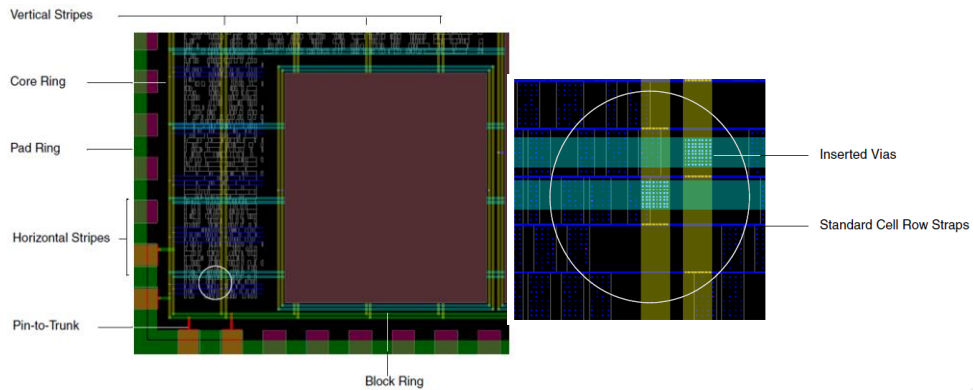
What Is the Power Router?



Use the Virtuoso Space-based Router's **Power Router** to construct the common power components.

Using the *Power Router*, you can construct the power components as seen in the example:

- Pad Ring, Core Ring, Block Ring, Pin-to-Trunk, Vertical/Horizontal Stripes, Standard Cell Row Straps, and Inserted Vias.



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Using the *Power Router*, you can construct the power components as seen in this example:

- Pad Ring, Core Ring, Block Ring, Pin-to-Trunk, Vertical/Horizontal Stripes, Standard Cell Row Straps, and Inserted Vias.

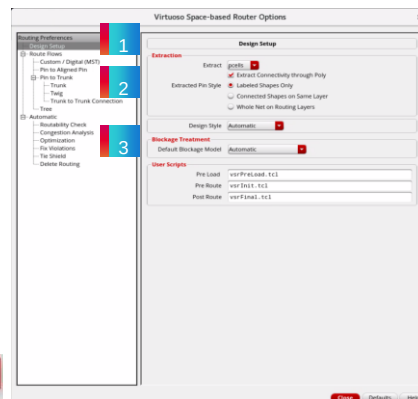
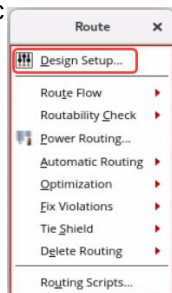
Invoking and Using the Virtuoso Space-based Router (VSR) Options Form



In the layout canvas, choose **Route – Design Setup** or click the **Design Setup** icon on the Virtuoso Space-based Router toolbar to open the Virtuoso Space-based Router Options form.

Using the *Virtuoso Space-based Router Options* form, you can specify the following top-level routing features:

1. Design Setup
2. Route Flows
3. Automatic



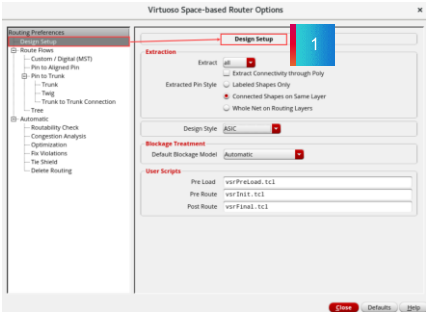
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Virtuoso Space-based Router Options – Design Setup

Click the *Design Setup* icon on the toolbar or choose **Route – Design Setup** to display the *Design Setup* subform in the Virtuoso Space-based Router Options form.

Using the *Design Setup* subform, you can specify the following options: *Extraction*, *Design Style*, *Blockage Treatment*, and *User Scripts*.

Extraction	Opening Virtuoso does not extract your design by default. You must set extractor depth (<i>Options – Layout XL</i>) and extract your design (<i>Connectivity – Extract Layout</i>). The extraction options in the Virtuoso Space-based Router Options form lets you control the instances that are extracted using router specific extraction.
Design Style	The design style determines which steps in the Wire Assistant flow will run and also affects the heuristics of global and detail routes. Therefore, the value of design style needs to reflect the style of design you are routing. By default, the design style that was last specified is selected in the drop-down list.
Blockage Treatment	These options let you alter the treatment of the blockages. <i>Default Blockage Model</i> : Ensures that all blockages have proper spacing properties and that the blockages model the metal on the edge of the block properly.
User Scripts	An existing routing flow can be captured in the form of Tcl routing commands in a Tcl script. You can run the Tcl scripts to view routing results.



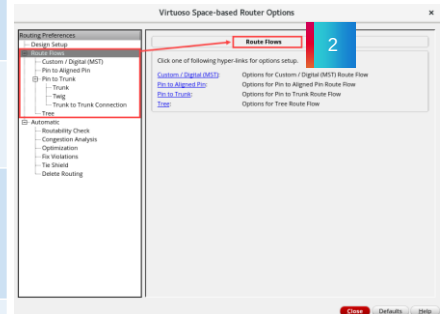
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Virtuoso Space-based Router Options – Route Flows

Click **Route Flows**. This displays the hyperlinks that let you quickly navigate to the available routing flows.

The router supports the routing flows: *Custom / Digital (MST)*, *Pin to Aligned Pin*, *Pin to Trunk*, and *Tree*.

Custom / Digital (MST)	The <i>Custom / Digital (MST)</i> routing style results in a connection with the shortest overall length. This routing style is usually used for congestion-driven designs. If there is no trunk, the <i>Custom / Digital (MST)</i> router routes the nets.
Pin to Aligned Pin	The <i>Pin to Aligned Pin</i> routing style is used to connect only the aligned pins on the same layer of net. The pins that are not aligned are not routed. This routing style is usually used when devices are placed in rows to connect pins that are aligned.
Pin to Trunk	The <i>Pin to Trunk</i> routing style is used to connect an individual pin to a trunk. Only the connections that are within the scope of a trunk are routed. The remaining opens are completed by using the <i>Custom / Digital (MST)</i> routing style or manually. This routing style is usually used when a spine structure is required.
Tree	The <i>Tree</i> Route flow lets you automatically route advanced node device-level designs with a minimum set of options.

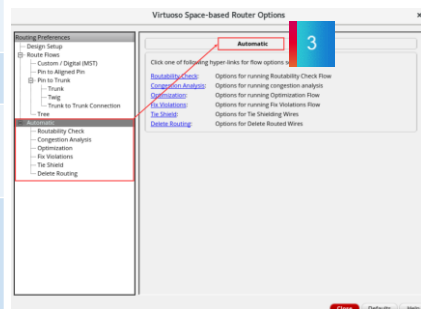


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Virtuoso Space-based Router Options – Automatic

Click **Automatic**. This displays the hyperlinks that let you quickly navigate to the other automatic routing flow subforms. The following routing flows are available in the *Automatic* subform: *Routability Check*, *Congestion Analysis*, *Optimization*, *Fix Violations*, *Tie Shield*, and *Delete Routing*.

Routability Check	The <i>Routability Check</i> feature is used to debug the design and determine potential routing issues that can prevent the router from achieving optimal results.
Congestion Analysis	Click <i>Congestion Analysis</i> from the <i>Automatic</i> tree under <i>Routing Preferences</i> . To perform Congestion Analysis, use the congestion analysis options displayed in the Congestion Analysis subform.
Optimization	Click <i>Optimization</i> from the <i>Automatic</i> tree under <i>Routing Preferences</i> . To optimize routing in the design, select any of the options from the subform.
Fix Violations	Using the <i>Fix Violations</i> flow, you can perform all spacing-related checks including merged shapes and fix the violation operations, for all nets, selected nets, or a selected area. Click <i>Fix Violations</i> from the <i>Automatic</i> tree under <i>Routing Preferences</i> . You can choose to fix any of the routing violations.
Tie Shield	The <i>Tie Shield</i> command is used for tying the shielding wires together and to bring the connected shielding wires to the nearest power source, which can be part of a power pin, power ring, or trunk. The <i>Tie Shield</i> command is not used for routing the entire shielding net. It is recommended that before tying the shielding wires together to the routed power net, power routing is performed.
Delete Routing	<i>Delete Routing</i> is an interactive command where you explicitly ask for the routing to be deleted. Clicking <i>Delete Routing</i> from the <i>Automatic</i> tree under <i>Routing Preferences</i> option deletes all the routing in FIXED constraints. However, the routes in the LOCKED constraint are not deleted.



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Using the Power Router Flow

Use the power routing tasks to construct the common power components for your design.

The power routing tasks you can use are:

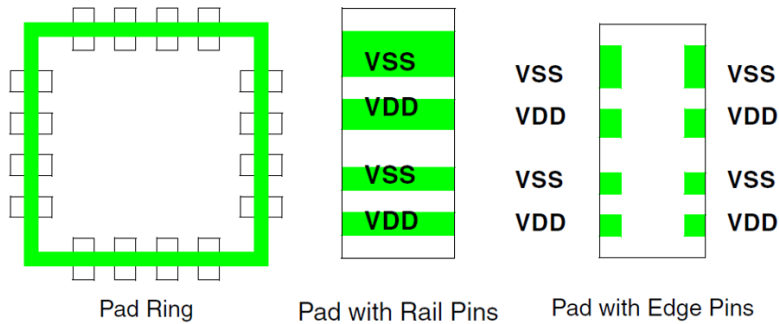
- Adding a Pad Ring
- Adding Core Rings
- Adding Block Rings
- Connecting Pin To Trunk
- Adding Stripes
- Adding Standard Cell Row Straps
- Trimming Stripes
- Inserting Vias
- Tie Shielding



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Adding a Pad Ring

Use the **Pad Ring** routing to add pads in the design.



Pad rings are routed between pads on the periphery of the design.
You can route to rail pins and/or edge pins on pads.

- The power routing tasks:
- Adding a Pad Ring
 - Adding Core Rings
 - Adding Block Rings
 - Connecting Pin To Trunk
 - Adding Stripes
 - Adding Standard Cell Row Straps
 - Trimming Stripes
 - Inserting Vias
 - Tie Shielding



Adding a Pad Ring

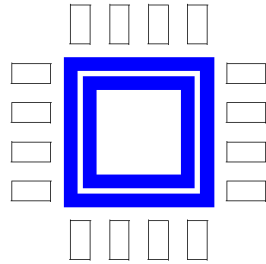
Use the Pad Ring routing to add pads in the design.

Pad rings are routed between pads on the periphery of the design.

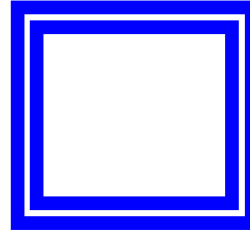
You can route to rail pins and/or edge pins on pads.

Adding Core Rings

Use the **Core Ring routing** to add core rings in the design.



Core Ring Created Around Design But Inside Pads



Core Ring Created Around Entire Design - No Pads

If the design has pads, rings are added around the core of the design.
If there are no pads, rings are added around the entire design.

- The power routing tasks:
- Adding a Pad Ring
 - **Adding Core Rings**
 - Adding Block Rings
 - Connecting Pin To Trunk
 - Adding Stripes
 - Adding Standard Cell Row Straps
 - Trimming Stripes
 - Inserting Vias
 - Tie Shielding



Adding Core Rings

Use the Core Ring routing to add core rings in the design.

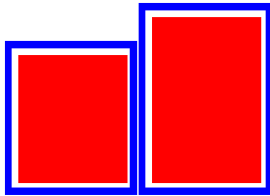
If the design has pads, rings are added around the core of the design.

If there are no pads, rings are added around the entire design.

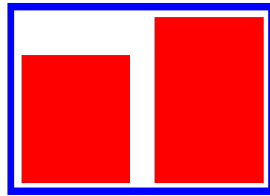
Adding Block Rings

Use the **Block Ring** routing to add wires in the design.

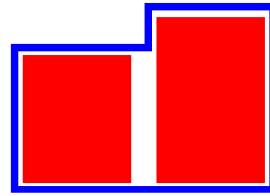
- The power routing tasks:
- Adding a Pad Ring
 - Adding Core Rings
 - **Adding Block Rings**
 - Connecting Pin To Trunk
 - Adding Stripes
 - Adding Standard Cell Row Straps
 - Trimming Stripes
 - Inserting Vias
 - Tie Shielding



Block Rings Created Around Individual Blocks



Block Ring Created Around A Pair of Blocks Without Contouring



Contoured Block Ring Created Around A Pair of Blocks

Block rings are wires around one or more block instances on a set of specified layers with a specific offset between the boundaries of the blocks and the power rings.

Rings can follow the contour of the blocks or form a rectangle around the blocks.



Adding Block Rings

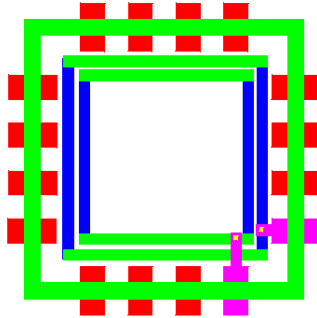
Use the Block Ring routing to add wires in the design.

Block rings are wires around one or more block instances on a set of specified layers with a specific offset between the boundaries of the blocks and the power rings.

Rings can follow the contour of the blocks or form a rectangle around the blocks.

Connecting Pin To Trunk

Use the **Pin To Trunk routing** to connect an individual pin to a trunk.



Power pins on blocks and power pads are connected to existing power rings or rails.
This example shows pin-to-trunk connections between pad pins and the core rings in the lower right corner.

- The power routing tasks:
- Adding a Pad Ring
 - Adding Core Rings
 - Adding Block Rings
 - **Connecting Pin To Trunk**
 - Adding Stripes
 - Adding Standard Cell Row Straps
 - Trimming Stripes
 - Inserting Vias
 - Tie Shielding



Connecting Pin To Trunk

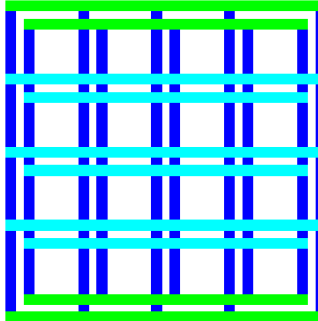
Use the Pin to Trunk routing to connect an individual pin to a trunk.

Power pins on blocks and power pads are connected to existing power rings or rails.

This example shows pin-to-trunk connections between pad pins and the core rings in the lower right corner.

Adding Stripes

Use the **Stripes** routing to add power mesh to the core area in the design.



You specify parameters such as which layers to use for the given nets, the clearance required between nets, pin clearance, the location and interval for the stripes.

Horizontal and Vertical Stripes in this example form a power mesh inside the Core Ring.

In a later step, vias are inserted to connect the stripes within the mesh to the rings.

- The power routing tasks:
- Adding a Pad Ring
 - Adding Core Rings
 - Adding Block Rings
 - Connecting Pin To Trunk
 - **Adding Stripes**
 - Adding Standard Cell Row Straps
 - Trimming Stripes
 - Inserting Vias
 - Tie Shielding



Adding Stripes

Use the Stripes routing to add power mesh to the core area in the design.

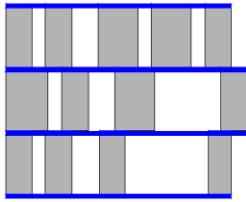
You specify parameters such as which layers to use for the given nets, the clearance required between nets, pin clearance, the location and interval for the stripes.

Horizontal and Vertical Stripes in this example form a power mesh inside the Core Ring.

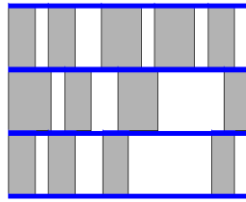
In a later step, vias are inserted to connect the stripes within the mesh to the rings.

Adding Standard Cell Row Straps

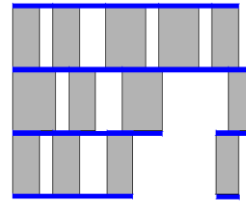
Use the **Cell Rows** routing to add straps along aligned power pins of standard cells.



Standard Cell Row Straps
to the Last Cell in Each
Row (Default)



Standard Cell Row Straps
to the End of Rows



Standard Cell Row Straps
With
Maximum Row Gap Size

Straps are added along aligned power pins of standard cells.

The straps can be extended to the end of rows or to the last cell in a physical row.

- The power routing tasks:
- Adding a Pad Ring
 - Adding Core Rings
 - Adding Block Rings
 - Connecting Pin To Trunk
 - Adding Stripes
 - **Adding Standard Cell Row Straps**
 - Trimming Stripes
 - Inserting Vias
 - Tie Shielding



Adding Standard Cell Row Straps

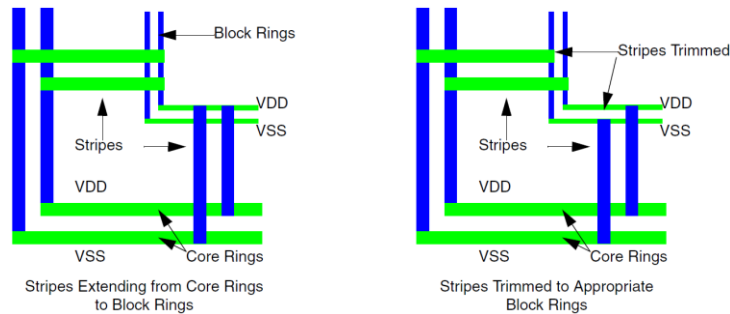
Use the Cell Rows routing to add straps along aligned power pins of standard cells.

Straps are added along aligned power pins of standard cells.

The straps can be extended to the end of rows or to the last cell in a physical row.

Trimming Stripes

Use the `rtePowerRouteTrimStripes` SKILL® function to trim the stripes.



Stripes are trimmed back to an intersecting ring on the same net, if any.
You can run this command after all stripes and rings have been created.

- The power routing tasks:
- Adding a Pad Ring
 - Adding Core Rings
 - Adding Block Rings
 - Connecting Pin To Trunk
 - Adding Stripes
 - Adding Standard Cell Row Straps
 - **Trimming Stripes**
 - Inserting Vias
 - Tie Shielding



Trimming Stripes

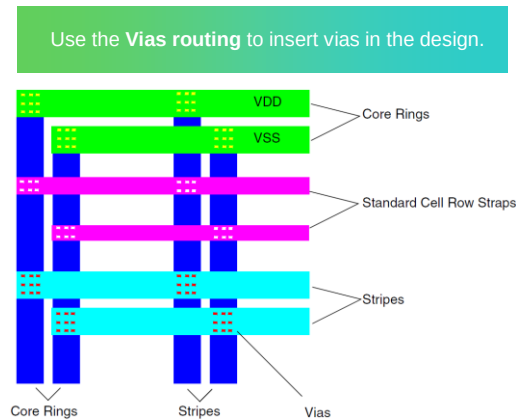
Use the `rtePowerRouteTrimStripes` function to trim the stripes.

Stripes are trimmed back to an intersecting ring on the same net, if any.

You can run this command after all stripes and rings have been created.

For more information, refer to the *Virtuoso Layout Suite SKILL Reference, Product Version IC23.1*.

Inserting Vias



- The power routing tasks:
- Adding a Pad Ring
 - Adding Core Rings
 - Adding Block Rings
 - Connecting Pin To Trunk
 - Adding Stripes
 - Adding Standard Cell Row Straps
 - Trimming Stripes
 - **Inserting Vias**
 - Tie Shielding

The current power routing topology does not support inserting vias instantaneously except connecting pin-to-trunk. Therefore, vias are added manually to provide interlayer connections at intersections of rings, stripes and straps. In this example, vias are inserted to connect stripes and straps to the core rings in the design.



Inserting Vias

Use the Vias routing to insert vias in the design.

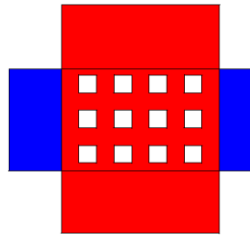
The current power routing topology does not support inserting vias instantaneously except connecting pin-to-trunk.

Therefore, vias are added manually to provide interlayer connections at intersections of rings, stripes and straps.

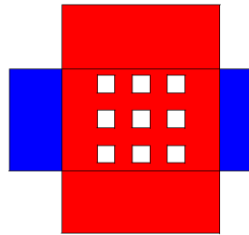
In this example, vias are inserted to connect stripes and straps to the core rings in the design.

Inserting Vias: With Cut Arrays

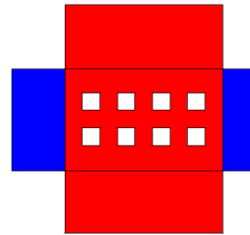
These examples show different cut arrays that can be created for the same via by changing command arguments.



Maximum Number of Rows and Columns (Rectangle, default)



Equal Number of Rows and Columns (Square)



Specific Number of Rows and Columns

By default, cut arrays are inserted with the maximum number of rows and columns for the size of the via area.

- The power routing tasks:
- Adding a Pad Ring
 - Adding Core Rings
 - Adding Block Rings
 - Connecting Pin To Trunk
 - Adding Stripes
 - Adding Standard Cell Row Straps
 - Trimming Stripes
 - **Inserting Vias**
 - Tie Shielding



Inserting Vias: With Cut Arrays

These examples show different cut arrays that can be created for the same via by changing command arguments.

By default, cut arrays are inserted with the maximum number of rows and columns for the size of the via area.

Tie Shielding

Use the **Tie Shield routing** to tie the shield wires to the shield nets in the design.

The power routing tasks:	
•	Adding a Pad Ring
•	Adding Core Rings
•	Adding Block Rings
•	Connecting Pin To Trunk
•	Adding Stripes
•	Adding Standard Cell Row Straps
•	Trimming Stripes
•	Inserting Vias
•	Tie Shielding

- The *Tie Shield* command is used for tying the shielding wires together and to bring the connected shielding wires to the nearest power source which can be part of a power pin, power ring, or trunk.
- The *Tie Shield* command is not used for routing the entire shielding net.
- It is recommended that before tying the shielding wires together to the routed power net, power routing is performed.



Tie Shielding

Use the Tie Shield Routing to tie the shield wires to the shield nets in the design.

- The *Tie Shield* command is used for tying the shielding wires together and to bring the connected shielding wires to the nearest power source which can be part of a power pin, power ring, or trunk.
- The *Tie Shield* command is not used for routing the entire shielding net.
- It is recommended that before tying the shielding wires together to the routed power net, power routing is performed.

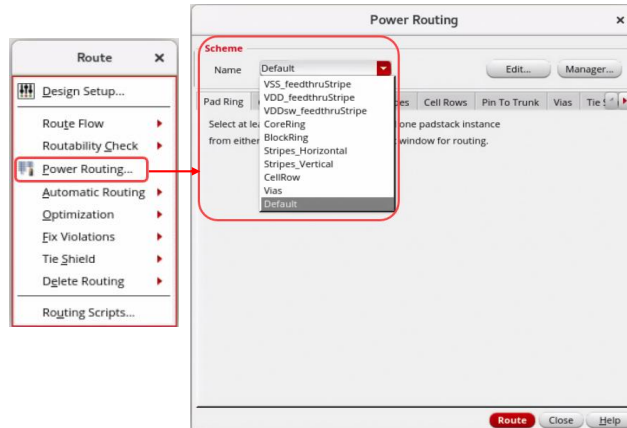
For more information, refer to the *Virtuoso Space-based Router User Guide, Product Version IC23.1*.

Invoking the Power Router



In the layout canvas, choose the **Route – Power Routing** menu command to display the Power Routing form.

Using the *Power Routing* form, you can create power components in your design.



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Invoking the Power Router

In the layout canvas, choose the **Route – Power Routing** menu command to display the Power Routing form.

Using the *Power Routing* form, you can create power components in your design.

The Power Routing Form



The *Power Routing* form has tabs that enable you to set up options for each power routing operation.

Name	The scheme name for the power routing operation.
Edit	Enables you to view or change the scheme definition.
Manager	Displays the <i>Power Routing Scheme Manager</i> form.
Route	Executes power routing with the options in the scheme.



The Power Routing Form

The *Power Routing* form has the following tabs, namely: *Name*, *Edit*, *Manager*, and *Route*.

These tabs enable you to set up the options for each power routing operation.

Name is the scheme name for the power routing operation.

Clicking the *Edit* button enables you to view or change the scheme definition.

Clicking the *Manager* button displays the *Power Routing Scheme Manager* form.

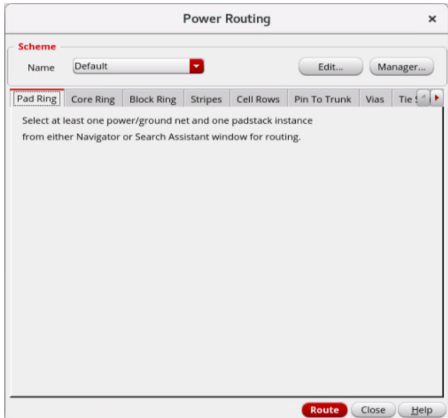
Clicking the *Route* button executes the power routing with the options in the scheme.

Using the Tabs in the Power Routing Form

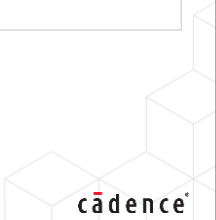


Use the tabs in the **Power Routing** form to set up the options for each power routing operation.

Different available tabs in Power Routing form are listed here:



- | | |
|---|--------------|
| 1 | Pad Ring |
| 2 | Core Ring |
| 3 | Block Ring |
| 4 | Stripes |
| 5 | Cell Rows |
| 6 | Pin To Trunk |
| 7 | Vias |
| 8 | Tie Shield |



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Power Routing Form: Pad Ring

In the *Power Routing* form, click on the **Pad Ring** tab.

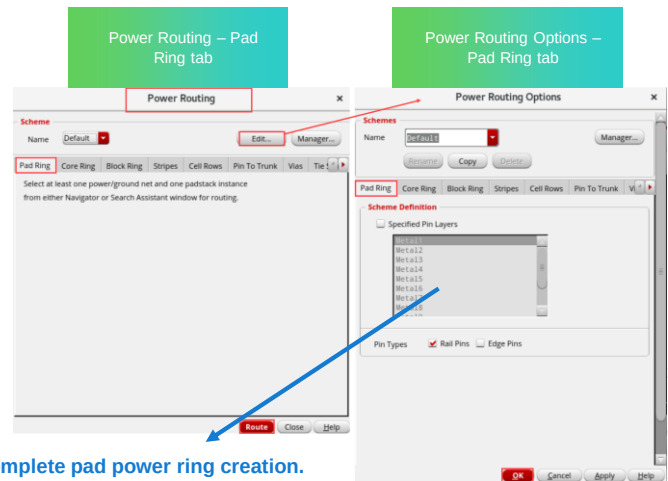
- *Pad Ring* routing adds a ring between pad instances on the periphery of the design.

Clicking the **Edit** button opens the *Power Routing Options* form.

- The **Pad Ring** tab of the *Power Routing Options* form lets you specify the pin layers and pin types to which to connect pad rings for the scheme.

Pad Power Ring

Select power net(s) interactively, specify the power pin routing layer and its routing type.



Note: Pin routing type needs to be validated in the layout to avoid incomplete pad power ring creation.

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Power Routing Form: Core Ring

In the *Power Routing* form, click on the **Core Ring** tab.

- *Core Ring routing* adds a ring around the core of a design that is surrounded by pad instances or I/O pins.

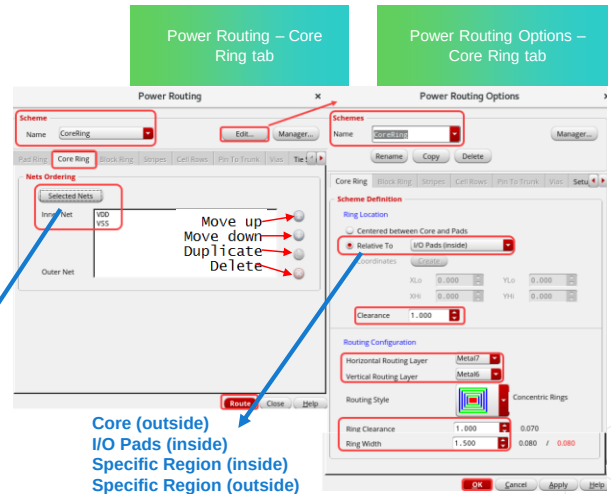
Clicking the **Edit** button opens the *Power Routing Options* form.

- The **Core Ring** tab of the *Power Routing Options* form lets you specify the core ring location and configuration for the scheme.

Core Power Ring

- Select power net(s) interactively and arrange the net order.
- The position and style of rings can be defined within the power routing option.

Once power nets are selected interactively, the user can import nets by clicking "Selected Nets" to arrange the net rings' order and duplicate or remove nets. Note: Duplicate net names are allowed.



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Power Routing Form: Block Ring

In the *Power Routing* form, click on the **Block Ring** tab.

- *Block Ring routing* adds rings around block instances.

Clicking the **Edit** button opens the *Power Routing Options* form.

- The **Block Ring** tab of the *Power Routing Options* form lets you specify the ring and routing configuration for block rings.

Block Power Ring

Select power net(s) and instance(s) interactively and arrange the net order.

Once power nets are selected interactively, the user can import nets by clicking “Selected Nets” to arrange the net rings’ order and duplicate or remove nets.

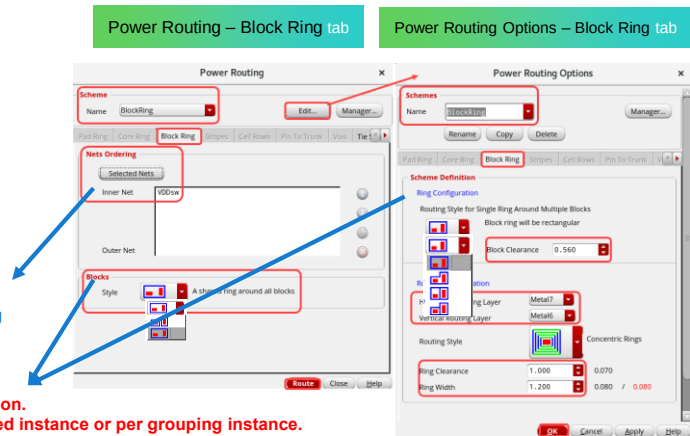
Note: Duplicate net names are allowed.

Do not confuse “Style” w/ “Ring Configuration” in Block Ring Scheme definition.

“Style” provides the user the option of creating one core ring per each selected instance or per grouping instance.

“Ring Configuration” triggers both contour and channel route options for grouped instance or individual instance:

1. No channel & contour route 2. Contour route only 3. Channel route only 4. Channel & contour.



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Power Routing Form: Stripes

In the *Power Routing* form, click on the **Stripes** tab.

- *Stripes routing* adds net stripes at regular intervals.

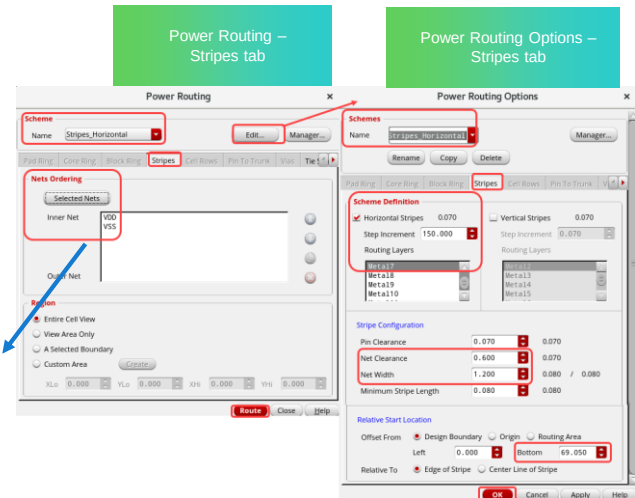
Clicking the **Edit** button opens the *Power Routing Options* form.

- The **Stripes** tab of the *Power Routing Options* form lets you specify the placement and configuration of the horizontal and vertical stripes.

Stripes

Select power net(s) and arrange the net order. Then, specify the placement of mesh nets.

Similar to core ring creation. However, the net ordering is L2R & B2T for Inner -> Outer net order with reference to the relative start location.



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Power Routing Form: Cell Row Straps

In the *Power Routing* form, click on the *Cell Rows* tab.

- *Cell Rows* routing adds straps along aligned pins of standard cells.

Clicking the *Edit* button opens the *Power Routing Options* form.

- The *Cell Rows* tab of the *Power Routing Options* form lets you specify routing layers, the cell row strap boundary, and whether the cell row straps should be extended to the nearest power rail.

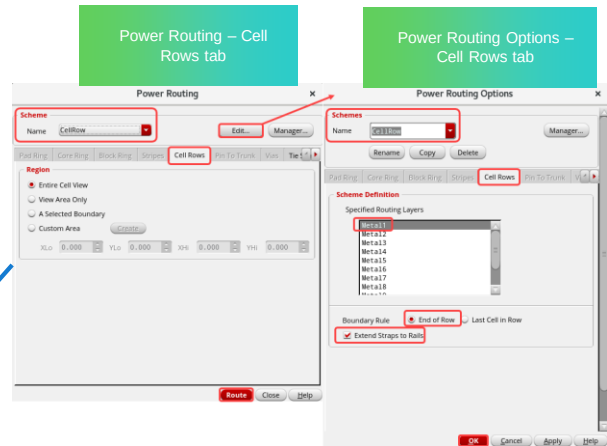
Cell Row Power Straps

Select power net(s) interactively and specify the routing area.

Specify the intended power straps routing area.

Note: In order for cell row power straps to create properly, the user has to make sure that:

1. "Row" placement characteristics reside within the layout view.
2. Layout is to be extracted prior to cell row power straps creation.



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Power Routing Form: Pin To Trunk

In the *Power Routing* form, click on the **Pin To Trunk** tab.

- *Pin-To-Trunk routing* allows you to add connections from the power pins of pad or macro instances to existing rings and rails. A macro instance is defined as any instance whose area is greater than 90% of the entire cell view area.

Clicking the **Edit** button opens the *Power Routing Options* form.

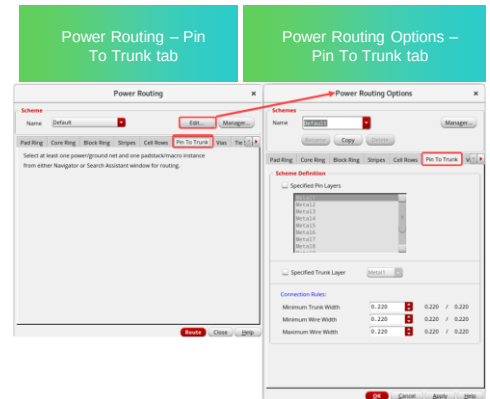
- The *Pin To Trunk* tab of the *Power Routing Options* form lets you specify pin layers, trunk layers, and connection rules.

Pin-To-Trunk Power Net Connections

- Select power net(s) interactively. Unlike the interactive p2t route in WA, the user is not required to compose the trunk prior to the pin-to-trunk route.
- This is the only feature in power routing that automatically

Note:

1. All power nets created through the power routing mechanism preserve trunk characteristics (wire:pin in ctu) for p2t power routing only.
2. This feature works for power nets only. For signal net p2t routing, refer to “compose trunk”. Intrinsically, interactive p2t and power p2t instantiate different routing tools.



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Power Routing Form: Via Insertion

In the *Power Routing* form, click on the **Vias** tab.

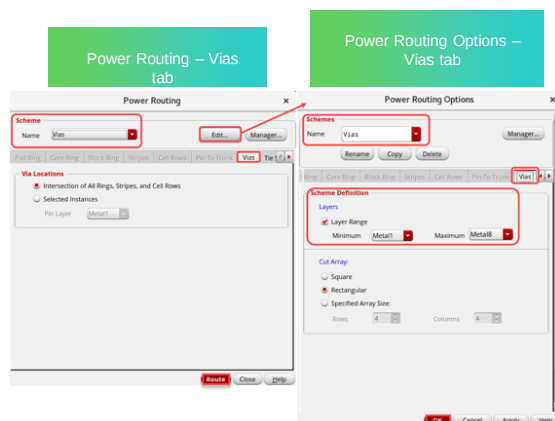
- *Vias routing* adds vias for interlayer connection at intersections of rings, rails, stripes, straps, and cell rows, or between instance pins and stripes.

Clicking the **Edit** button opens the *Power Routing Options* form.

- The **Vias** tab of the *Power Routing Options* form lets you specify layers and cut array types.

Via Insertion

- Besides p2t which automatically inserts vias after routing, other power net creation requires running via insertion step.
- User selects type of via insertion and via cut array types.



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Power Routing Form: Tie Shield

In the *Power Routing* form, click on the **Tie Shield** tab.

- *Tie Shield* wires to shield nets in the design.

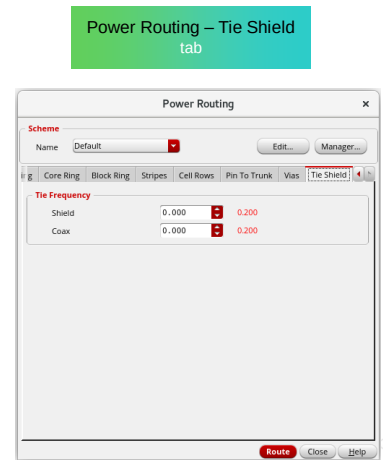
Tie Shield

In the Tie Frequency section, choose the distance between the ties.

- Choose the frequency of the *Shield* ties and the frequency of the *Coax* shield ties.
- Click **Route** to tie the shield wires to the nets.

Note:

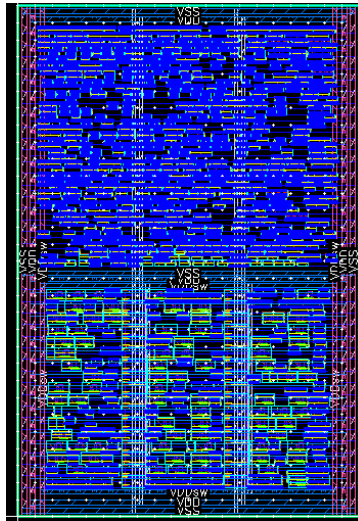
- The **Tie Shield** tab is a generic tool for connecting the shield nets (signal or power net type) to desired power nets.
- No *Edit* options form is provided.



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Example: Power Routed Design

An example of a power routed design.



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Routing the Power Components

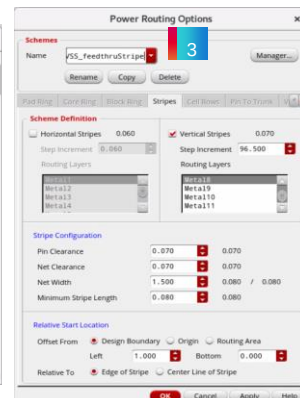
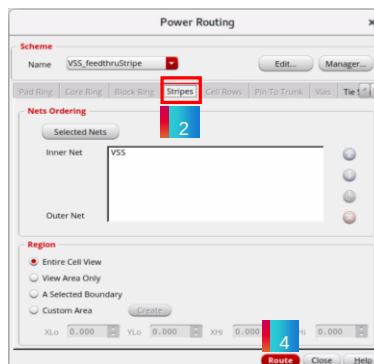
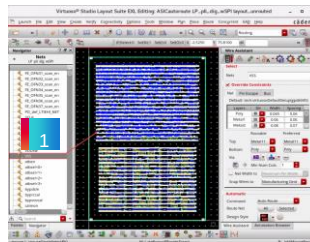


Using the **Power Routing** form, you can create power components in your design.

To route the power components in your design:

1. First, select at least one power or ground net before starting any power routing operation.
2. Then, in the *Power Routing* form, click the tab (**Stripes**) of the component you want to route.
3. Set all the options in the scheme subform for that power routing component.
4. Next, click the **Route** button at the bottom of the *Power Routing* form to perform that particular power routing operation.

Now, the *Power Routing* is performed for the selected power routing component.



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Routing the Power Components

Using the **Power Routing** form, you can create power components in your design.

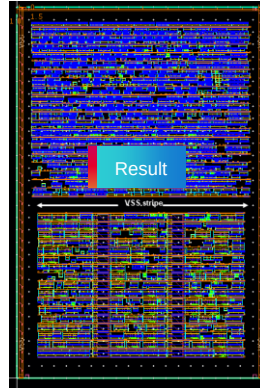
To route the power components in your design:

1. First, select at least one power or ground net before starting any power routing operation.
2. Then, in the *Power Routing* form, click the tab of the component you want to route.
3. Set all the options in the scheme subform for that power routing component.
4. Next, click the *Route* button at the bottom of the *Power Routing* form to perform that particular power routing operation.

Now, the *Power Routing* is performed for the selected power routing component.

Result: Routing the Power Components

Result: The *Power Routing* is performed for the selected power routing component.



The *Power Routing Options* form, which lets you view and edit the scheme package for each power routing option, can be accessed by choosing the *Edit* button from each of the tabs on the power routing form.

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Result: Routing the Power Components

Result: The *Power Routing* is performed for the selected power routing component.

In this example, after selecting the net VSS, from the Stripes tab, all the required options are set in the scheme subform and routing is performed.

You can see that, the Vertical Feedthru Stripe connection for the VSS pin is created.

The *Power Routing Options* form which lets you view and edit the scheme package for each power routing option can be accessed by choosing the *Edit* button from each of the tabs on the power routing form.

Loading the Schemes During Startup



You can load and set up the schemes automatically when starting **virtuoso** which you can use to route the power components in your design.

For each scheme you want to load, add the following SKILL command to the `.cdsinit` file in your working directory:

```
(load "mySchemeFilePath/mySchemeFileName")
```

Note: Here, *mySchemeFileName* is the file that will load all the schemes during startup.



Loading the Schemes During Startup

You can load and set up the schemes automatically when starting **virtuoso** which you can use to route the power components in your design.

For each scheme you want to load, add the following SKILL command to the `.cdsinit` file in your working directory:

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▪ (load "mySchemeFilePath/mySchemeFileName")
```

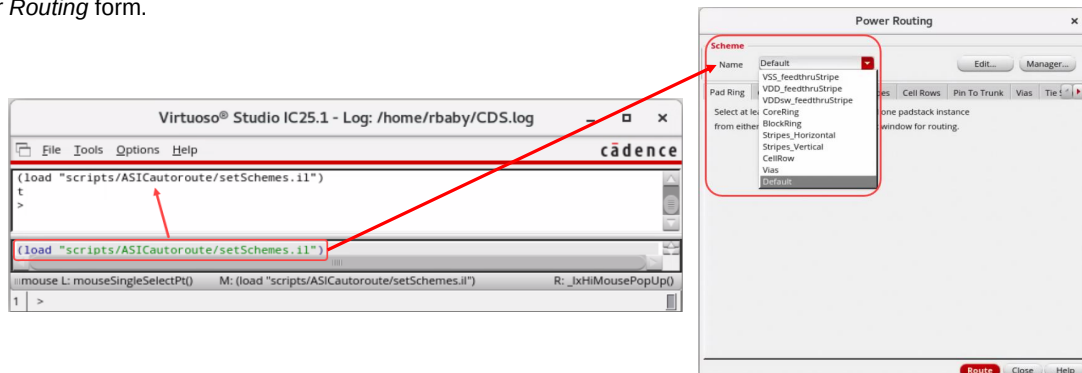
Note: Here *mySchemeFileName* is the file which will load all the schemes during startup.

Example: Loading the Schemes During Startup

In this example, to load the schemes automatically during startup, add the below SKILL code in your .cdsinit file:

```
(load "scripts/ASICautoroute/setSchemes.il")
```

Result: All the schemes in the *setSchemes.il* file are loaded automatically, which show up from the *Name* drop-down list box in the *Power Routing* form.



Example: Loading the Schemes During Startup

In this example, to load the schemes automatically during startup, add the below SKILL code in your .cdsinit file:

```
▪ (load "scripts/ASICautoroute/setSchemes.il")
```

Result: All the schemes in the *setSchemes.il* file are loaded automatically which show up from the *Name* drop-down list box in the *Power Routing* form.

Setting the Packages for the Scheme

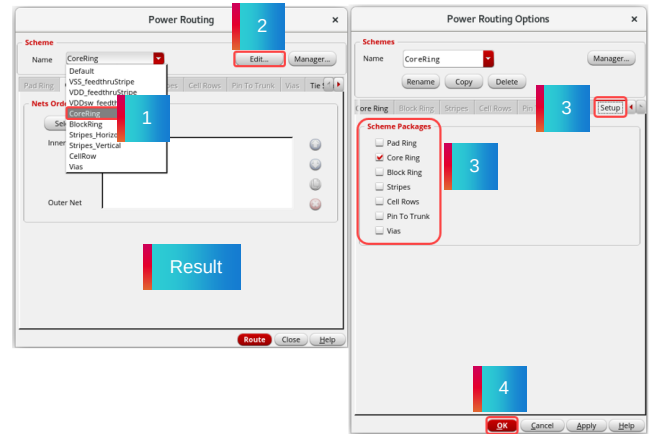


You can select and use the packages available in the scheme to route the power components in your design.

To select the packages available in the scheme. In the Power Routing form,

1. Select the scheme you want to use from the **Name** drop-down list box.
2. Then, click **Edit** to display the Power Routing Options form.
3. Next, from the **Setup** tab, in the **Scheme Packages** section, select the packages that are part of this scheme.
4. Click **OK** to apply the settings and close the Power Routing Options form.

Result: The selected Scheme Packages are applied.



Setting the Packages for the Scheme

You can select and use the packages available in the scheme to route the power components in your design.

To select the packages available in the scheme:

1. In the Power Routing form, select the scheme you want to use from the **Name** drop-down list box.
2. Then, in the Power Routing form, click **Edit** to display the Power Routing Options form.
3. Next, in the Power Routing Options form, from the **Setup** tab, in the **Scheme Packages** section, select the packages that are part of this scheme.
4. Click **OK** to apply the settings and close the Power Routing Options form.

Result: The selected Scheme Packages are applied.

Creating New Schemes



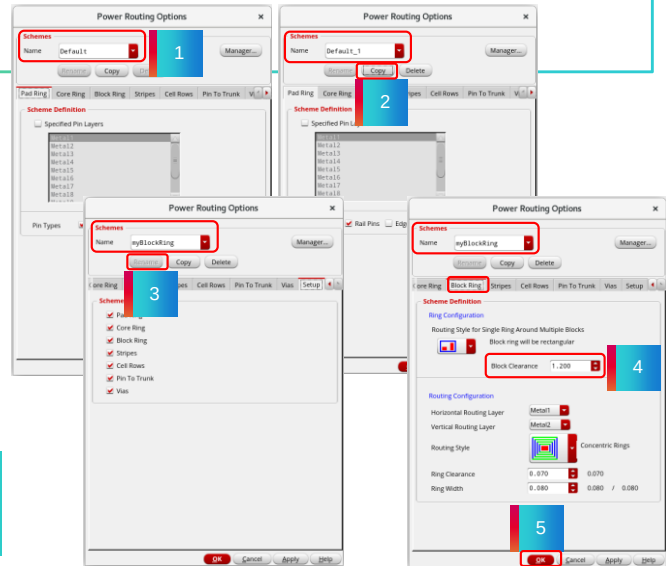
You can create new schemes for your power routing tasks to construct the common power components for your design.

To create a new scheme:

1. In the Power Routing Options form, ensure that the **Name** field displays the name of the scheme as *Default*.
2. Next, click the **Copy** button, copying scheme *Default* to scheme *Default_n*. The default name is the original scheme name with *_n*, where *n* is a number, automatically appended.
3. Now, optionally, you can rename the scheme by entering the new name in the **Name** field and clicking **Rename**. Specify the scheme name as, say for example, *myBlockRing*.
4. Click any tab, say for example, the **Block Ring** tab and make some changes to the settings.
5. Click **OK** to save the *myBlockRing* scheme to the memory.

A new scheme is created and saved in the memory.

You cannot use the name Default when renaming a scheme. Default is an internal scheme that should not be overwritten or removed. The form will not allow you to rename a scheme Default or change the Default scheme.



Creating New Schemes

You can create new schemes for your power routing tasks to construct the common power components for your design.

To create a new scheme:

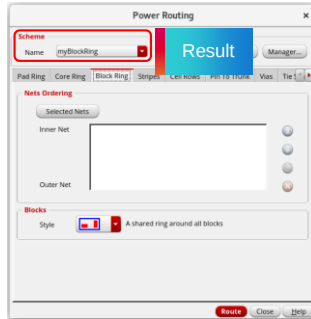
1. In the Power Routing Options form, ensure that the **Name** field displays the name of the scheme as *Default*.
2. Next, click the **Copy** button, copying scheme *Default* to scheme *Default_n*. The default name is the original scheme name with *_n*, where *n* is a number, automatically appended.
3. Now, optionally, you can rename the scheme by entering the new name in the **Name** field and clicking **Rename**. Specify the scheme name as, say for example, *myBlockRing*.
4. Click any tab, say for example, the **Block Ring** tab and make some changes to the settings.
5. Click **OK** to save the *myBlockRing* scheme to the memory.

A new scheme is created and saved in the memory.

You cannot use the name Default when renaming a scheme. Default is an internal scheme that should not be overwritten or removed. The form will not allow you to rename a scheme Default or change the Default scheme.

Result: Creating New Schemes

Result: A new scheme is created and saved in the memory.



Result: Creating New Schemes

Result: A new scheme is created and saved in the memory.

Deleting a Scheme



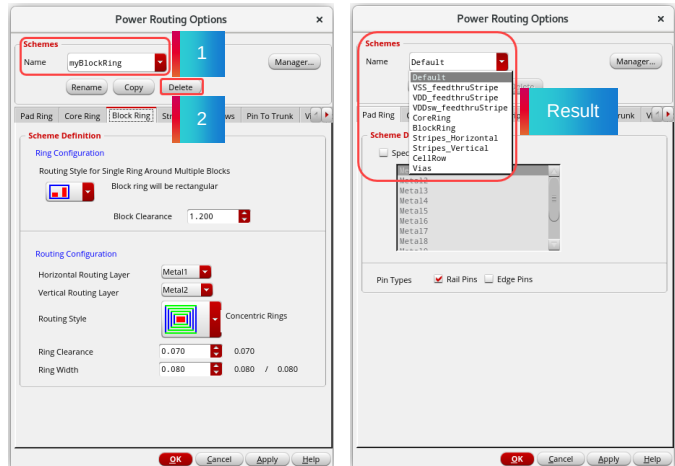
You can delete the schemes which are not required for your power routing tasks when constructing the common power components for your design.

To delete a scheme:

1. In the Power Routing Options form, display the name of the scheme to delete in the **Name** field.
2. Now, click the **Delete** button.

Result: The selected scheme is deleted from the **Schemes** section.

You cannot delete the Default scheme. Default is an internal scheme that should not be overwritten or removed. The form will not allow you to delete the Default scheme.



Deleting a Scheme

You can delete the schemes which are not required for your power routing tasks when constructing the common power components for your design.

To delete a scheme:

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Loading a Scheme



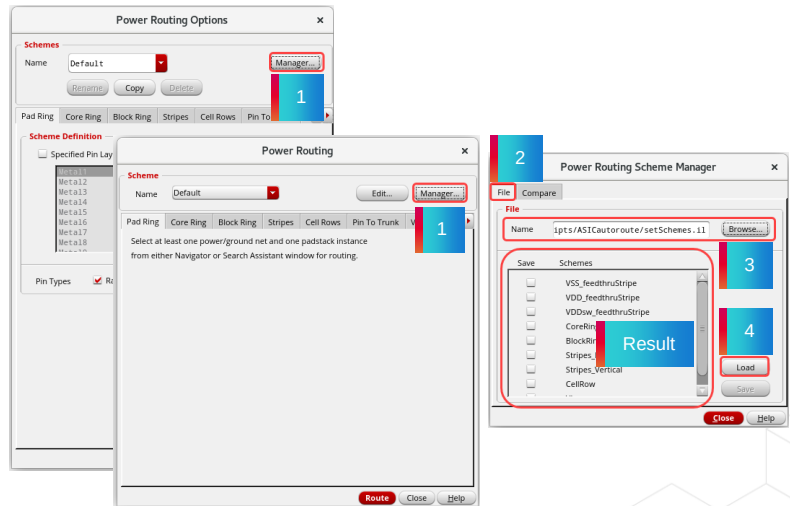
You can load a scheme from a file which you can use for your power routing tasks to construct the common power components for your design.

To load a scheme from a file:

1. Display the Power Routing Scheme Manager form by clicking the **Manager** button from either the Power Routing or Power Routing Options form.
2. Next, ensure you are in the **File** tab of the Power Routing Scheme Manager form.
3. Now, enter the name of the file containing the schemes in the **Name** field. Alternatively, you can use the **Browse** button to select the file containing the schemes.
4. Then, click the **Load** button to load all the schemes in the file into the power routing scheme database.

Result: All loaded schemes now appear in the Schemes section list box.

The form will not duplicate schemes with identical names. If you load a scheme having the same name as an existing scheme, the existing scheme is overwritten.



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Loading a Scheme

You can load a scheme from a file which you can use for your power routing tasks to construct the common power components for your design.

To load a scheme from a file:

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3. Now, enter the name of the file containing the schemes in the **Name** field. Alternatively, you can use the **Browse** button to select the file containing the schemes.
4. Then, click the **Load** button to load all the schemes in the file into the power routing scheme database.

Result: Now, all the loaded schemes appear in the list box of the *Schemes* section.

The form will not duplicate schemes with identical names. If you load a scheme having the same name as an existing scheme, the existing scheme is overwritten.

Saving a Scheme File



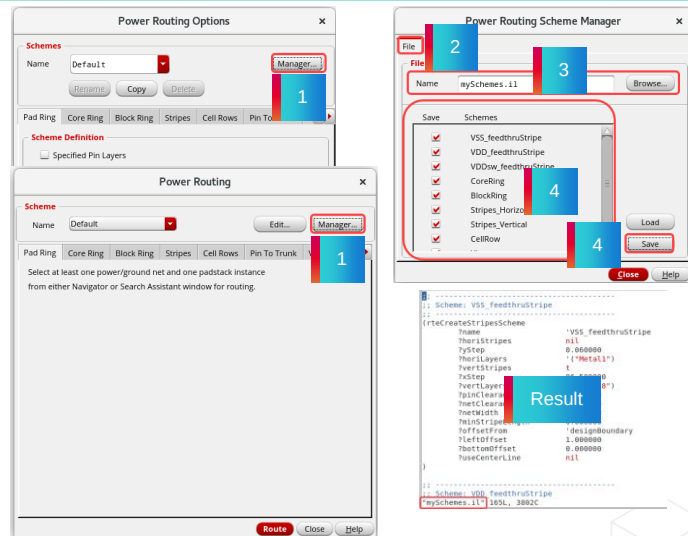
You can save the schemes to a file for reuse.

To save the schemes to a file:

1. Display the Power Routing Scheme Manager form by clicking the **Manager** button from either the Power Routing or Power Routing Options form.
2. Next, ensure you are in the **File** tab of the Power Routing Scheme Manager form. Also, see that the available schemes are listed in the **Schemes** list box.
3. Now, in the **Name** field, enter the name of the file to which you want to save the schemes. Alternatively, you can use the **Browse** button to select the filename.
4. Then, click the checkboxes of the schemes to save in the file. The **Save** button is enabled. Click the **Save** button.

Result: The selected schemes are now saved to the specified file name.

If you save schemes to an existing file, the original contents of the file are removed and replaced by the newly saved schemes.



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Saving a Scheme File

You can save the schemes to a file for reuse.

To save the schemes to a file:

1. Display the Power Routing Scheme Manager form by clicking the **Manager** button from either the Power Routing or Power Routing Options form.
2. Next, ensure you are in the **File** tab of the Power Routing Scheme Manager form. Also, see that the available schemes are listed in the **Schemes** list box.
3. Now, in the **Name** field, enter the name of the file to which you want to save the schemes. Alternatively, you can use the **Browse** button to select the filename.
4. Then, click the checkboxes of the schemes to save in the file. The **Save** button is enabled; Click the **Save** button.

Result: Now, the selected schemes are saved to the given file name.

If you save schemes to an existing file, the original contents of the file are removed and replaced by the newly saved schemes.

Comparing the Schemes

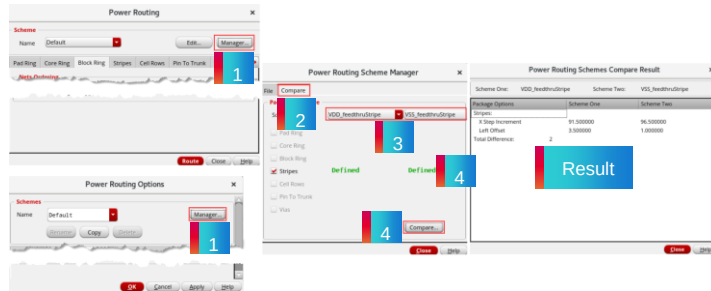


You can compare the packages defined in two schemes.

To compare the packages defined in two schemes:

1. Display the Power Routing Scheme Manager form by clicking the **Manager** button from either the Power Routing or Power Routing Options form.
2. Next, click the **Compare** tab in the Power Routing Scheme Manager form.
3. Now, select two schemes from the **Schemes** drop-down list boxes. The form indicates which packages are defined for each scheme.
4. Then, click the checkboxes of the scheme packages you want to compare and click the **Compare** button.

Result: Now, the comparison results are displayed in the *Power Routing Schemes Compare Result* form.



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Comparing the Schemes

You can compare the packages defined in two schemes.

To compare the packages defined in two schemes:

1. Display the Power Routing Scheme Manager form by clicking the **Manager** button from either the Power Routing or Power Routing Options form.
2. Next, click the **Compare** tab in the Power Routing Scheme Manager form.
3. Now, select two schemes from the **Schemes** drop-down list boxes. The form indicates which packages are defined for each scheme.
4. Then, click the checkboxes of the scheme packages you want to compare and click the **Compare** button.

Result: Now, the comparison results are displayed in the *Power Routing Schemes Compare Result* form.

Using Power Routing Functions



You can use the Virtuoso Space-based Router's **Power Routing Functions** for your power routing tasks to construct the common power components for your design.

- You can use the tabs in the *Power Routing* form to set up the options for each power routing operation.
- To create power routing using SKILL, you can use the power routing functions.

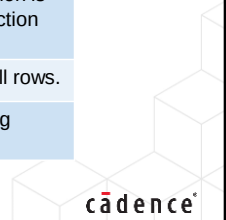


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Power Routing Functions

To create power routing using SKILL, use the power routing functions.

SKILL Function	Function Description
<i>rtePowerRouteBlockRing</i>	Executes block ring routing for a layout cellview.
<i>rtePowerRouteCellRow</i>	Executes cell row routing for a layout cellview.
<i>rtePowerRouteCoreRing</i>	Executes core ring power routing for a layout cellview.
<i>rtePowerRoutePadRing</i>	Executes pad ring power routing for a layout cellview.
<i>rtePowerRoutePinToTrunk</i>	Executes pin to trunk routing for a layout cellview. A trunk is usually a power net created by power routing commands such as pad ring, core ring, block ring, stripes, or cell rows.
<i>rtePowerRouteStripes</i>	Executes stripes power routing for a layout cellview.
<i>rtePowerRouteTrimStripes</i>	Trim existing stripes created by the <i>rtePowerRouteStripes</i> function. This function is commonly executed after running <i>rtePowerRouteStripes</i> . You can run the function either before or after <i>rtePowerRouteVialInsertion</i> .
<i>rtePowerRouteVialInsertion</i>	Executes via insertion between previously routed power rings, stripes, and cell rows.
<i>rtePowerRouteTieShield</i>	Ties shield wires to the shield nets in the entire design. Executed after creating shield nets either automatically or interactively.



Power Routing Functions

To create power routing using SKILL, use the power routing functions.

Using Power Routing Scheme Creation Functions



You can use the Virtuoso Space-based Router's **Power Routing Scheme Creation Functions** to create new schemes for your power routing tasks to construct the common power components for your design.

- You can create new schemes manually by setting up the options for each power routing operation.
- To create power routing schemes using SKILL, you can use the power routing scheme creation functions.



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Power Routing Scheme Creation Functions

To create power routing schemes using SKILL, use the power routing scheme creation functions.

SKILL Function	Function Description
<i>rteCreateBlockRingScheme</i>	Creates a block ring routing scheme.
<i>rteCreateCellRowsScheme</i>	Creates a cell rows routing scheme.
<i>rteCreateCoreRingScheme</i>	Creates a core ring routing scheme.
<i>rteCreatePadRingScheme</i>	Creates a pad ring routing scheme.
<i>rteCreatePinToTrunkScheme</i>	Creates a pin to trunk routing scheme. A trunk is usually a power net created by power routing commands such as pad ring, core ring, block ring, stripes, or cell rows.
<i>rteCreateStripesScheme</i>	Creates a stripes routing scheme.
<i>rteCreateViasScheme</i>	Creates a via insertion scheme.



Power Routing Scheme Creation Functions

To create power routing schemes using SKILL, use the power routing scheme creation functions.

Running Signal Routing

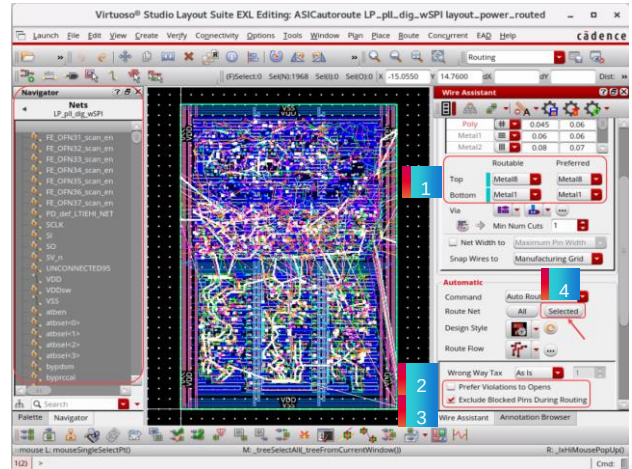


You can run the automatic signal routing using the Wire Assistant.

To run the signal routing, in the layout canvas, in the Wire Assistant, set the required values:

1. *Routable: Top: **Metal8**; Bottom: **Metal1**.*
2. *Prefer Violations to Opens – **Off**.*
3. *Exclude Blocked Pins During Routing – **On**.*
4. Click the **Selected** button in the Wire Assistant **Auto Routing** menu.

The design is auto-routed using the Wire Assistant.



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Running Signal Routing

You can run the automatic signal routing using the Wire Assistant.

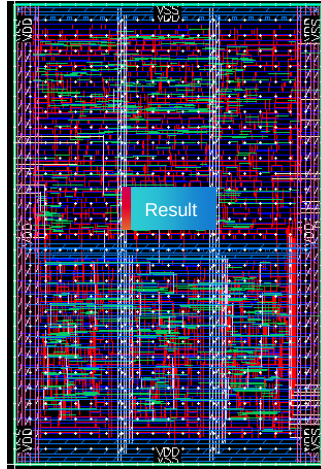
To run the signal routing, in the layout canvas, in the Wire Assistant, set the required values:

1. *Routable: Top: **Metal8**; Bottom: **Metal1**.*
2. *Prefer Violations to Opens – **Off**.*
3. *Exclude Blocked Pins During Routing – **On**.*
4. Click the **Selected** button in the Wire Assistant **Auto Routing** menu.

The design is auto routed using the Wire Assistant.

Result: Running Signal Routing

Result: The design is auto-signal routed using the Wire Assistant.



Result: Running Signal Routing

Result: The design is auto signal routed using the Wire Assistant.

Invoking the Connectivity Form

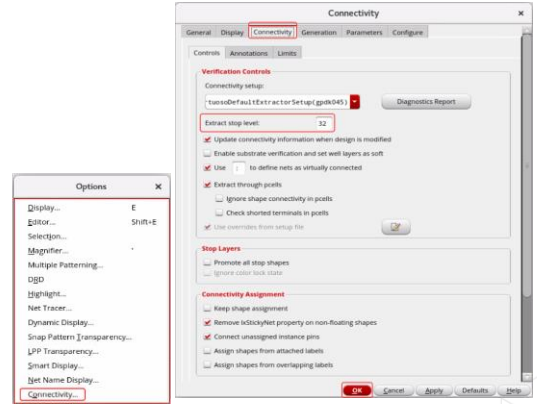


In the layout canvas, choose the **Options – Connectivity** menu command to open the Connectivity form.

Use the *Connectivity* form to set Layout XL options for either the current cellview or the Layout XL session.

In this example:

- In the **Connectivity** tab, *Extract stop level* is set as 32.
 - This option specifies how much of the design hierarchy the extractor considers when extracting the top-level design.



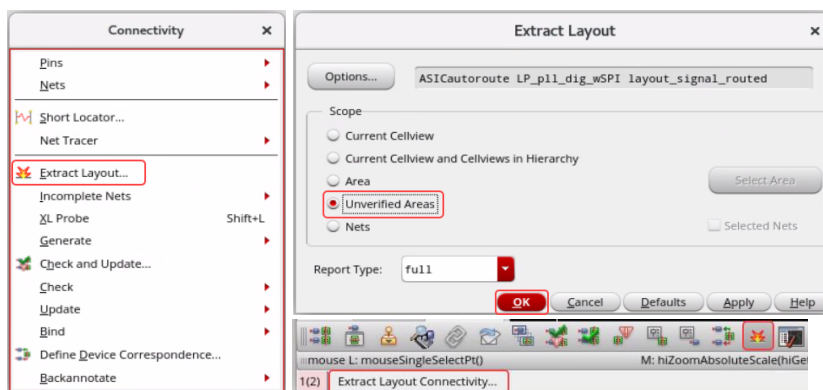
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Extracting the Design



In the layout canvas, choose the **Connectivity – Extract Layout** menu command or use the **Extract Layout Connectivity** icon in the Layout XL toolbar to open the Extract Layout form.

Use the *Extract Layout* form to run connectivity extraction on the layout design.



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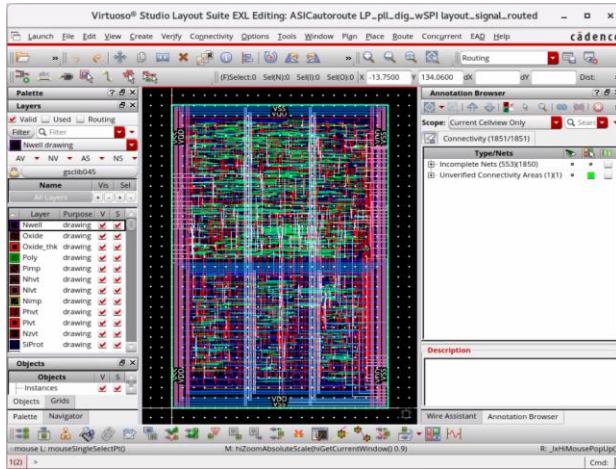
Extracting the Design

In the layout canvas, choose the **Connectivity – Extract Layout** menu command or use the **Extract Layout Connectivity** icon in the Layout XL toolbar to open the Extract Layout form.

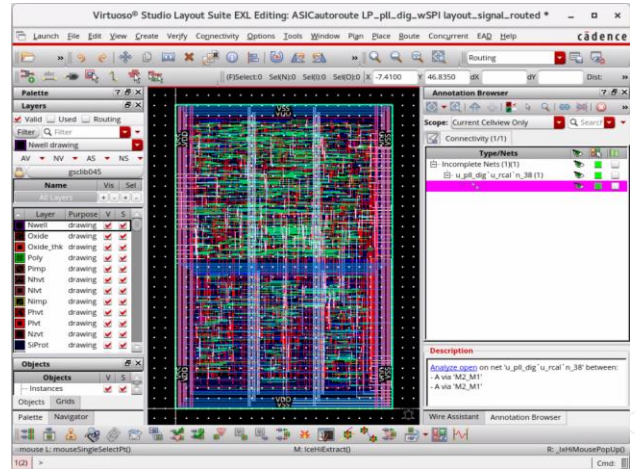
Use the *Extract Layout* form to run connectivity extraction on the layout design.

Result: Extracting the Design

The design before extraction



The design after extraction



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Result: Extracting the Design

You see the design before extraction and after extraction here.

Fixing the Connectivity Errors

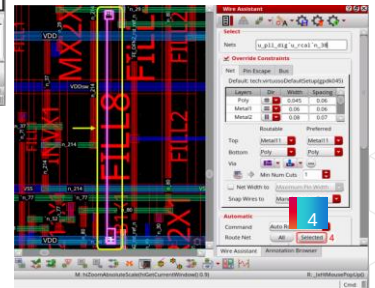
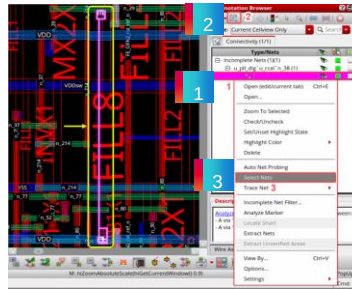


You can fix the connectivity errors in the design using the Annotation Browser and Wire Assistant.

To fix the connectivity errors, in the layout design, set the following:

1. First, select the incomplete nets in the Annotation Browser assistant.
2. Then, click the **Zoom To Selected** button in the Annotation Browser assistant.
3. Next, **right-click** and choose **Select Nets** in the Annotation Browser assistant.
4. Now, click the **Selected** button in the Wire assistant **Auto Routing** menu.

The connectivity errors are removed.



Fixing the Connectivity Errors

You can fix the connectivity errors in the design using the Annotation Browser assistant and Wire assistant.

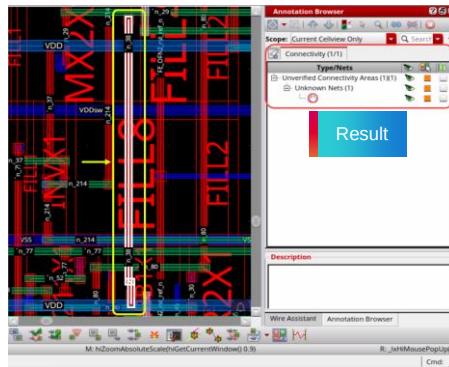
To fix the connectivity errors, in the layout design, set the following:

1. First, select the incomplete nets in the Annotation Browser assistant.
2. Then, click the **Zoom To Selected** button in the Annotation Browser assistant.
3. Next, **right-click** and choose **Select Nets** in the Annotation Browser assistant.
4. Now, click the **Selected** button in the Wire Assistant **Auto Routing** menu.

The connectivity errors are removed.

Result: Fixing the Connectivity Errors

Result: The connectivity errors are removed after auto-routing the design.

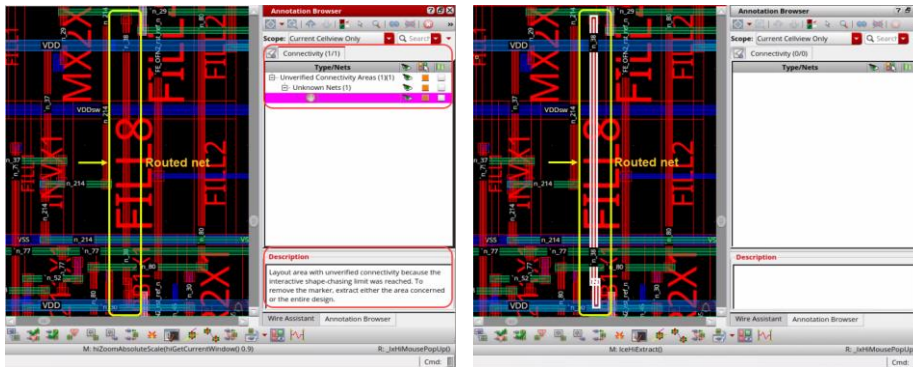


Result: Fixing the Connectivity Errors

Result: The connectivity errors are removed after auto routing the design.

Result: Fixing the Connectivity Errors (continued)

Result: The unverified connectivity markers seen in the Annotation Browser assistant are removed after running the connectivity extraction in the design.



Result: Fixing the Connectivity Errors

Result: The unverified connectivity markers seen in the Annotation Browser assistant are removed after running the connectivity extraction in the design.

Quiz: Virtuoso Space-based Router



You can use the Virtuoso Space-based Router's ----- Router to construct the common power components in your design.

Enter the answer in the blank field.

Answer: Power

You can use the Virtuoso Space-based Router's Power Router to construct the common power components in your design.



Answer: Power

You can use the Virtuoso Space-based Router's Power Router to construct the common power components in your design.

Quiz: Pad Ring Routing



Pad Ring routing adds a ring between pad instances on the periphery of the design.

- A. True
- B. False

Answer: A

Pad rings are routed between pads on the periphery of the design. You can route to rail pins and/or edge pins on pads.



Answer: A

Pad rings are routed between pads on the periphery of the design. You can route to rail pins and/or edge pins on pads.

Quiz: The Power Routing Form



What are the tabs in the Power Routing form which you can use to set up options for each power routing operation?

Select all that apply.

- A. Name
- B. Edit
- C. Manager
- D. Route
- E. Save
- F. All of the above

Answer: A, B, C, and D

You can use the Name, Edit, Manager, and Route tabs in the Power Routing form to set up options for each power routing operation.



Answer: A, B, C, and D

You can use the Name, Edit, Manager, and Route tabs in the Power Routing form to set up options for each power routing operation.

Quiz: The Power Routing Options Form



The Power Routing Options form which lets you view and edit the scheme package for each power routing option can be accessed by choosing the ---- button from each of the tabs on the power routing form.

Enter the answer in the blank field.

Answer: Edit

The Power Routing Options form which lets you view and edit the scheme package for each power routing option can be accessed by choosing the Edit button from each of the tabs on the power routing form.



Answer: Edit

The Power Routing Options form which lets you view and edit the scheme package for each power routing option can be accessed by choosing the Edit button from each of the tabs on the power routing form.

Quiz: The Power Routing Options Form



Which tab under the Power Routing Options form can you use to enable/disable certain scheme packages in power routing options?

Select all that apply.

- A. Setup
- B. Manager
- C. Edit
- D. Scheme
- E. None of the above

Answer: A

You can use the Setup tab under the Power Routing Options form to enable/disable certain scheme packages in power routing options.



Answer: A

You can use the Setup tab under the Power Routing Options form to enable/disable certain scheme packages in power routing options.

Quiz: Renaming a Scheme



You can use the name Default when renaming a scheme in the Power Routing Options form.

- A. True
- B. False

Answer: B

You cannot use the name Default when renaming a scheme. Default is an internal scheme that should not be overwritten or removed. The form will not allow you to rename a scheme Default or change the Default scheme.



Answer: B

You cannot use the name Default when renaming a scheme. Default is an internal scheme that should not be overwritten or removed. The form will not allow you to rename a scheme Default or change the Default scheme.

Module Summary

This module covered:

- Implement a full power routing structure using Virtuoso Space-based Router (VSR).
- Create rings, meshes, straps, and trunk connections.
- Manage routing schemes for repeatable flows.
- Insert vias and tie shields.
- Run signal routing and extract the design.
- Fix connectivity issues and verify the final layout.



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Labs



Lab 3-1 Power Routing Using VSR

Lab 3-2 Signal Routing

Lab 3-3 Connectivity Verification and DRCs

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Module 4

Course Conclusions

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Summary

In this course, you learned to

- Analyze the constraint-driven flow
- Set up and run power routing



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References

Refer to these links for more details on the topics discussed in this course.

- [Virtuoso Unified Custom Constraints User Guide Product Version IC25.1](#)
- [Virtuoso Unified Custom Constraints SKILL Reference Product Version IC25.1](#)
- [Virtuoso Space-based Router User Guide Product Version IC25.1](#)
- [Virtuoso Layout Suite SKILL Reference Product Version IC25.1](#)



- [Virtuoso Unified Custom Constraints User Guide Product Version IC25.1](#)
- [Virtuoso Unified Custom Constraints SKILL Reference Product Version IC25.1](#)
- [Virtuoso Space-based Router User Guide Product Version IC25.1](#)
- [Virtuoso Layout Suite SKILL Reference Product Version IC25.1](#)



Module 5

Next Steps

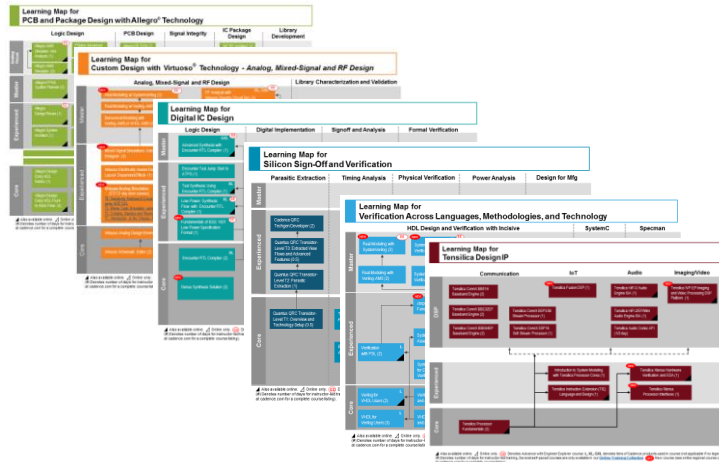
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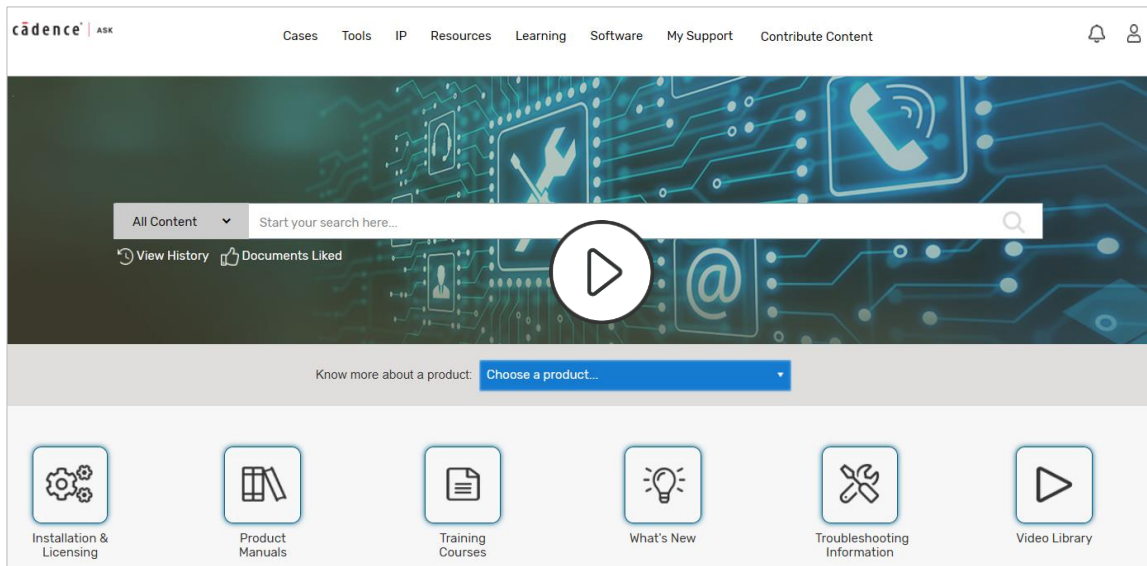
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Wrap Up

- Complete Post Assessment, if provided
- Complete the Course Evaluation
- Get a Certificate of Course Completion
- Complete the Course's Digital Badge exam to earn your Badge

Thank you!



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